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(54) **ENGINEERED CPN1 CONSTRUCTS AND VARIANTS**

(71) Applicant: **Vertex Pharmaceuticals Incorporated**,
Boston, MA (US)

(72) Inventors: **Aaron AGUHOB**, South San Francisco,
CA (US); **Natacha LE MOAN**, San
Francisco, CA (US); **Grant E.**
BLOUSE, Burlingame, CA (US); **Arzu**
SANDIKCI, Berkeley, CA (US); **Ryan**
PAVLOVICZ, Seattle, WA (US);
Amanda LOSHBAUGH, Seattle, WA
(US); **Yifan SONG**, Seattle, WA (US);
Timothy MYLES, South San
Francisco, CA (US)

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2319/31 (2013.01); **C07K 2319/50** (2013.01)

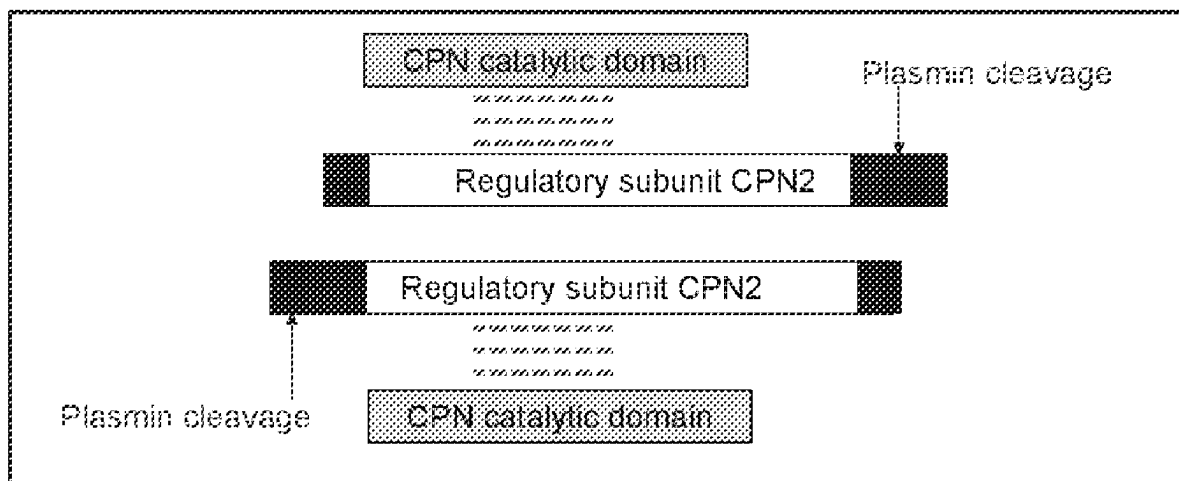
(57)

ABSTRACT

Provided herein are carboxypeptidase N catalytic subunit (CPN1) variants, comprising at least one modification with respect to a wild type carboxypeptidase N1 of the M14 family, wherein the variants have at least one improved characteristic as compared to the wild type CPN1. Also provided herein are fusion constructs comprising CPN1, or variants thereof. Also provided herein are methods of making and using such variants and constructs. The variant and constructs provided herein may be useful for treating diseases or conditions associated with dysregulation of the complement system.

Specification includes a Sequence Listing.

CPN heterotetramer



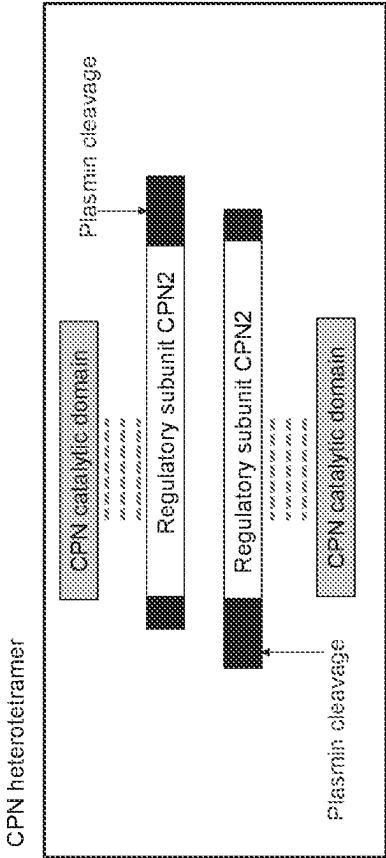


FIG. 1A

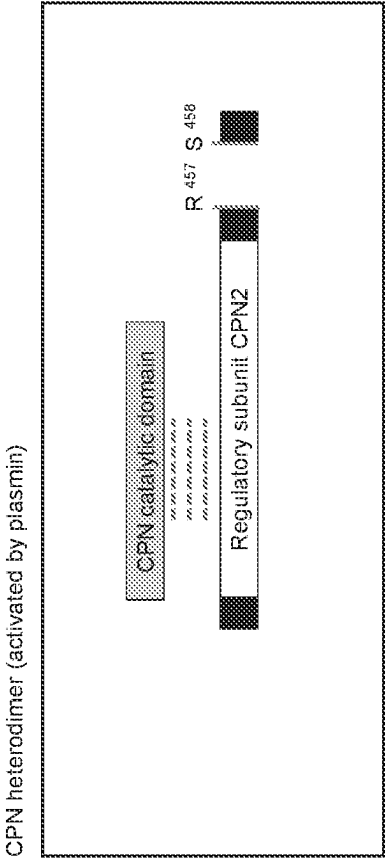


FIG. 1B

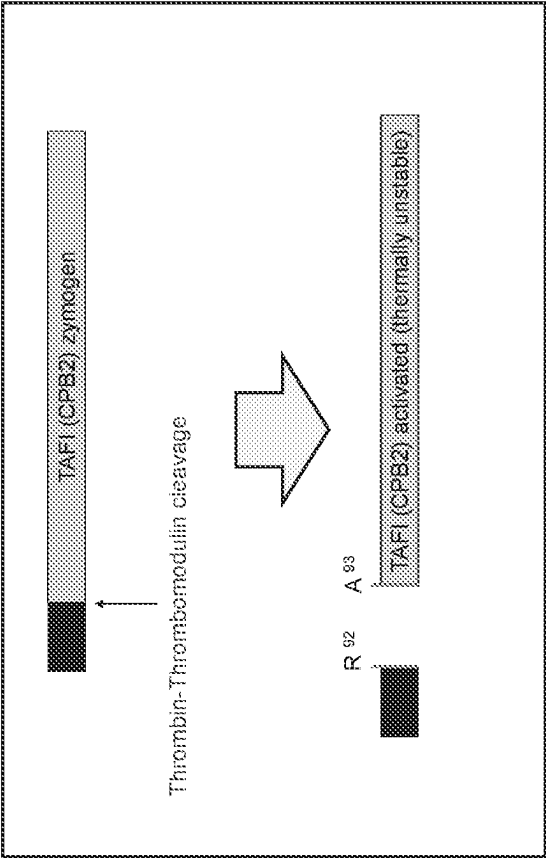


FIG. 2

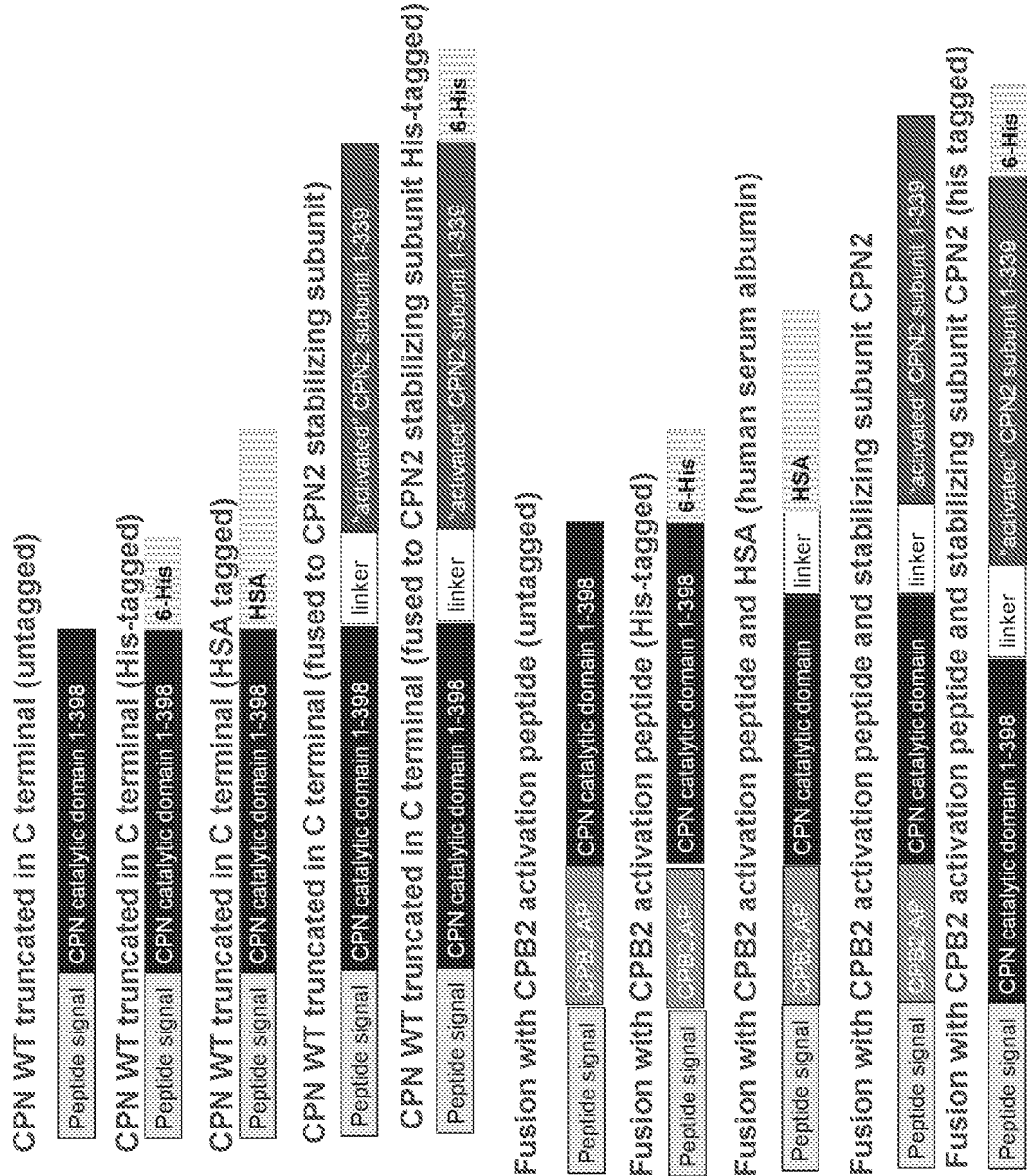


FIG. 3

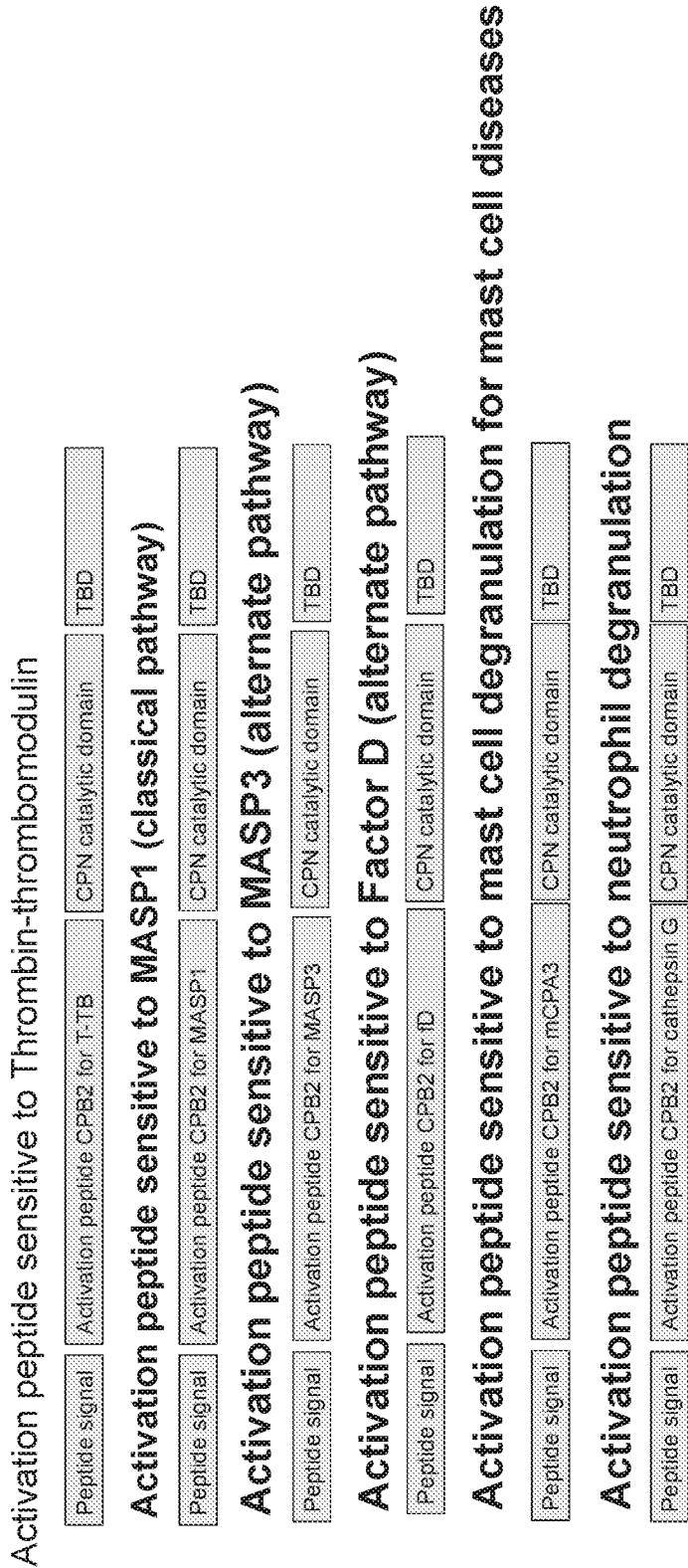


FIG. 4A

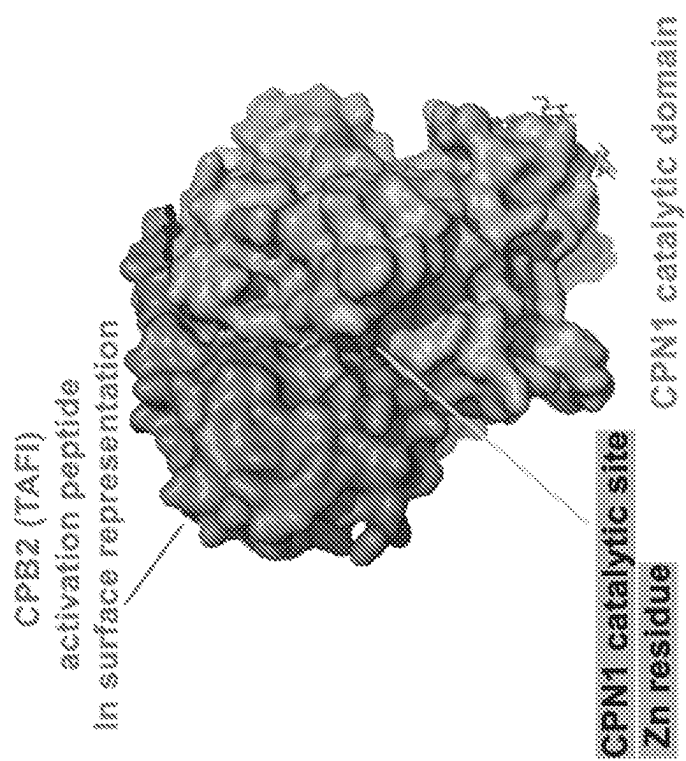


FIG. 4B

Fusion with CPB2 activation peptide (untagged)



FIG. 4C

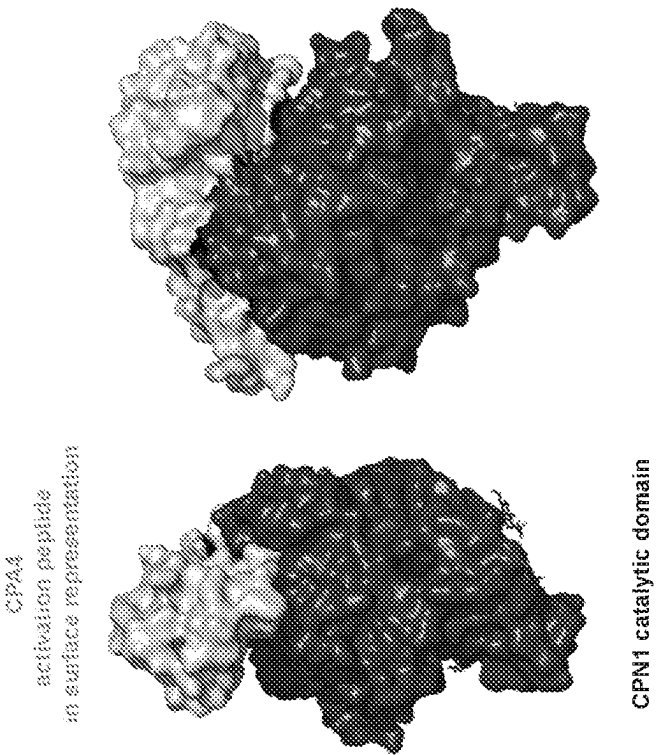


FIG. 4E

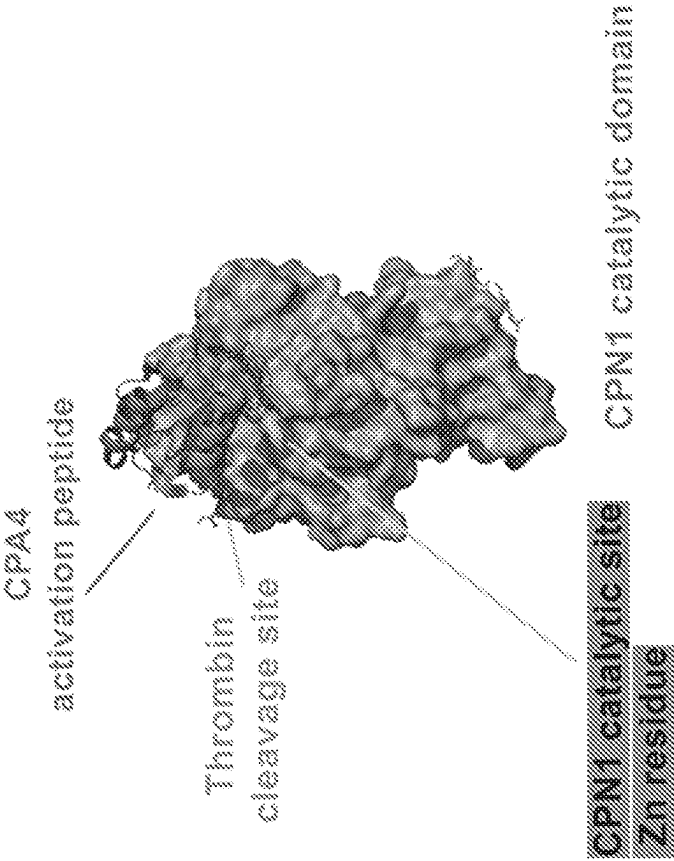


FIG. 4D

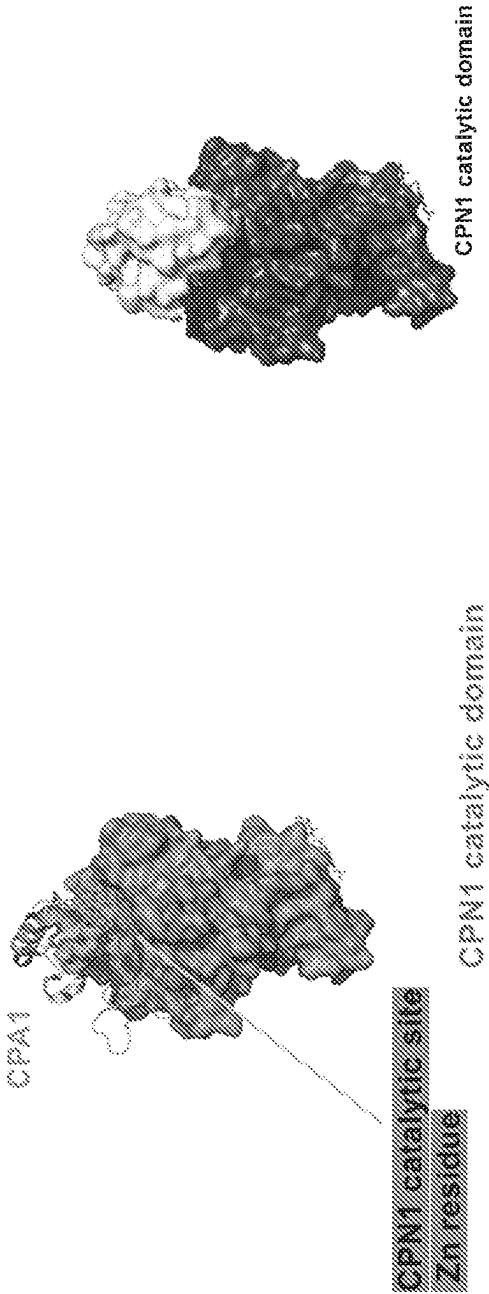


FIG. 4F

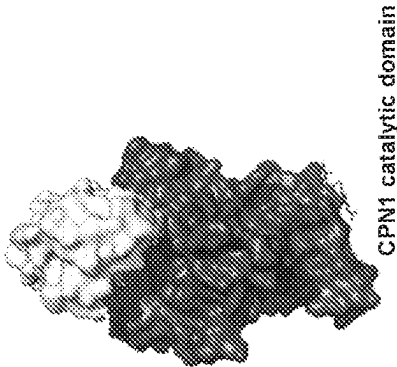


FIG. 4G

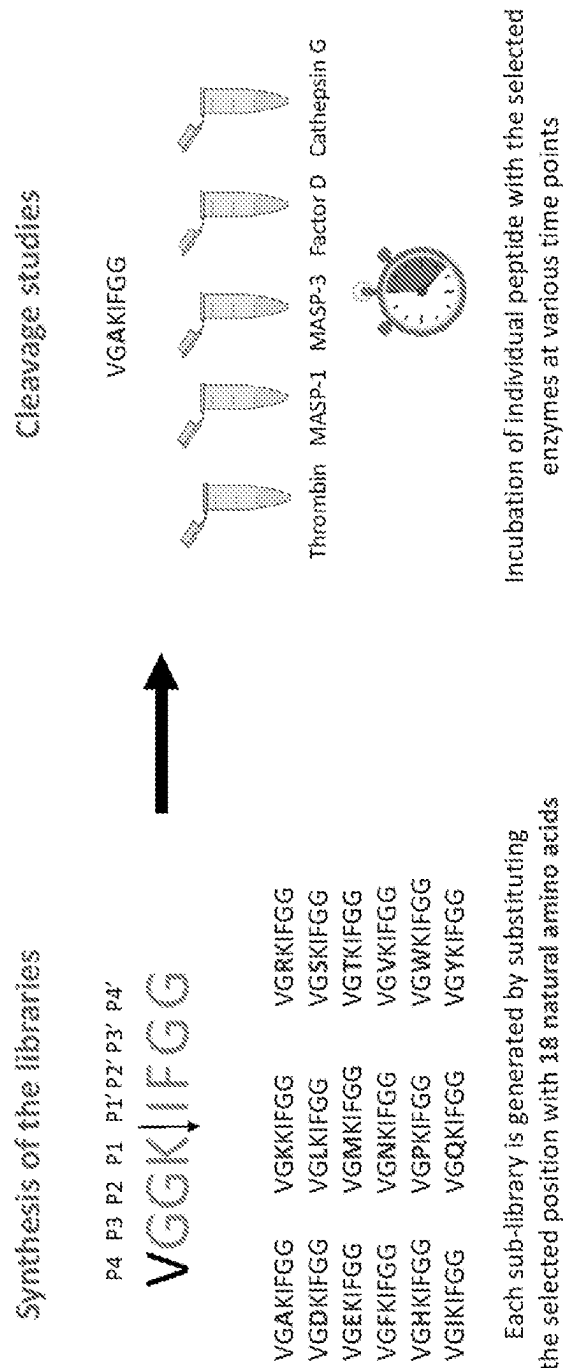


FIG. 5

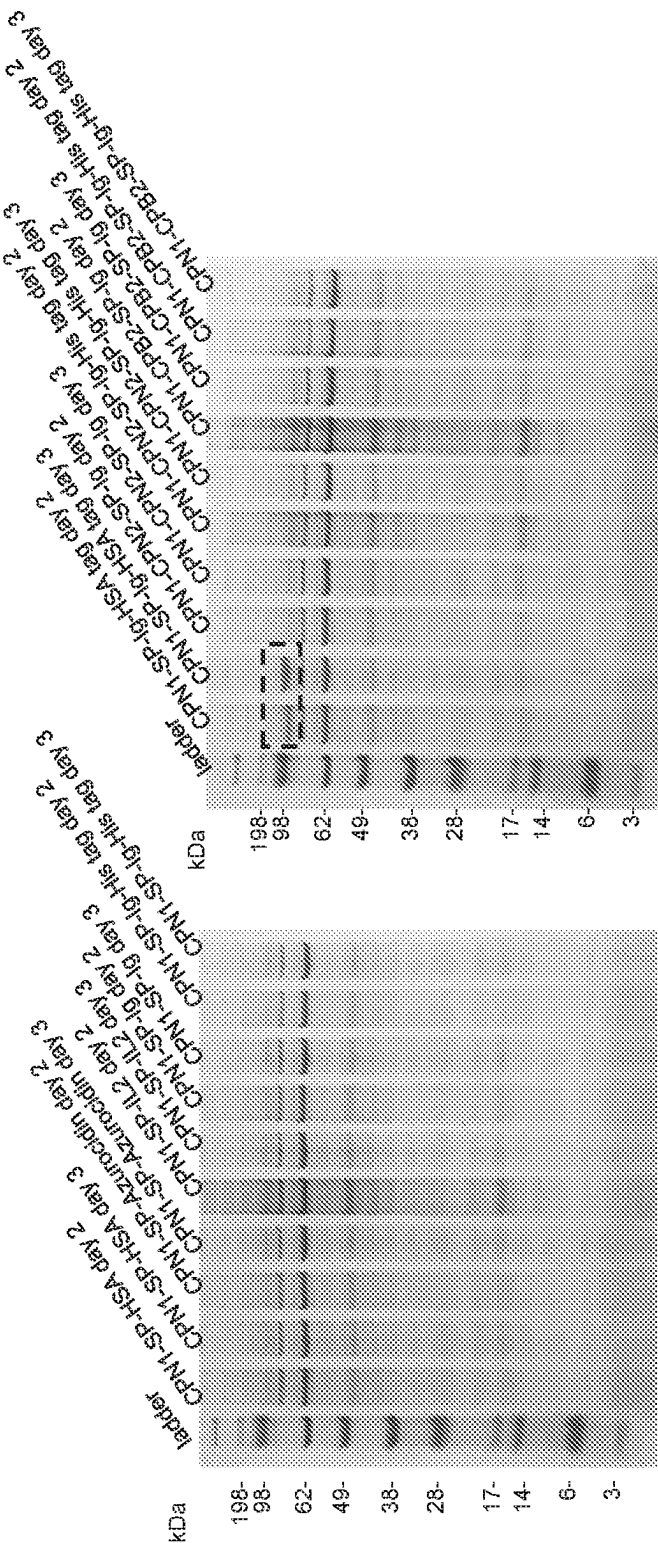


FIG. 6B

FIG. 6A

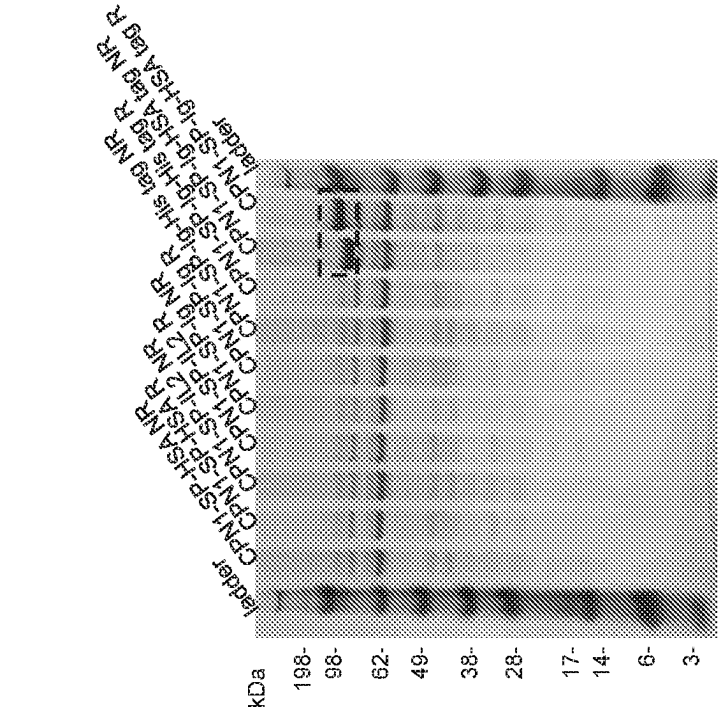


FIG. 6D

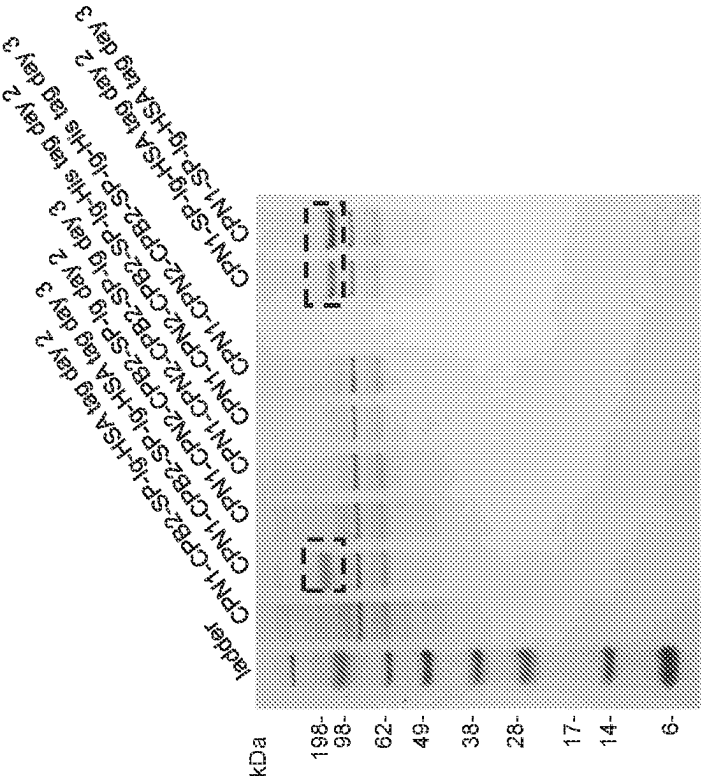


FIG. 6C

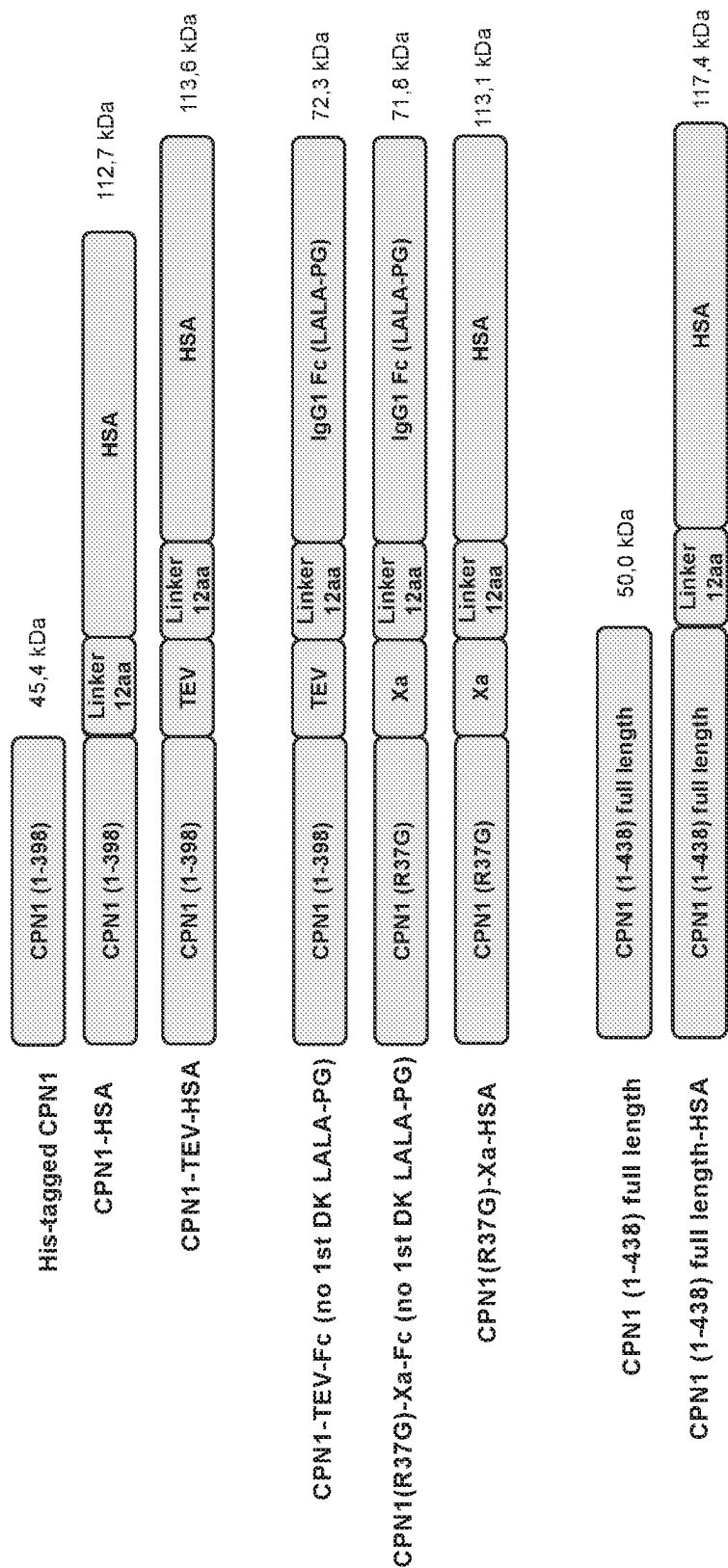


FIG. 6E

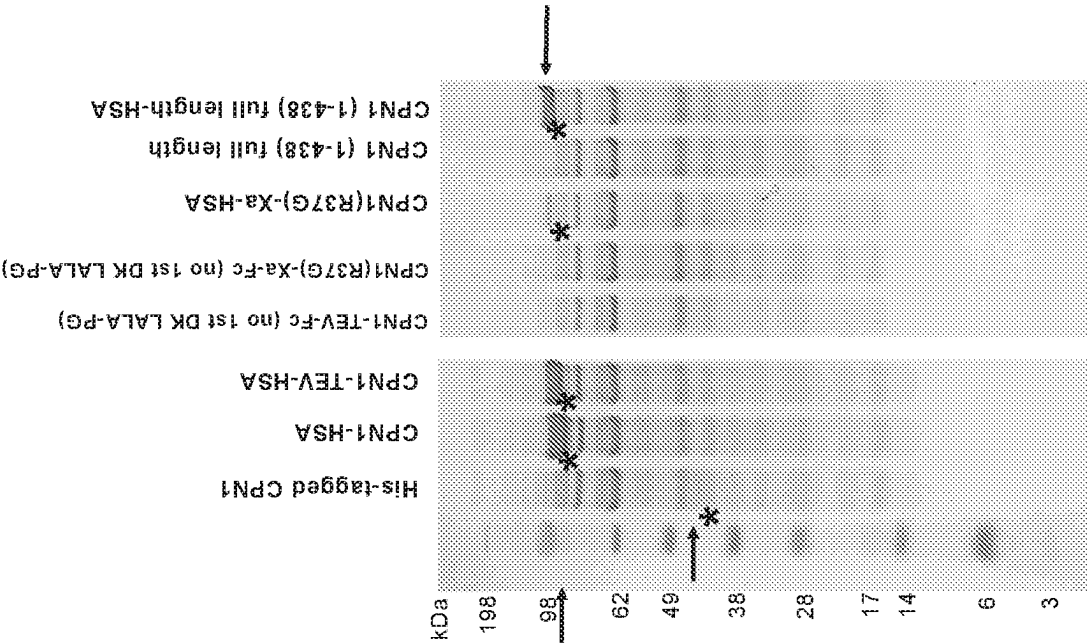


FIG. 6F

His-Sumo-CPN1-HSA

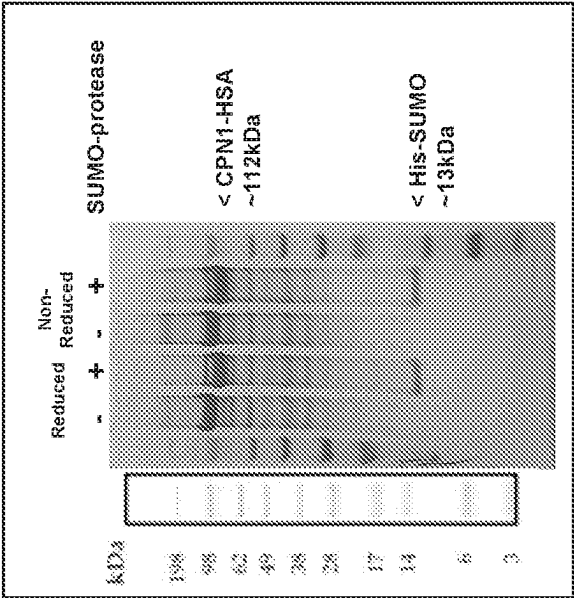


FIG. 6H



His-Sumo-CPN1-HSA

FIG. 6G

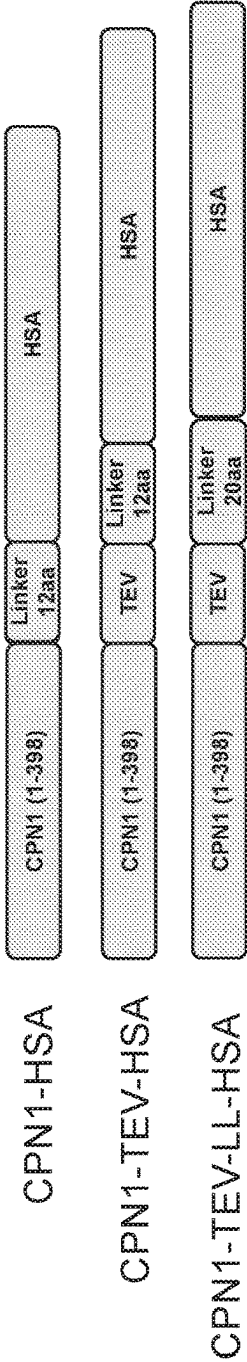


FIG. 6I

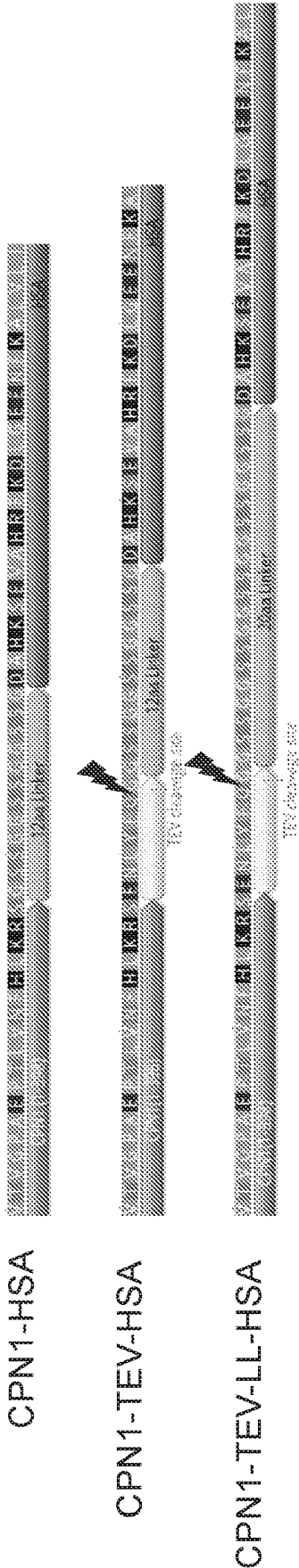


FIG. 6J

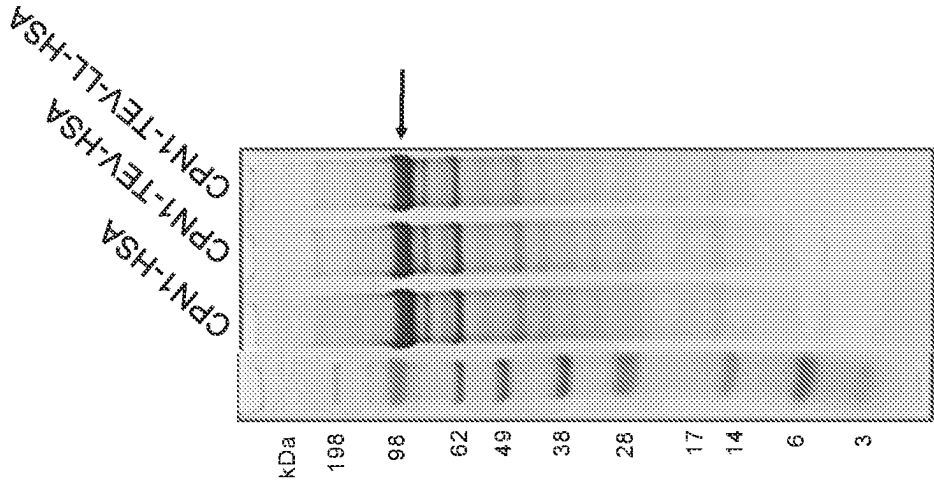


FIG. 6K

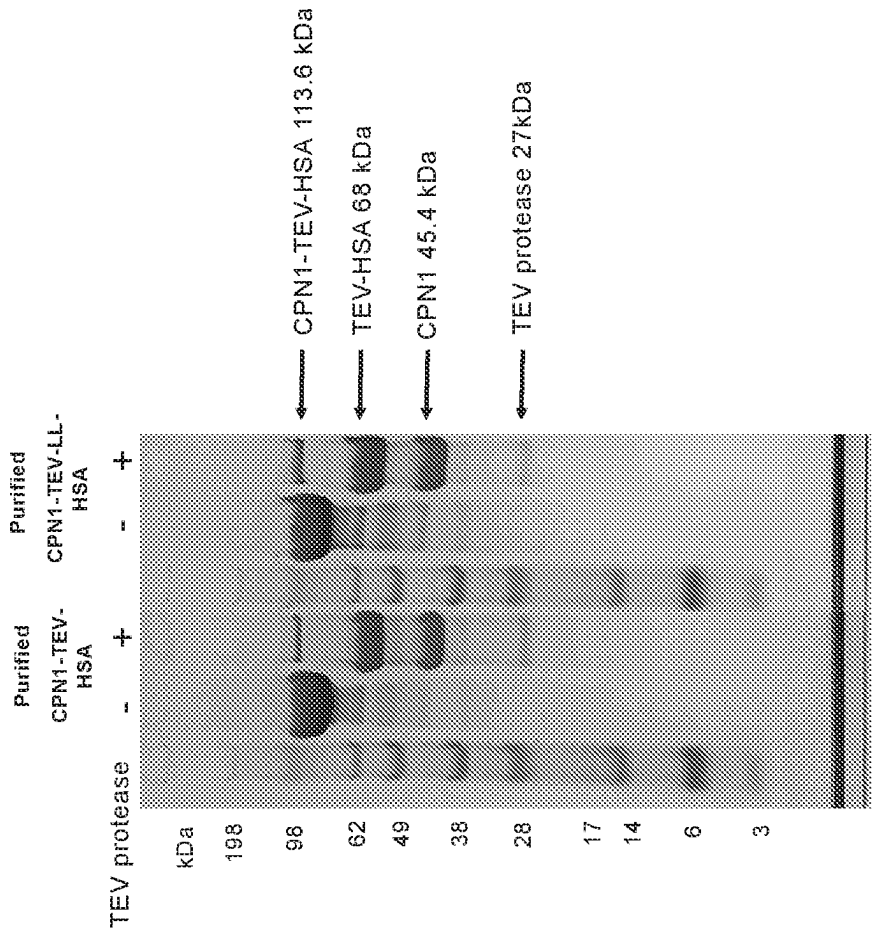


FIG. 6L

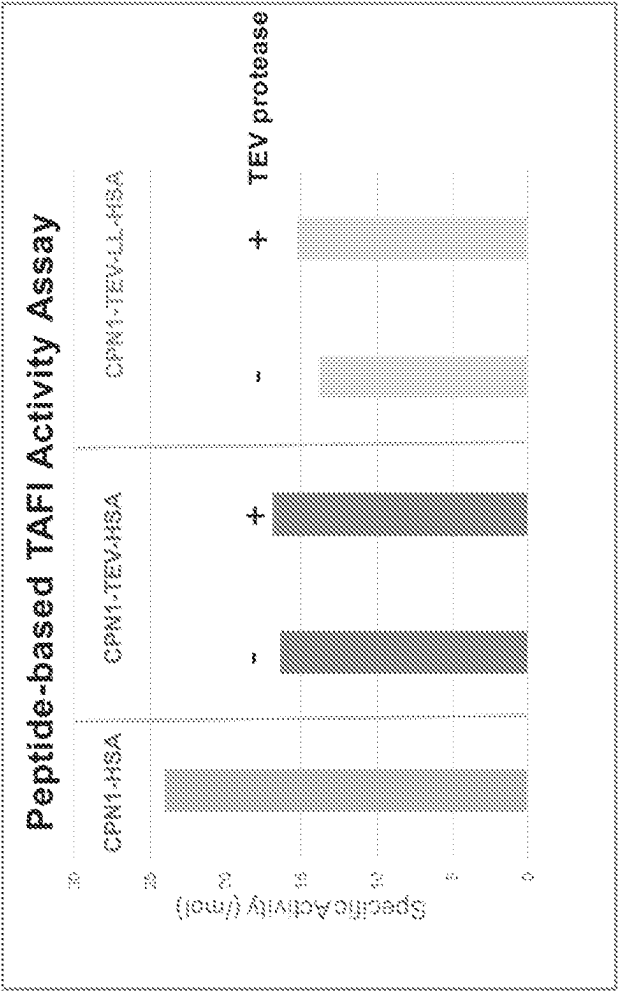


FIG. 6M

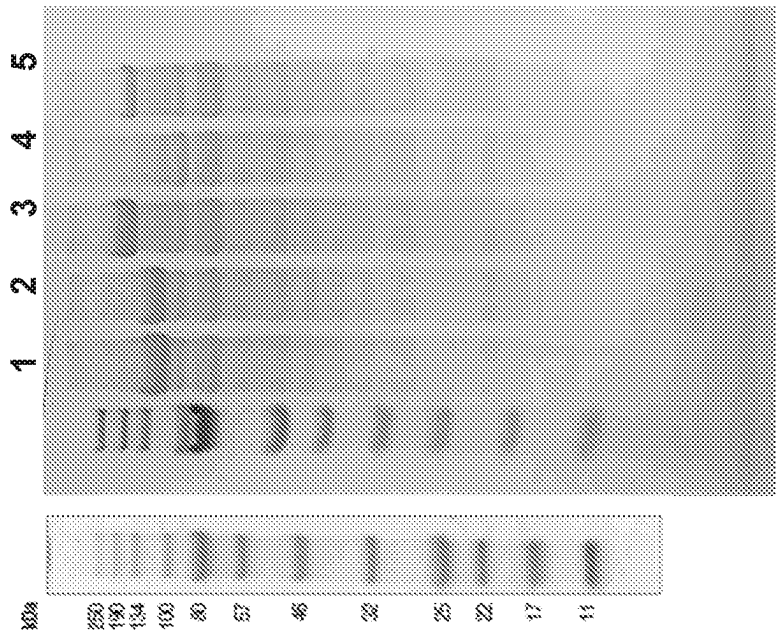


FIG. 60

Lane	Name	Expected MW	Expression Notes
1	CPN1 (1-398)-HSA	112,740	+++
2	CPN1-HSA (+TEV)	113,590	+++
3	MBP-CPN1-HSA (+TEV)	154,372	+++
4	MBP-CPN1 (+TEV)	87,090	+
5	GST-CPN1-HSA (+TEV)	140,442	++

FIG. 6N

Lane	Content	MW (Da)	Expression Level
1	Ladder		
2	CPN1-CPN2_fusion_1 untagged		
3	CPN1-CPN2_fusion_2 untagged	93716	-
4	CPN1-CPN2_fusion_3 untagged		
5	CPN1-CPN2_fusion_4 untagged		
6	CPN1-CPN2_fusion_5 untagged		
7	CPN1-CPN2_fusion_6 untagged		
8	CPN1-CPN2_fusion_7 untagged		
9	CPN1-CPN2_fusion_8 untagged	95490	-
10	CPN1-CPN2_fusion_9 untagged	93184	-
11	CPN1-CPN2_fusion_10 untagged	94224	-
12	stability_design_1 untagged	47509	-
13	stability_design_2 untagged	45408	-
14	stability_design_3 untagged	453780	-
15	stability_design_4 untagged	45322	-

FIG. 6P

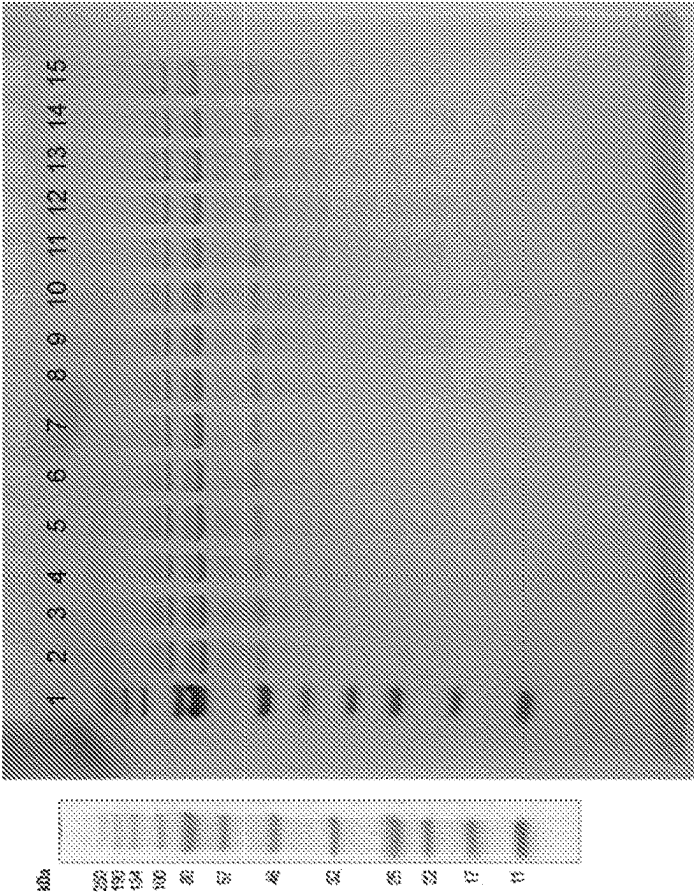


FIG. 6Q

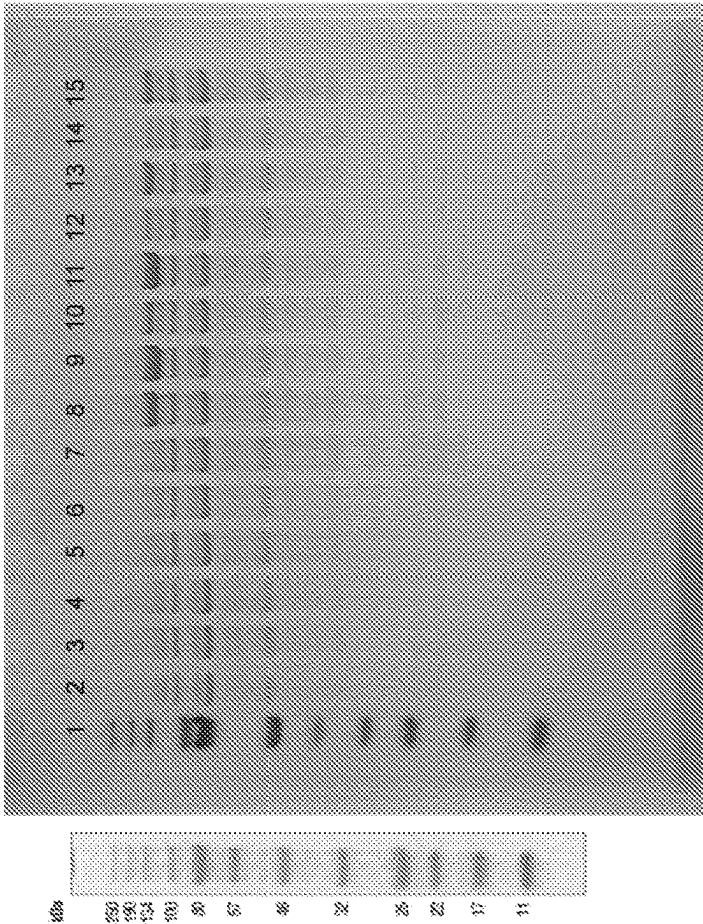


FIG. 6S

Lane	Content	MW	Expression Level	
1	Ladder			
2	stability_design_5	untagged	45489	-
3	stability_design_6	untagged	45404	-
4	stability_design_7	untagged	45430	-
5	stability_design_8	untagged	45430	-
6	stability_design_9	untagged	45483	-
7	stability_design_10	untagged	45444	-
8	stability_design_1	HSA	112696	++
9	stability_design_2	HSA	112690	+++
10	stability_design_3	HSA	112563	+
11	stability_design_4	HSA	112604	+++
12	stability_design_5	HSA	112771	+
13	stability_design_6	HSA	112686	++
14	stability_design_7	HSA	112712	-
15	stability_design_8	HSA	112712	+

FIG. 6R

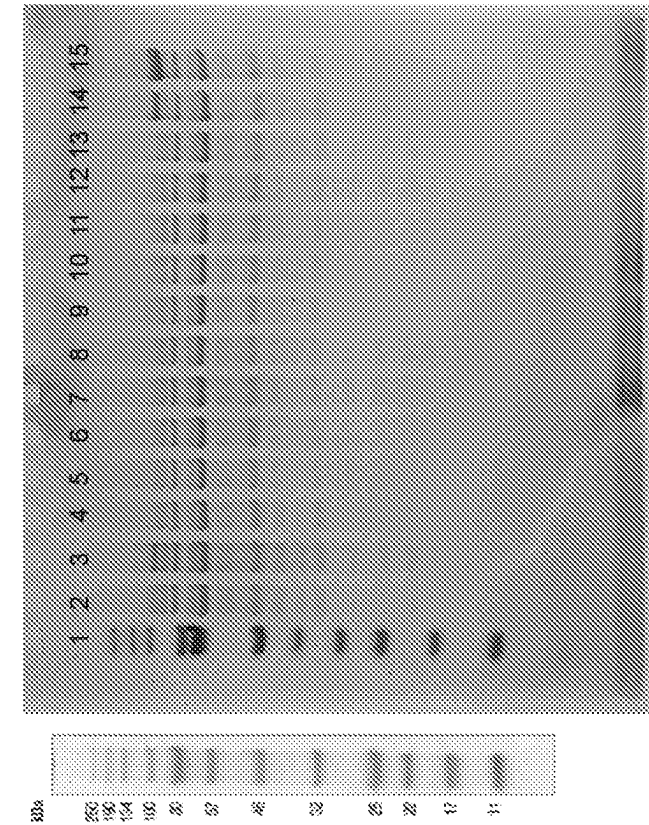


FIG. 6U

Lane	Content	MW	Expression Level
1	Ladder		
2	stability_design_9	HSA	112875
3	stability_design_10	HSA	112726
4	CPN1-CPN2_fusion_1	His-SUMO	107527
5	CPN1-CPN2_fusion_2	His-SUMO	106473
6	CPN1-CPN2_fusion_3	His-SUMO	107729
7	CPN1-CPN2_fusion_4	His-SUMO	107428
8	CPN1-CPN2_fusion_5	His-SUMO	112547
9	CPN1-CPN2_fusion_6	His-SUMO	114014
10	CPN1-CPN2_fusion_7	His-SUMO	107208
11	CPN1-CPN2_fusion_8	His-SUMO	108247
12	CPN1-CPN2_fusion_9	His-SUMO	105942
13	CPN1-CPN2_fusion_10	His-SUMO	106981
14	cat_var_1	HSA	
15	cat_var_2	HSA	

FIG. 6T

Lane	Content	MW	Expression Level
1	Ladder		
2	cat_var_2 HSA		++
3	Cat_var_3 HSA		++
4	Dead_1 TEV-HSA		+
5	Dead_2 TEV-HSA		+
6	Dead_3 TEV-HSA		+
7			
8	CPN1	45421	
9	CPN1 HSA	112740	++
10	CPN1 HSA		+++
11	CPN1 TEV-HSA	113.590	++
12			
13	Ladder		
14	CPN1 HSA	112740	++
15			

FIG. 6V

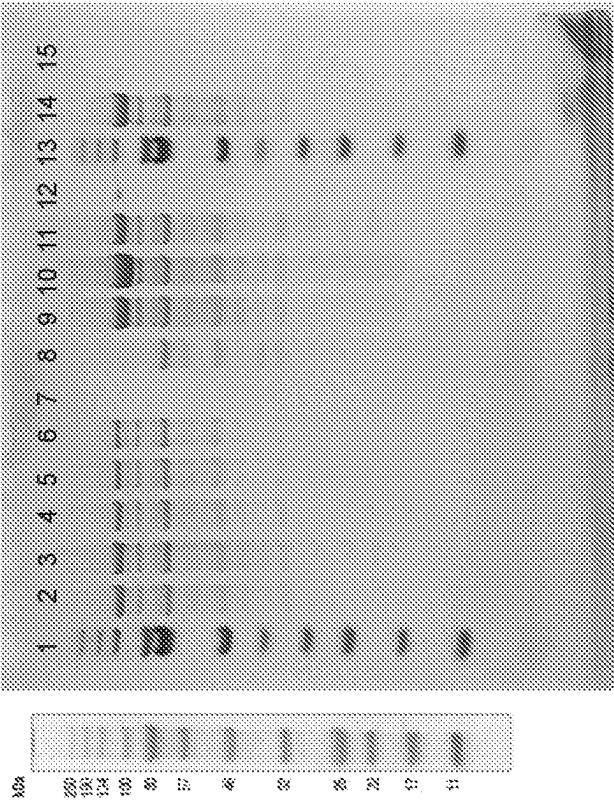


FIG. 6W

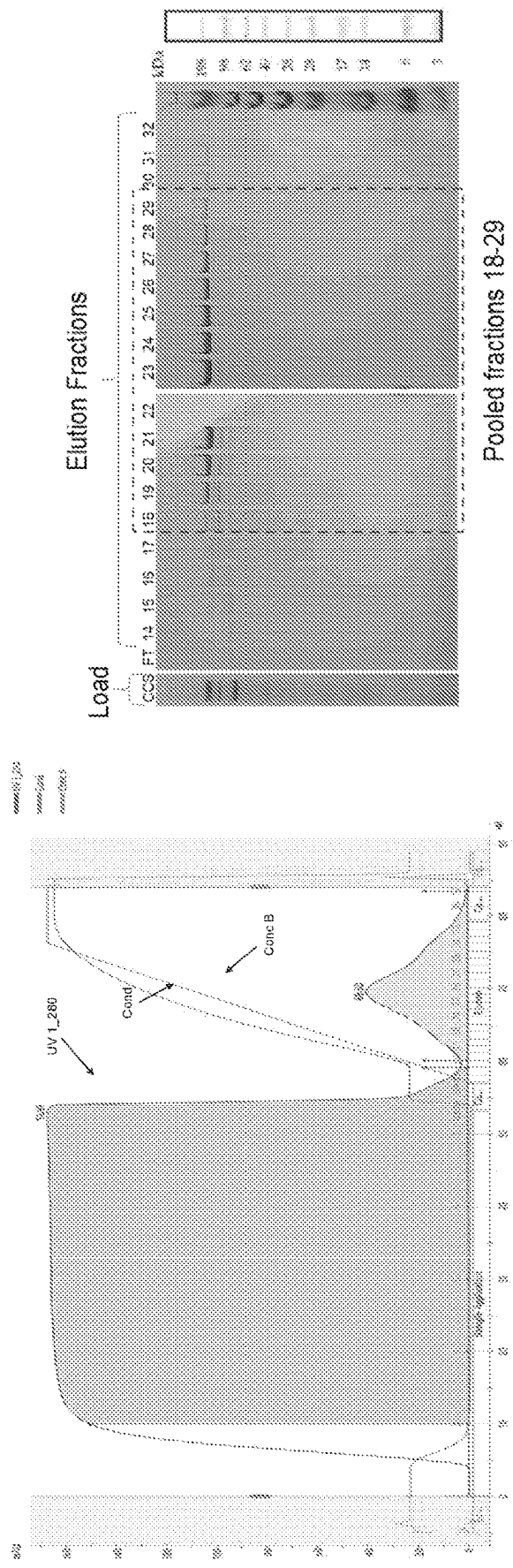


FIG. 7A

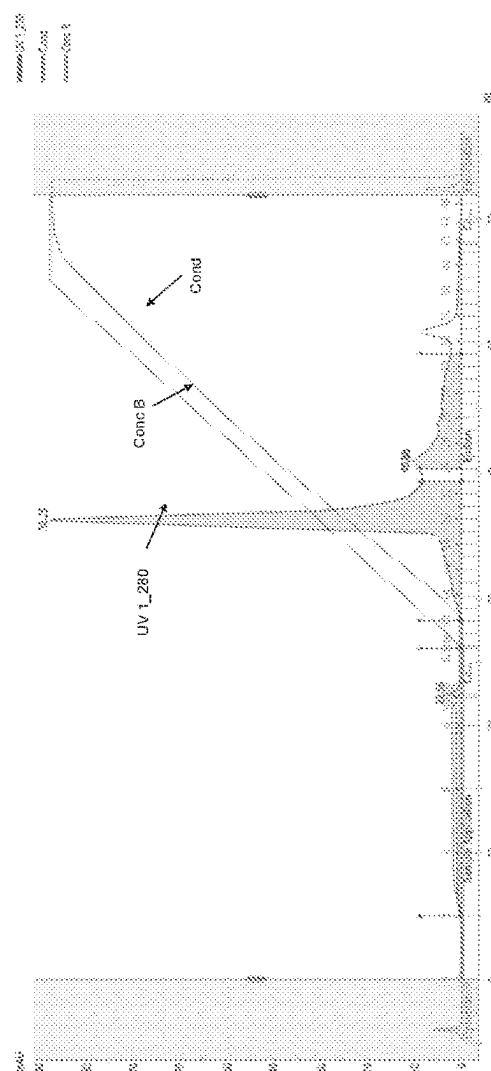
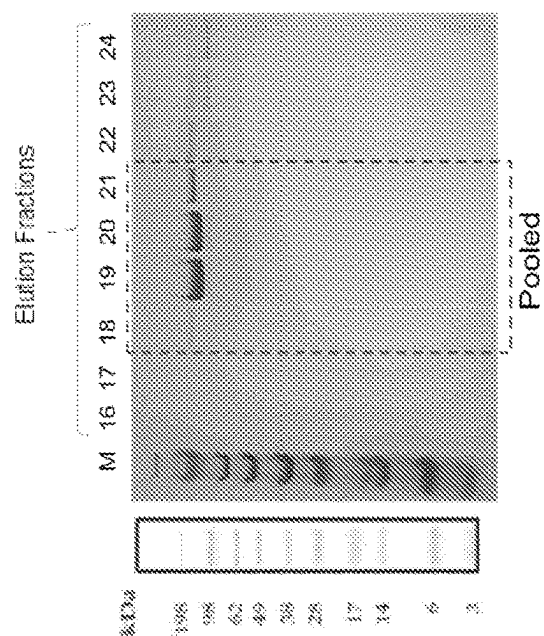


FIG. 7B

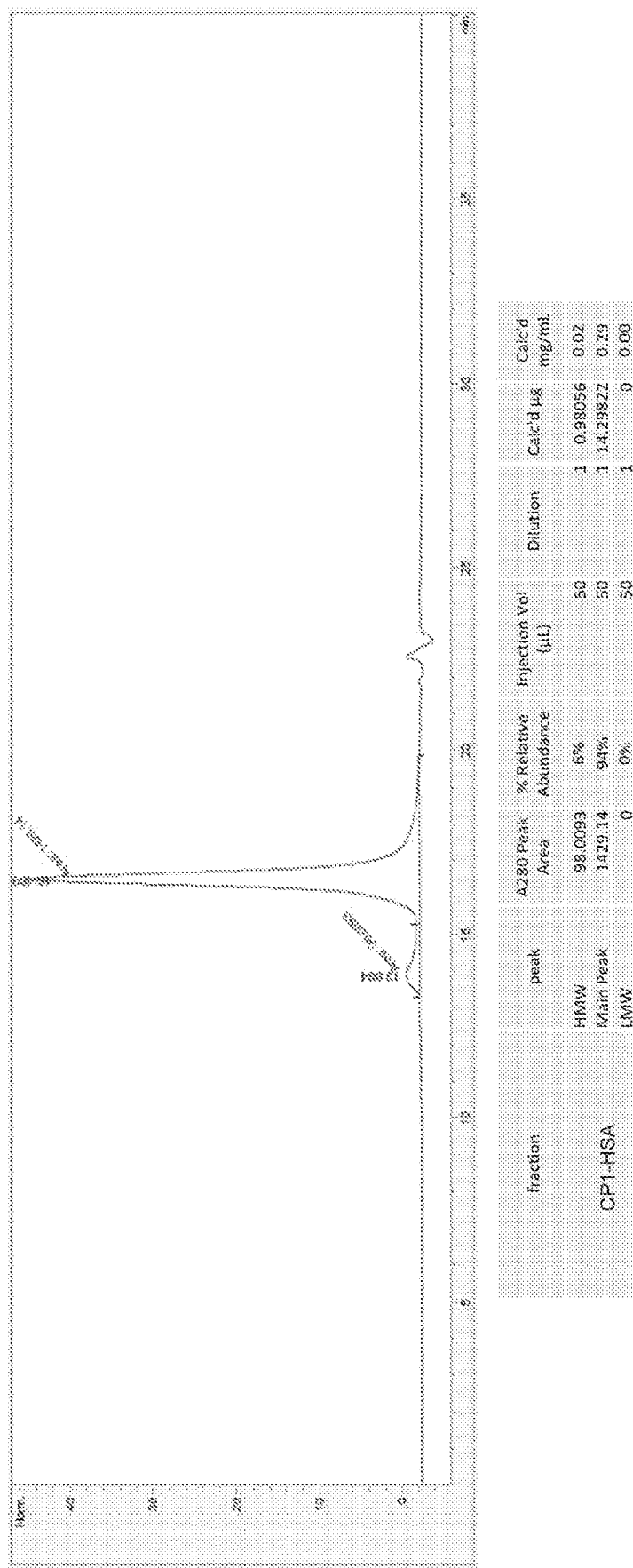


FIG. 7C

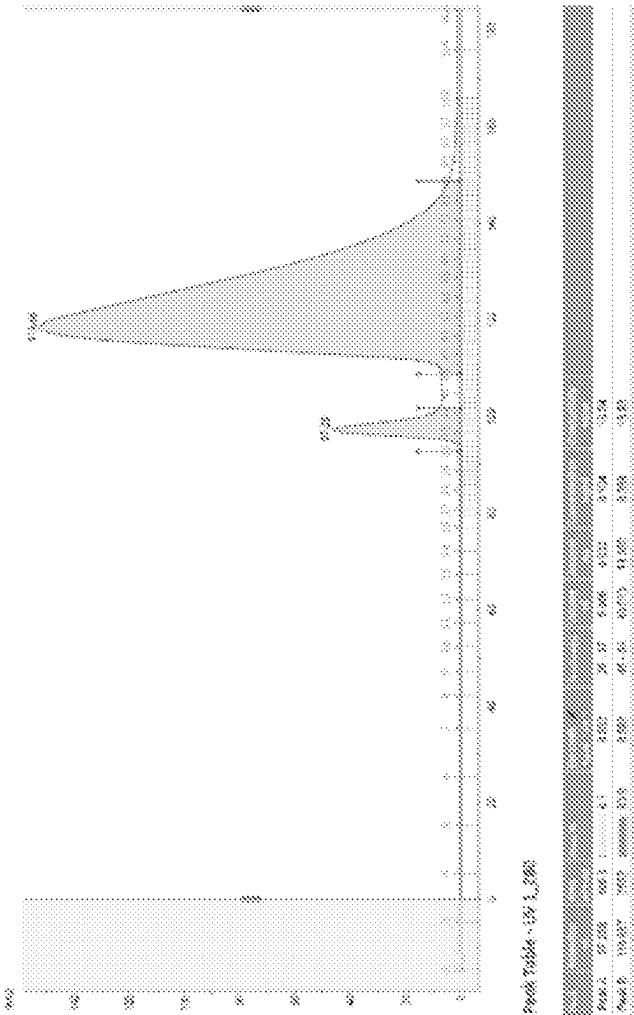


FIG. 7E

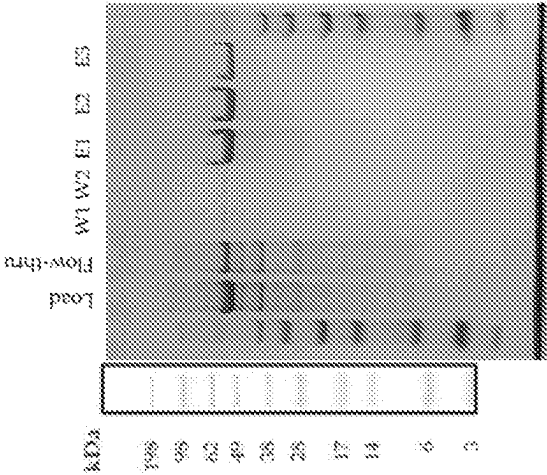
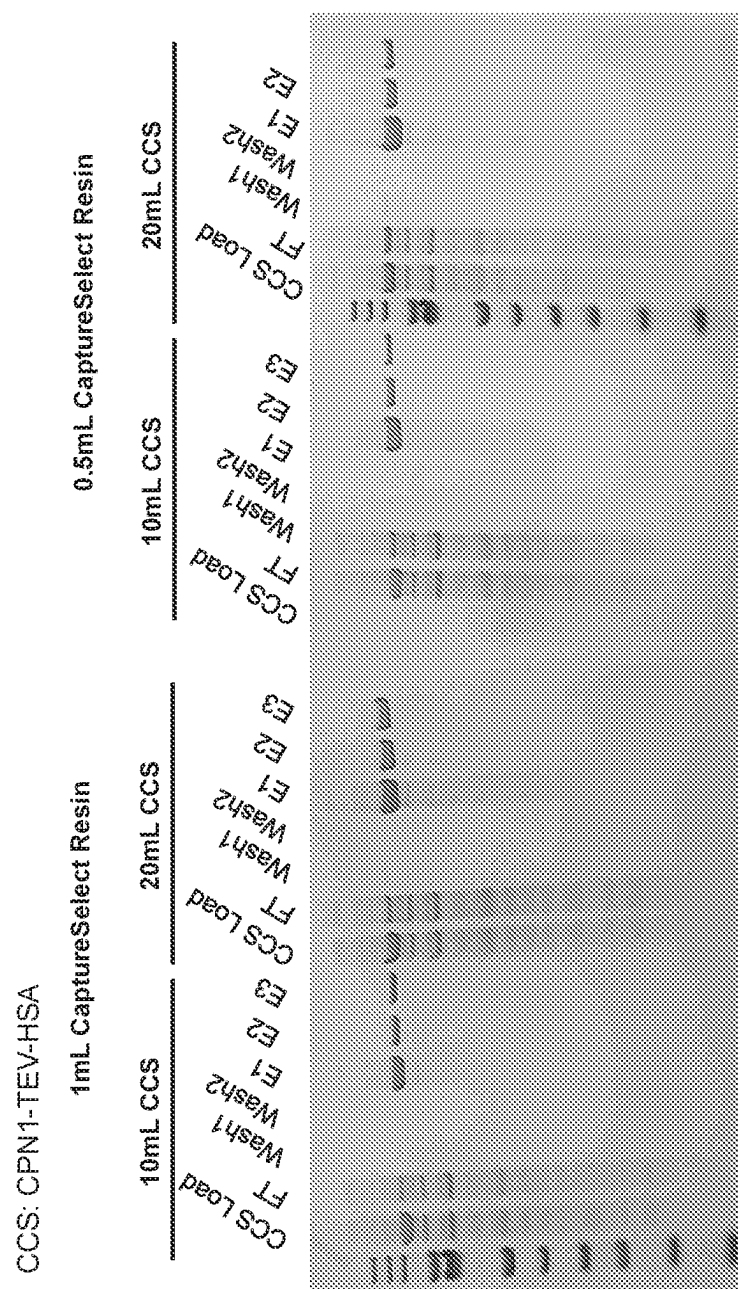


FIG. 7D



LEGE

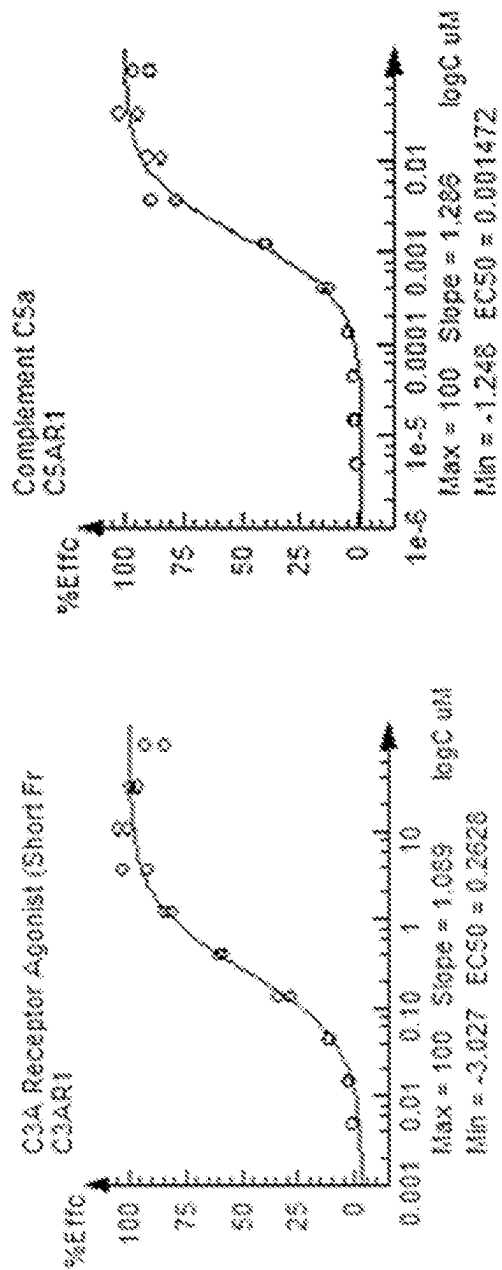
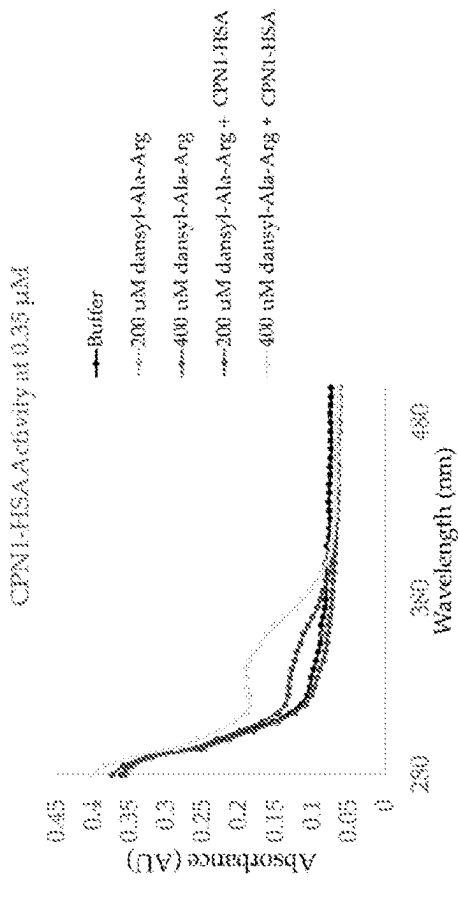


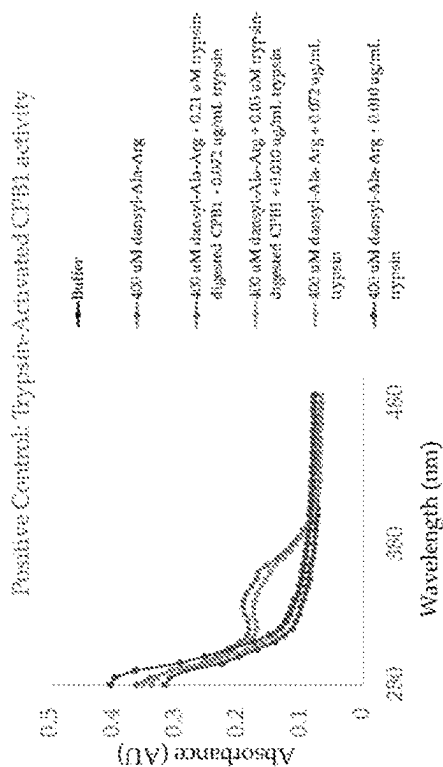
FIG. 8A

FIG. 8B



Sample	OD340	[dansyl-Ala-OH] (μ M):
Buffer	0.0954	10
200 μ M dansyl-Ala-Arg	0.0983	17
400 μ M dansyl-Ala-Arg	0.0908	-2
CPN1-HSA + 200 μ M dansyl-Ala-Arg	0.1183	92
CPN1-HSA + 400 μ M dansyl-Ala-Arg	0.1891	244

FIG. 8D



Sample	OD340	[dansyl-Ala-OH] (μ M):
Trypsin-digested CPB1 (0.21 μ M) + 400 μ M dansyl-Ala-Arg + 0.072 μ g/mL trypsin	0.1889	244
Trypsin-digested CPB1 (0.01 μ M) + 400 μ M dansyl-Ala-Arg + 0.010 μ g/mL trypsin	0.171	199
400 μ M dansyl-Ala-Arg + 0.072 μ g/mL trypsin	0.1043	32
400 μ M dansyl-Ala-Arg + 0.010 μ g/mL trypsin	0.1085	43

FIG. 8C

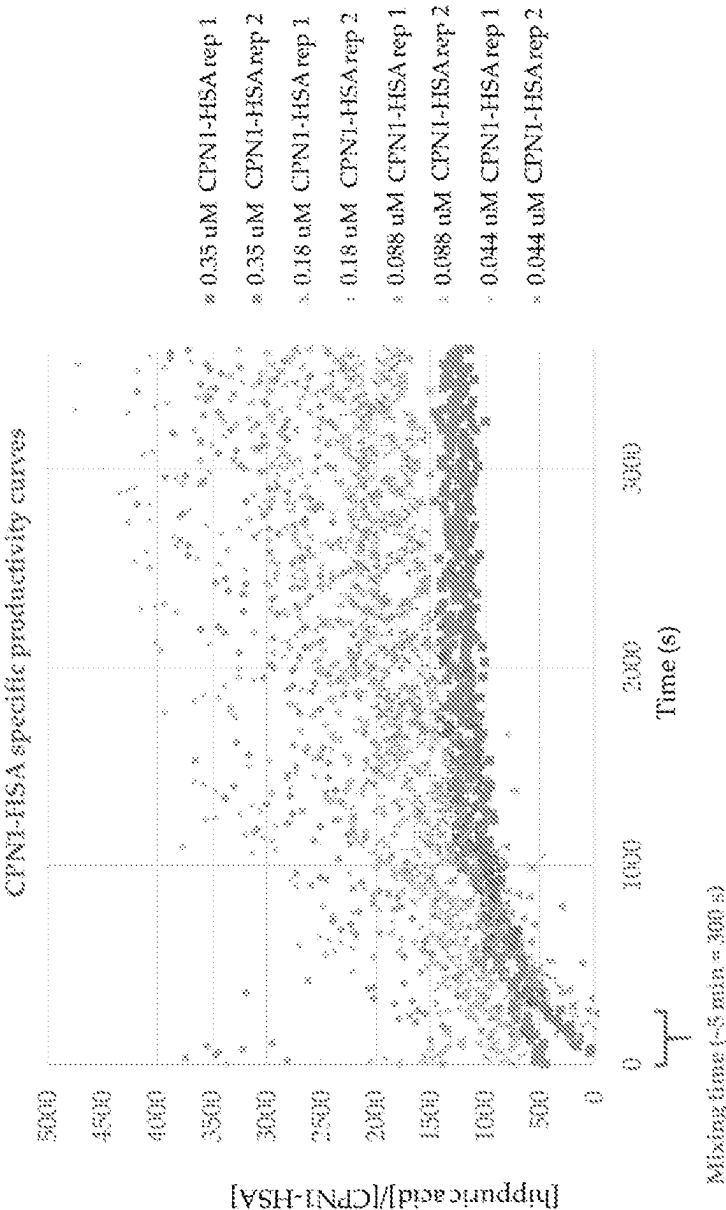


FIG. 8E

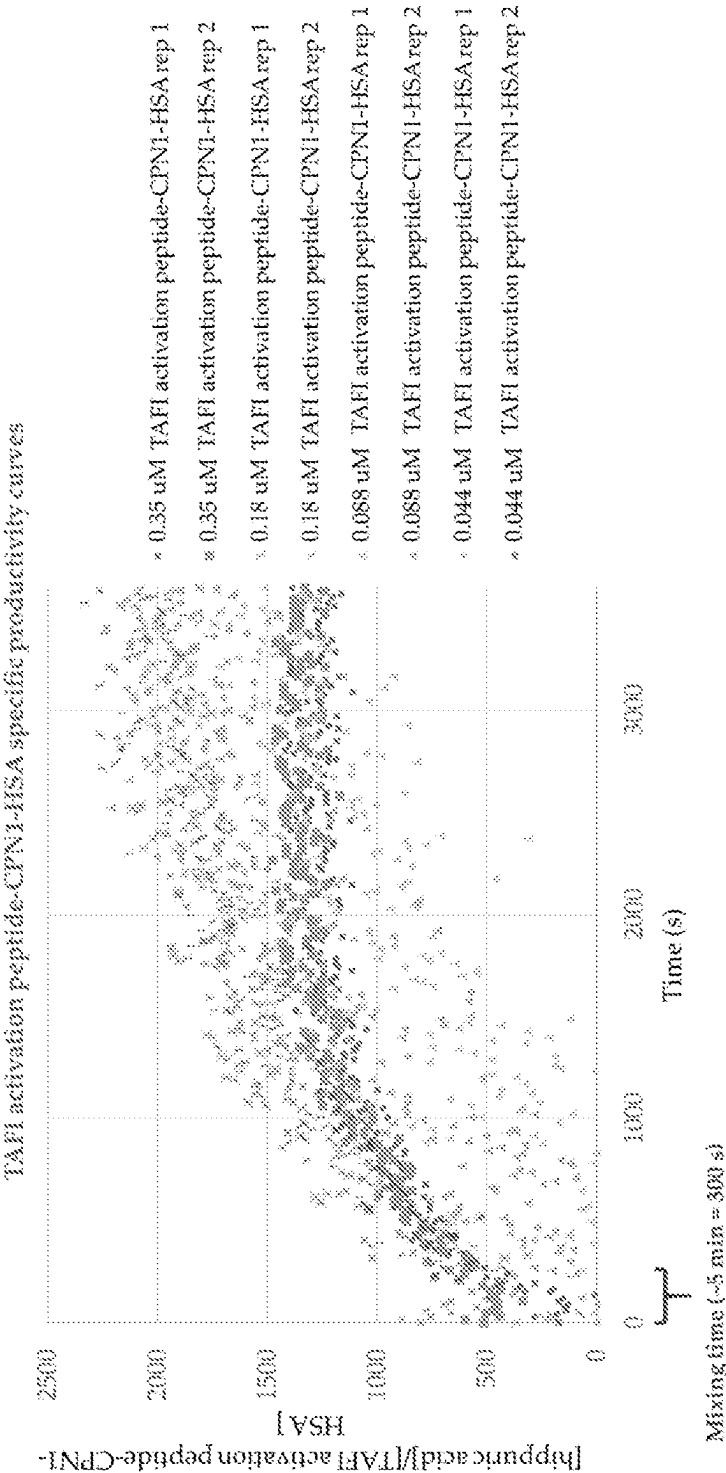


FIG. 8F

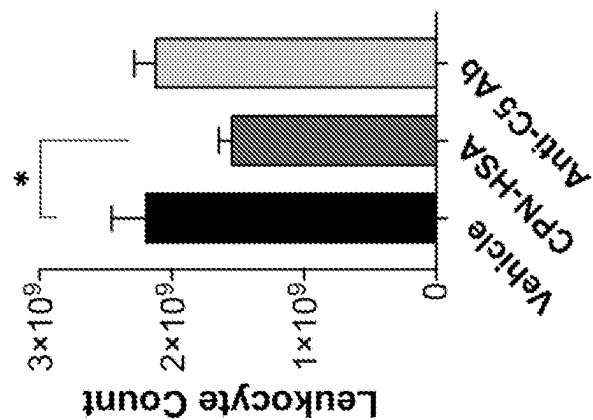


FIG. 9B

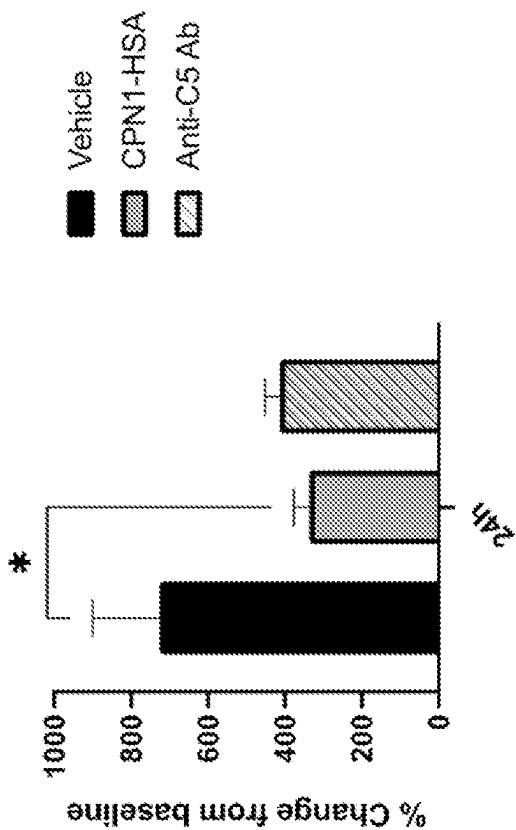


FIG. 9A

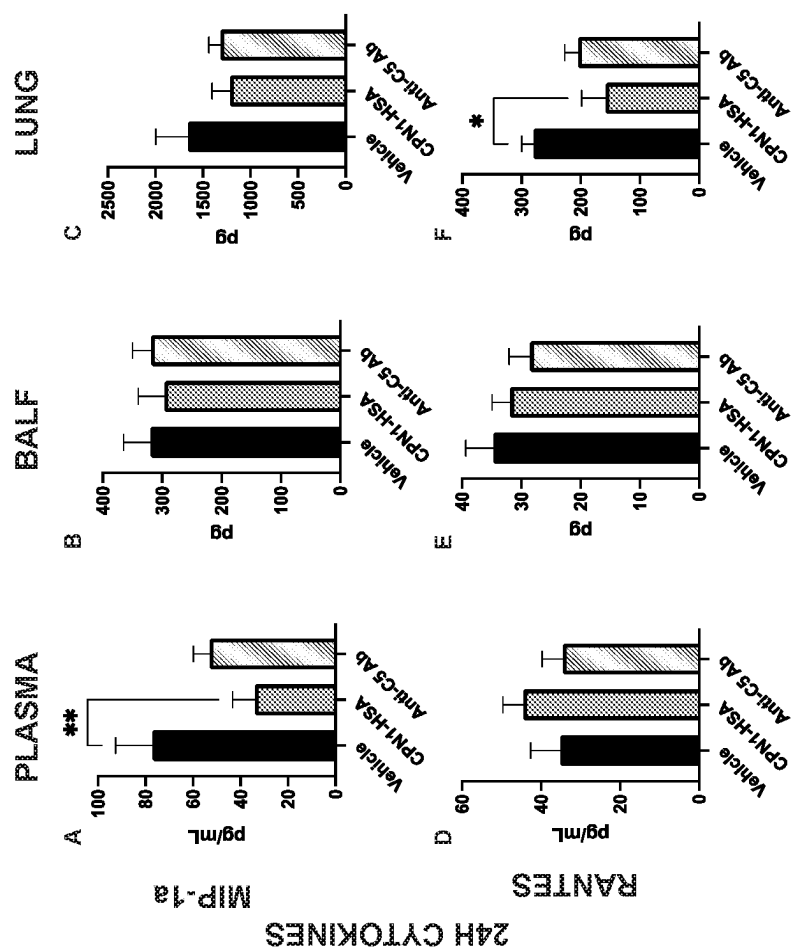


FIG. 10

CPN1-HSA						
No. animals analyzed	Terminal half-life (hours)	T _{max} (hours)	C _{max} (ng/mL)	MRT (hours)	AUC 0-inf (ng/mL x hours)	AUC 0-t (ng/mL x hours)
3 of 8	2	0.9	12,000	8.8	62,100	55,200

FIG. 11A

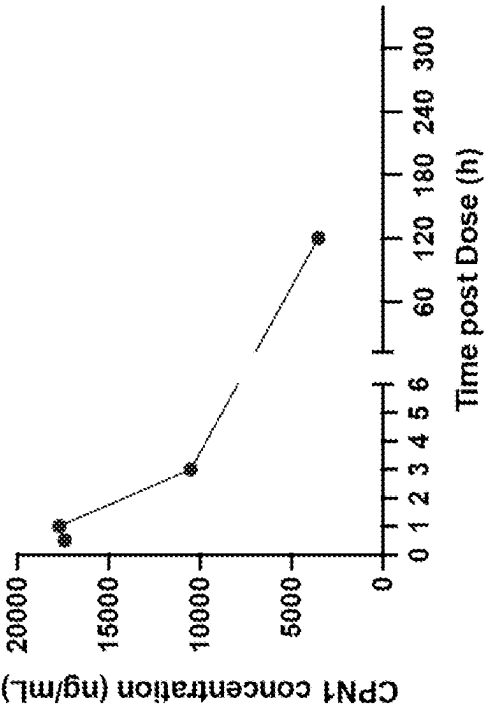


FIG. 11C

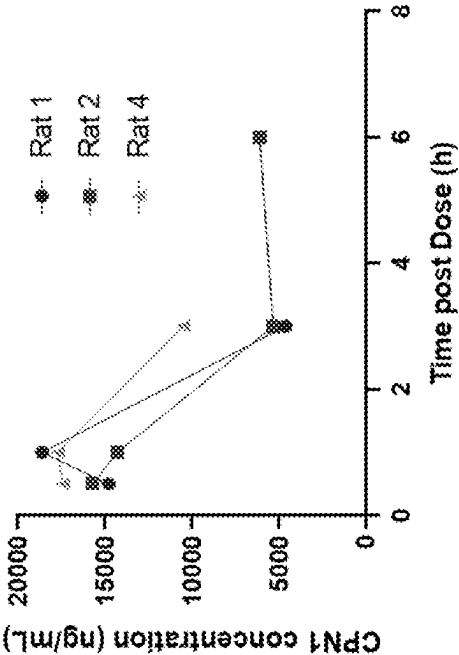


FIG. 11B

Table 1. Experimental Design and Group Assignments

Group	Test Article	Exposure Route	Concentration (mg/mL)	Dose (mg/kg)	Dose Volume (mL/kg)	Animals	
						Male	Female
1	CP6	IV	1.13	2	1.77	2	1
2	CP6	SC	1.13	6	5.31	2	1

Table 2. Blood Collection Schedule

Sample Type	Volume	Tube Type	Timepoints*
Plasma	1 mL	K ₂ EDTA	Baseline, immediately post-dose, and 0.5, 3, 6, 12, 24, 48, 72, 120, 168 and 240 hours post-dose
Serum	1.5 mL	SST	Baseline and 3, 24, 72 and 240 hours post-dose

*Blood collection times are approximate. Actual collection times will be documented in the study files.

FIG. 12

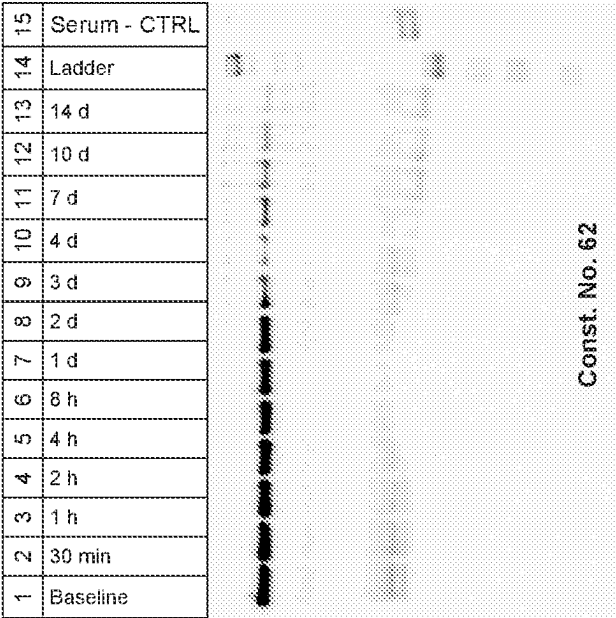


FIG. 13B

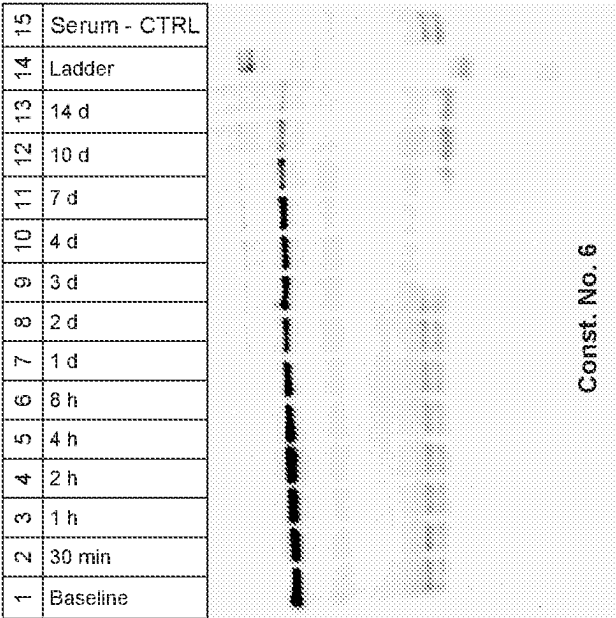


FIG. 13A

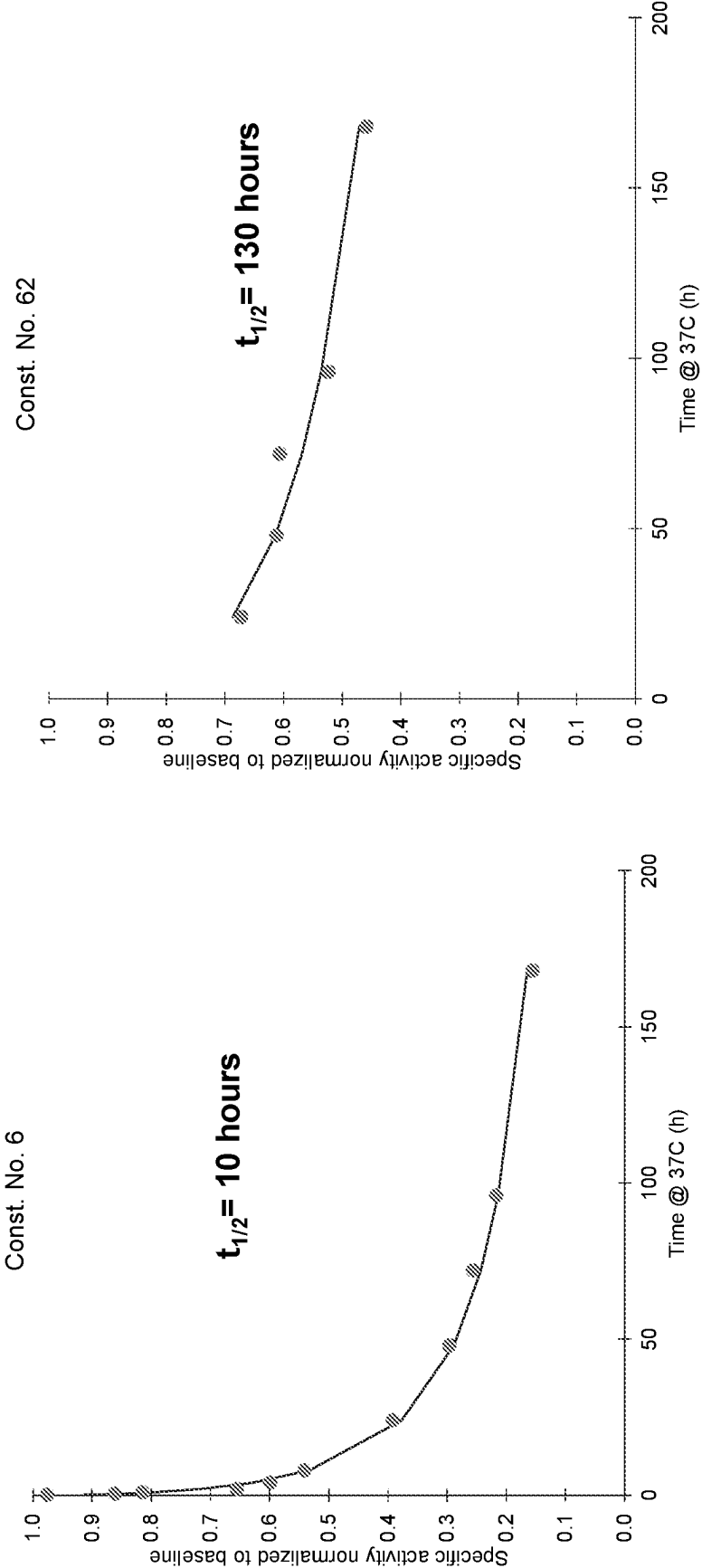


FIG. 13D

FIG. 13C

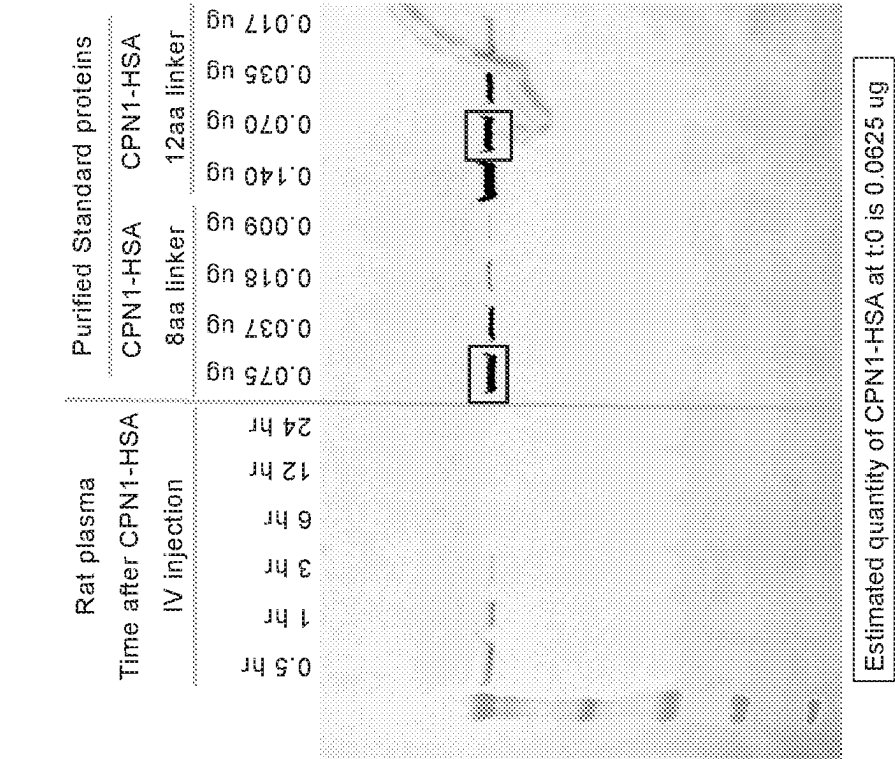


FIG. 14B

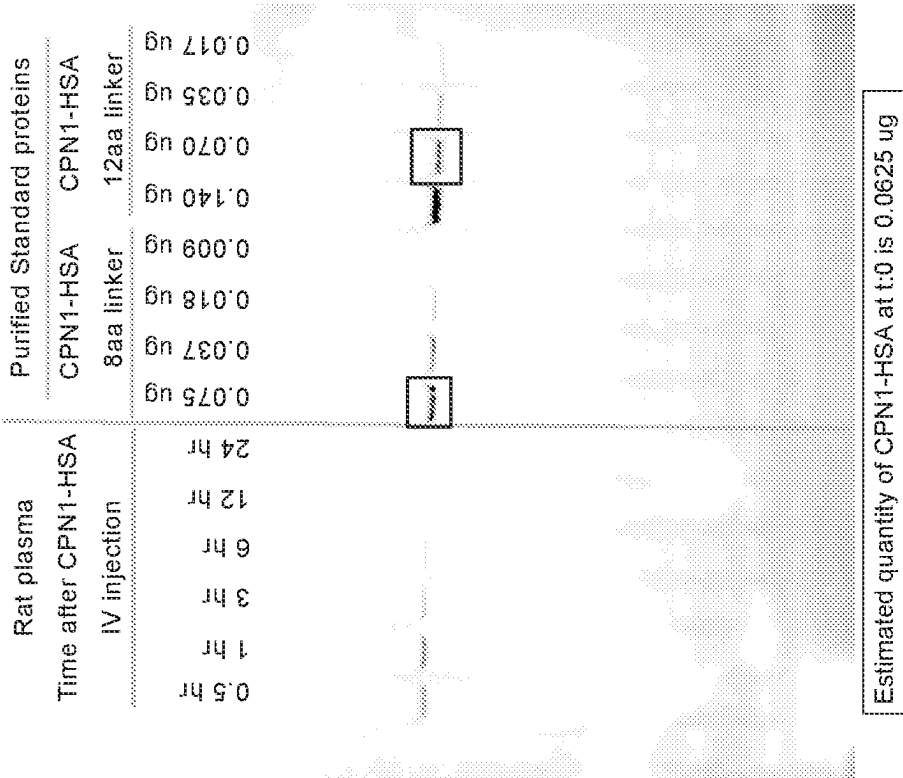


FIG. 14A

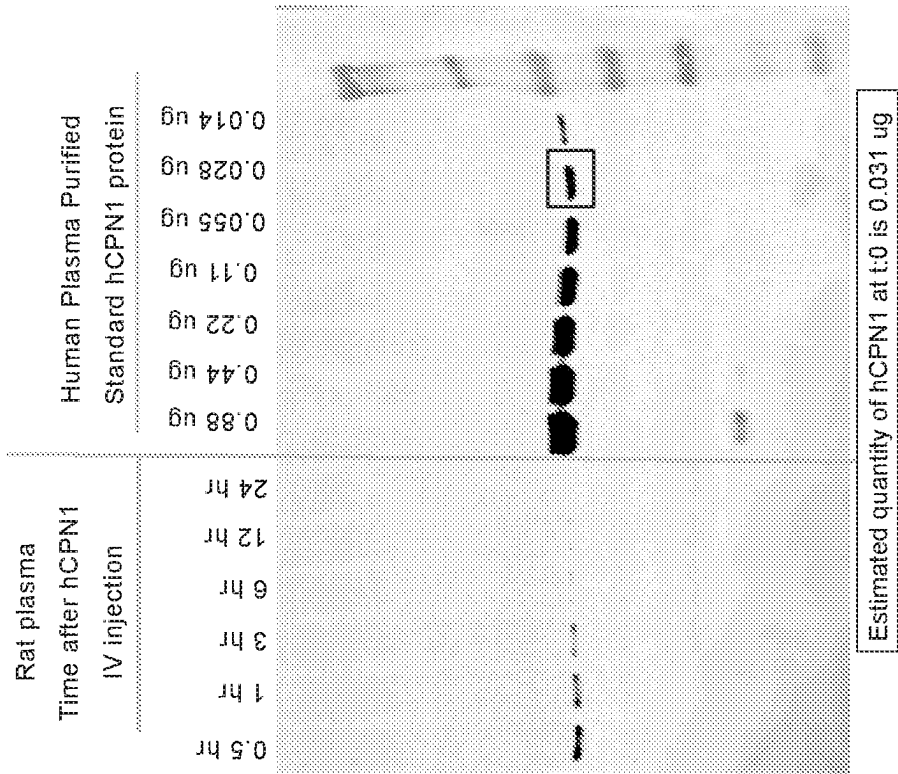


FIG. 14D

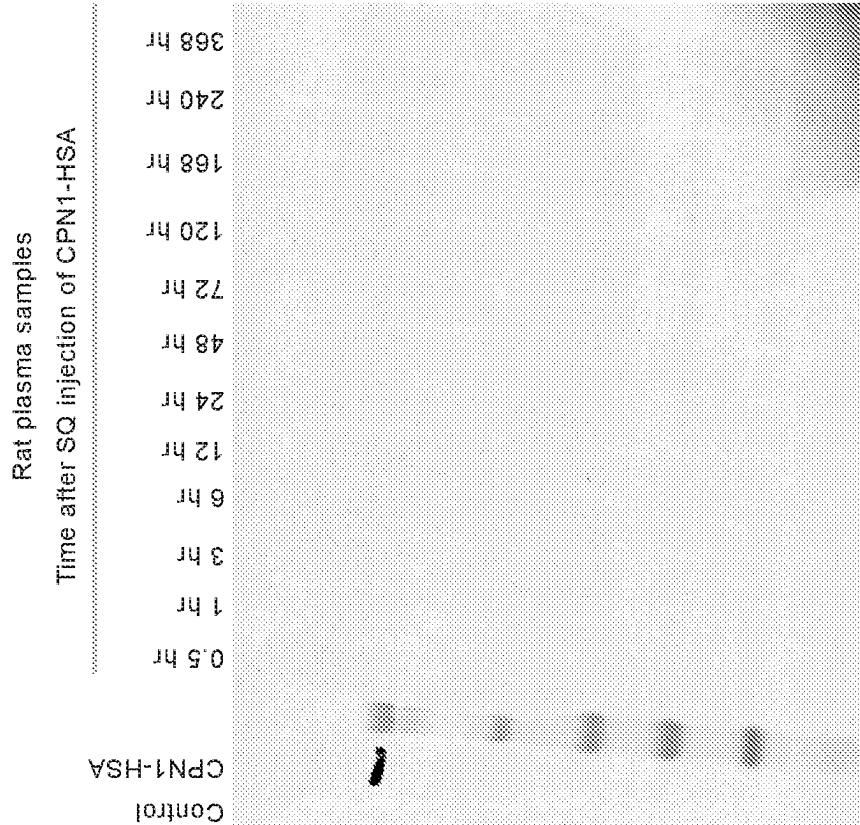


FIG. 14C

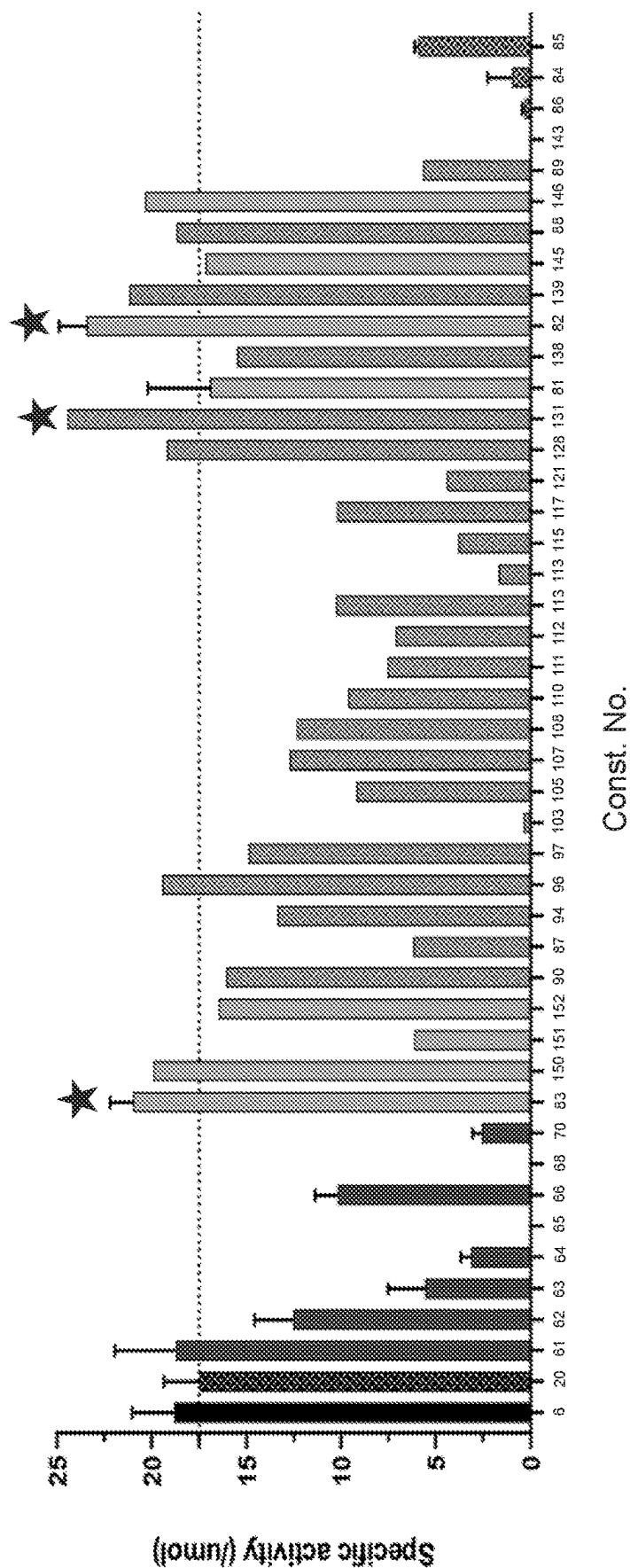


FIG. 15

ENGINEERED CPN1 CONSTRUCTS AND VARIANTS

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Application No. 63/142,462 filed on Jan. 27, 2021, and U.S. Provisional Application No. 63/222,929 filed on Jul. 16, 2021, the contents of which are incorporated herein by reference in their entireties.

REFERENCE TO SEQUENCE LISTING

[0002] An electronic version of the Sequence Listing is filed herewith, the contents of which are incorporated by reference in their entirety. The electronic file was created on Jan. 27, 2022, is 3.29 megabytes in size, and is titled CTBI_005_02WO_SeqList_ST25.txt.

BACKGROUND

[0003] The complement system includes the classical, alternative, and lectin pathways, and is tightly controlled by a number of regulators. One such regulator is carboxypeptidase N (CPN), a plasma enzyme of the metalloprotease class and M14B subfamily. CPN cleaves basic amino acids from the C-terminal end of bioactive peptides and proteins, leading to their inactivation. CPN has a role in preventing the buildup of peptides that are involved in signaling and regulation of inflammation, or controlling blood pressure, thus regulation of the levels of these peptides is necessary. CPN is synthesized in the liver and secreted into the bloodstream in a constitutively active form, while another similar carboxypeptidase, carboxypeptidase B2 (CPB2), circulates in plasma as a zymogen, and requires activation by the thrombin-thrombomodulin complex or plasmin. Normally, CPN circulates in the blood as a hetero-tetramer consisting of two 83 kDa (CPN2) domains, each flanked by a 48 to 55 kDa catalytic (CPN1) domain.

[0004] While CPN targets a number of substrates, CPN plays a significant role in complement regulation by targeting C3a and C5a, which are generated during complement activation. C3a stimulates macrophages and is implicated in B cell antibody response. Due to the expression of C3a receptor (C3aR) at the surface of endothelial cells and platelets, C3a can also be involved in platelet function and thrombus formation. C5a is a chemotactic factor for leukocytes and activates neutrophils, basophils and mast cells and therefore can be involved in expansion of T helper type 1 (Th1) cells and suppression of regulatory T cells (Treg). CPN inactivates C3a and C5a by cleaving one or more amino acids from their C-terminal ends. Like CPN, CPB2 also has several physiological substrates, including C3a and C5a. Continuous regulation of C3a and C5a levels is necessary to maintain a pro- and anti-inflammatory balance in the complement system. For example, a genetic deficiency in CPN or CPB2 can result in the exacerbation of pathological symptoms of complement disorders such as hemolytic-uremic syndrome (HUS) and cobra venom factor (CVF) challenge. Provided herein are compositions and methods to address the dysfunction and/or dysregulation in the complement system.

SUMMARY

[0005] In one aspect provided herein are variants of a carboxypeptidase N catalytic subunit (CPN1) comprising at least one modification with respect to a wild type CPN1, wherein the variant has at least one improved characteristic as compared to the wild type CPN1.

[0006] In another aspect, provided herein are fusion constructs comprising a carboxypeptidase N catalytic subunit (CPN1) or variant thereof. Exemplary fusion constructs are provided in Table 2 and Table 5.

[0007] In another aspect provided herein is a method of treating a disease or condition in a subject in need thereof, comprising administering to the subject any one of the CPN1 variants or fusion constructs of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1A depicts a schematic diagram of a CPN tetramer made up of two heterodimers, and FIG. 1B depicts a single heterodimer of the tetramer.

[0009] FIG. 2 depicts a schematic diagram of CPB2 (TAFI), showing both the zymogen and activated forms.

[0010] FIG. 3 depicts schematic diagrams of a wild type CPN with its catalytic subunit (CPN1) alone, with its catalytic subunit (CPN1) plus its regulatory subunit (CPN2), and various exemplary fusion constructs comprising other components.

[0011] FIG. 4A depicts schematic diagrams of exemplary fusion constructs of the disclosure comprising various activation peptides to alter the sensitivity of the constructs to certain complement components.

[0012] FIGS. 4B-4C depict the structure of a CPN1-CPB2 fusion, showing the CPB2 activation peptide (TAFI activation peptide) masking the catalytic site of CPN1, and a schematic diagram of the fusion, respectively.

[0013] FIGS. 4D-4E depict the structure of a CPN1-CPA4 fusion, and a surface representation of the same, respectively.

[0014] FIGS. 4F-4G depict the structure of a CPN1-CPA1 fusion, and a surface representation of the same, respectively.

[0015] FIG. 5 depicts a general schematic diagram of a screening process for activation peptides (SEQ ID NOs: 45-63), and examples of libraries for the activation peptide screening.

[0016] FIGS. 6A-6C depict Coomassie staining from SDS-PAGE analysis showing various fusion constructs (chimeras) expressed in Expi293 cells. FIG. 6D shows Coomassie staining from SDS-PAGE analysis of CPN1-containing fusion construct expression, using a similar process as depicted in FIGS. 6A-6C, with five constructs. FIGS. 6E-6H show exemplary CPN1-containing fusion constructs and their corresponding SDS-PAGE expression analysis. FIGS. 6I-6J shows exemplary CPN1-HSA fusion constructs with and without TEV protease cleavage sites (SEQ ID NOs: 64-66), and FIG. 6K shows their corresponding SDS-PAGE expression analysis. FIG. 6L shows SDS-PAGE analysis of an exemplary CPN1-HSA fusion construct in the presence or absence of TEV protease. FIG. 6M shows the results of a peptide-based TAFI activity assay for exemplary CPN1-HSA constructs in the presence or absence of TEV protease. FIGS. 6N-6O shows SDS-PAGE expression analysis of

exemplary CPN1-containing constructs. FIGS. 6P-6W shows SDS-PAGE expression analysis of exemplary CPN1-containing constructs.

[0017] FIGS. 7A-7C depict various chromatography and SDS-PAGE results from a two-step purification of exemplary fusion constructs. FIGS. 7D-7F show SDS-PAGE and SEC chromatograms for exemplary CPN1 variants purified by affinity purification.

[0018] FIGS. 8A-8B depict examples of data obtained from a cell-based screening assay used to evaluate construct activity on C3a and C5a by measuring the level of activation of C3aR and C5aR by C3a and C5a. FIGS. 8C-8D show activities of exemplary fusion constructs in the Dansyl-Ala-Arg Activity assay. FIGS. 8E-8F show activities of exemplary fusion constructs in the Hippuryl-Arg Activity assay.

[0019] FIGS. 9A-9B show results of CPN1-HSA efficacy after intravenous administration in an in vivo rodent model of complement activation. FIG. 9A shows Pulmonary Congestion Index (PenH) % change from baseline. FIG. 9B shows leukocyte infiltration in lung 24 hours after CPN1-HSA treatment.

[0020] FIGS. 10A-10F show cytokine and chemokine effects after intravenous administration of CPN1-HSA in an in vivo rodent model of complement activation.

[0021] FIGS. 11A-11C depict in vivo pharmacokinetics profiles of CPN1-HSA intravenously administered to rats. FIG. 11A depicts PK parameter estimates for three animals assessed. FIG. 11B depicts the PK profiles of three animals analyzed, and the representative PK profile is depicted in FIG. 11C.

[0022] FIG. 12 depicts the regimen for determining the pharmacokinetics profile of CPN1-HSA intravenously or subcutaneously administered to cynomolgus macaques. Table 1 (of FIG. 12, not to be confused with the Table 1 of the Detailed Description) summarizes the dose levels of the test articles and vehicles. Table 2 (of FIG. 12, not to be confused with the Table 1 of the Detailed Description) summarizes the blood collection schedule.

[0023] FIGS. 13A-13D depict the results of serum stability assays for CPN1-HSA. FIGS. 13A & 13B depict Western blots for two exemplary CPN1-HSA constructs. FIGS. 13C and 13D depict half-life curves for exemplary CPN1-HSA constructs in serum.

[0024] FIGS. 14A-14D depict the results of CPN1-HSA stability in plasma isolated from rats following intravenous or subcutaneous administration of CPN1-HSA. FIGS. 14A & 14B depict two different rats' plasma samples. FIG. 14C depicts CPN1-HSA stability following subcutaneous injection. FIG. 14D depicts CPN1-HSA stability following intravenous injection of human plasma purified CPN1.

[0025] FIG. 15 depicts the specific activity of exemplary fusion constructs as determined through a TAFI assay.

BRIEF DESCRIPTION OF THE TABLES

[0026] The following Tables are provided as a part of the application.

[0027] Table 1 depicts the amino acid sequences of exemplary components comprising the constructs of the disclosure.

[0028] Table 2 depicts exemplary fusion constructs with components in the N and C termini of the core molecule specified.

[0029] Table 3A depicts exemplary modification strings of core molecule hCPN1 (1-398) (SEQ ID NO: 7).

Amino acid substitutions, deletions, insertions, etc. are noted in conventional format.

[0030] Table 3B depicts exemplary modification strings of core molecule hCPN1 (1-320) (SEQ ID NO: 8). Amino acid substitutions, deletions, insertions, etc. are noted in conventional format.

[0031] Table 3C depicts exemplary modification strings of core molecule hCPN1 (1-438) (SEQ ID NO: 6). Amino acid substitutions, deletions, insertions, etc. are noted in conventional format.

[0032] Table 4A depicts exemplary modifications of individual residues of core molecule hCPN1 (1-398) (SEQ ID NO: 7). Amino acid substitutions, deletions, insertions, etc. are noted in conventional format.

[0033] Table 4B depicts exemplary modifications of individual residues of core molecule hCPN1 (1-320) (SEQ ID NO: 8). Amino acid substitutions, deletions, insertions, etc. are noted in conventional format.

[0034] Table 4C depicts exemplary modifications of individual residues of core molecule hCPN1 (1-438) (SEQ ID NO: 6). Amino acid substitutions, deletions, insertions, etc. are noted in conventional format.

[0035] Table 5 depicts the signal sequences and SEQ IDs of exemplary fusion constructs of the disclosure. The signal sequences are optional, and useful for the expression the fusion constructs.

DETAILED DESCRIPTION

[0036] The disclosure provides compositions and methods useful for modulating the signaling and regulation of the complement system. Specifically, complement system modulation can be observed by providing variants of carboxypeptidase N catalytic subunit (CPN1), and fusion constructs comprising CPN1, and variants thereof. Provided herein are CPN1 variants, and CPN-1 containing fusion constructs, that are more active, and/or more stable in circulation than wild type CPN. Such modulation can include an increase in cleavage and inactivation of C3a and/or C5a, thus reducing complement-mediated inflammation, and reducing the amplification of the complement pathways. For example, some carboxypeptidase variants can alter levels of regulators within the complement system, which includes C3a or C5a. In some embodiments, the variants and fusion constructs provided herein can act on the classical pathway of the complement system, or on the alternative pathway of the complement system, or on the lectin pathway of the complement system, or on one or more pathways. The disclosure also provides methods of making and using these variants and fusion constructs, for example in treating a disease or condition associated with complement dysregulation, e.g. treating an overactive complement response.

Carboxypeptidase N Variants

[0037] FIG. 1A depicts a schematic diagram of a CPN tetramer made up of two heterodimers. FIG. 1B depicts a single heterodimer of the tetramer. CPN endogenously exists as a tetramer, having two identical catalytic subunits called CPN1 ("CPN catalytic domain" in FIGS. 1A-1B), and two identical regulatory subunits called CPN2 ("regulatory subunit CPN2" in FIGS. 1A-1B) which act to stabilize the

CPN1 subunits. Each heterodimer includes a CPN1 domain and a CPN2 domains. CPN1 is also referred to as a CPN catalytic domain herein.

[0038] In some embodiments, provided herein are CPN variants, such variants comprise one or more modifications with respect to a wild type CPN, and are referred to herein as “CPN variants.” As used herein, a “modification” to a wild type CPN includes: a deletion of one or more amino acid residues, a deletion of one or more CPN domains, a substitution of one or more amino acid residues in one or more domains, a substitution of one or more CPN domains, an insertion of one or more amino acid residues in one or more domains, an insertion of one or more CPN domains, a swapping of one or more CPN domains, an insertion of one or more domains from a protein or any other component that is not CPN, and a fusion to a protein or component that is not CPN. For example, such components (collectively referred to herein as a “non-CPN domain”) can be, but are not limited to, other members of the carboxypeptidase family such as CPB2, or a half-life extender. Such CPN variants fused to one or more non-CPN domains, such as domains from other members of the carboxypeptidase family, may be referred to herein as “CPN chimeras” or “CPN fusion proteins” or “CPN fusion constructs.”

[0039] In some embodiments, provided herein are variants of the CPN catalytic domain (also referred to herein as CPN1), such variants comprise one or more modifications with respect to a wild type CPN1, and are referred to herein as “CPN1 variants.” As used herein, a “modification” to a wild type CPN1 includes one or more of: a deletion of one or more amino acid residues, a substitution of one or more amino acid residues, and an insertion of one or more amino acid residues. As used herein, a “CPN1 variant” is any CPN1-derived polypeptide having a modification to a wild type CPN1. As used herein, a “wild type CPN1” is a naturally-occurring CPN1 that is not a disease-causing CPN1. In some embodiments, the wild type CPN1 is a human CPN1. A CPN1 variant may also be referred to as a CPN variant, that is all CPN1 variants are CPN variants, but the opposite is not the case.

[0040] Table 1 provides SEQ ID NO: 6, the full length human wild type CPN1. Table 1 provides the amino acid sequences of human wild-type CPN1 (SEQ ID NO: 1) and CPB2 (SEQ ID NO: 2), inclusive of their signal sequences. The peptide signal sequence of each are indicated in bolded letters. The terms “peptide signal”, “signal peptide”, and “signal sequence” are used interchangeably herein. The underlined residues and slash in the amino acid sequence of CPB2 presented in SEQ ID NO: 2 shows the thrombin-thrombomodulin (T-TB) cleavage sequence. Also provided in Table 1 is an exemplary CPN1 variant fused to an activation peptide T-TB from CPB2, shown in SEQ ID NO: 3.

[0041] Table 1 provides a number of human CPN1 amino acid sequences, some of which are truncation variants of full length human wild type CPN1 of SEQ ID NO: 6. The sequences of Table 1 include: SEQ ID NO: 1, which constitutes full length human wild type CPN1 with a signal peptide (referred to herein as hCPN1 (1-438) with a signal peptide); SEQ ID NO: 6, which constitutes full length human wild type CPN1, (referred to herein as hCPN1 (1-438); SEQ ID NO: 7, which constitutes the N-terminal amino acids 1-398 of SEQ ID NO: 6 (referred to herein as hCPN1 (1-398), a CPN1 truncation variant); SEQ ID NO:

11, which constitutes the N-terminal amino acids 1-397 of SEQ ID NO: 6 (referred to herein as hCPN1 (1-397), a CPN1 truncation variant); SEQ ID NO: 12, which constitutes the N-terminal amino acids 1-396 of SEQ ID NO: 6 (referred to herein as hCPN1 (1-396), a CPN1 truncation variant); SEQ ID NO: 8, which constitutes the N-terminal amino acids 1-320 of SEQ ID NO: 6 (referred to herein as hCPN1 (1-320), a CPN1 truncation variant). One or more of these hCPN1 sequences are further modified to generate the CPN1 variants of the disclosure, as exemplified in Tables 3A-4C and 4A-4C. These hCPN1 sequences, including variants thereof, are also used as core molecules for the fusion constructs of the disclosure, as provided in Tables 2 and 5.

[0042] In some embodiments, the CPN1 variants provided herein are useful for treatment of a subject in need thereof. As used herein, the terms “patient” or “subject” refer to any vertebrate including, without limitation, humans and other primates (e.g., chimpanzees, cynomolgus monkeys, and other apes and monkey species), farm animals (e.g., cattle, sheep, pigs, goats and horses), domestic mammals (e.g., dogs and cats), laboratory animals (e.g., rabbits, rodents such as mice, rats, and guinea pigs), and birds (e.g., domestic, wild and game birds such as chickens, turkeys and other gallinaceous birds, ducks, geese, and the like). In some embodiments, the subject is a mammal. In exemplary embodiments, the subject is a human.

[0043] In some embodiments, provided are CPN2 variants; a CPN2 variant may comprise one or more modifications with respect to hCPN1 (1-456), SEQ ID NO: 10.

[0044] In some embodiments, the CPN1 variant of the disclosure comprises at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity to SEQ ID NO: 6.

[0045] In some embodiments, the CPN1 variant of the disclosure comprises at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity to SEQ ID NO: 7.

[0046] In some embodiments, the CPN1 variant of the disclosure comprises at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity to SEQ ID NO: 8.

[0047] In some embodiments, the CPN1 variant of the disclosure comprises at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity to SEQ ID NO: 11.

[0048] In some embodiments, the CPN1 variant of the disclosure comprises at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity to SEQ ID NO: 12.

[0049] In some embodiments, the CPN1 variant of the disclosure comprises SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 11, or SEQ ID NO: 12, comprising

one of the modification strings selected from the group consisting of the modification strings provided in Table 4A, Table 4B, and Table 4C.

[0050] In some embodiments, the CPN1 variant of the disclosure comprises SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 11, or SEQ ID NO: 12, comprising one of the modification strings selected from the group consisting of the modification strings provided in Table 3A, Table 3B, and Table 3C.

[0051] In some embodiments, a CPN1 variant comprise one or more modifications with respect to hCPN1 (1-438), SEQ ID NO: 6. Table 4C of the disclosure provides exemplary modifications of hCPN1 (1-438), SEQ ID NO: 6, listed singly. Accordingly, a CPN1 variant of the disclosure may comprise one or more of the modifications provided in Table 4C. Table 3C of the disclosure provides exemplary modification strings of hCPN1 (1-438), SEQ ID NO: 6. Accordingly, a CPN1 variant of the disclosure may comprise one of the modification strings provided in Table 3C.

[0052] In some embodiments, a CPN1 variant comprise one or more modifications with respect to hCPN1 (1-398), SEQ ID NO: 7. Table 4A of the disclosure provides exemplary modifications of hCPN1 (1-398), SEQ ID NO: 7 listed singly. Accordingly, a CPN1 variant of the disclosure may comprise one or more of the modifications provided in Table 4A. Table 3A of the disclosure provides exemplary modification strings of hCPN1 (1-398), SEQ ID NO: 7. Accordingly, a CPN1 variant of the disclosure may comprise one of the modification strings provided in Table 3A.

[0053] In some embodiments, a CPN1 variant comprise one or more modifications with respect to hCPN1 (1-397), SEQ ID NO: 11.

[0054] In some embodiments, a CPN1 variant comprise one or more modifications with respect to hCPN1 (1-396), SEQ ID NO: 12.

[0055] In some embodiments, a CPN1 variant is hCPN1 (1-320) (also referred to herein as hCPN1 (1-320) Delta TT), SEQ ID NO: 8. The amino acid sequence of hCPN1 (1-320), SEQ ID NO: 8, is a C-terminal truncation of the transthyretin (TT) domain of hCPN1 (1-398), SEQ ID NO: 7. In some embodiments, a CPN1 variant comprise one or more modifications with respect to hCPN1 (1-320), SEQ ID NO: 8. Table 4B of the disclosure provides exemplary modifications of hCPN1 (1-320), SEQ ID NO: 8, listed singly. Accordingly, a CPN1 variant of the disclosure may comprise one or more of the modifications provided in Table 4B. Table 3B of the disclosure provides exemplary modification strings of hCPN1 (1-320), SEQ ID NO: 8. Accordingly, a CPN1 variant of the disclosure may comprise one of the modification strings provided in Table 3B.

[0056] In some embodiments, the CPN1 variants provided herein are in an active form. In other embodiments the CPN1 variants are provided in an inactive, zymogen form.

Improved Characteristics of the CPN1 Variants

[0057] The CPN1 variants provided herein may modulate activity of the complement system, and have at least one improved characteristic as compared to a wild type CPN1. In some embodiments, the CPN1 variants provided herein have at least one improved characteristic as compared to the wild type CPN1, wherein the wild type CPN1 is a human CPN1. In some embodiments, the improved characteristic includes, but is not limited to, an increase or a decrease in

any one or more of: half-life, activity, potency, affinity for one or more substrates, sensitivity, cofactor affinity, stability, and catalytic capability.

[0058] In some embodiments, the CPN1 variants do not require a CPN2 for activity. In some embodiments, the CPN1 variants exhibit a higher affinity for C3a, C5a, or both C3a and C5a. In some embodiments the CPN1 variants are part of a fusion construct comprising additional N-terminal and C-terminal domains. In some embodiments, the CPN1 variants are non-immunogenic.

[0059] In some embodiments, the at least one improved characteristic comprises an increase in affinity for one or more substrates, wherein at least one substrate is C3a. In some embodiments, the at least one improved characteristic comprises an increase in affinity for one or more substrates, wherein at least one substrate is C5a. In some embodiments, the CPN1 variants provided herein are inactive towards the C5adesArg and C3adesArg.

[0060] In some embodiments, the increase in activity of a CPN1 variant of the disclosure, as compared to a wild type CPN1, comprises an increase in the cleavage of C3a and/or C5a.

[0061] In some embodiments, the increase in activity of a CPN1 variant of the disclosure, as compared to a wild type CPN1, comprises an increased k_{cat}/K_M ($M^{-1} s^{-1}$) for cleavage of C3a and/or C5a.

[0062] In some embodiments, the increased k_{cat}/K_M ($M^{-1} s^{-1}$) for cleavage of C3a and/or C5a exhibited by a CPN1 variant is about 2-fold, about 3-fold, about 4-fold, about 5-fold, about 6-fold, about 7-fold, about 8-fold, about 9-fold, about 10-fold greater than that of the wild type CPN1.

[0063] In some embodiments, the increase in activity of a CPN1 variant of the disclosure, as compared to a wild type CPN1, comprises an increase in k_{cat} with a decrease in K_M . Generally, an increase in k_{cat} with a decrease in K_M is an increase in affinity, and therefore, in some embodiments, the increase in activity comprises an increase in affinity for a substrate.

[0064] In some embodiments, the increase in activity of a CPN1 variant of the disclosure, as compared to a wild type CPN1, comprises decrease in K_D (nM) value for cleavage of C3a and/or C5a. Generally, a decrease in K_D value is an increase in affinity, and therefore, in some embodiments, the increase in activity comprises a decrease in K_D value. In some embodiments, the decreased K_D (nM) value is about 2-fold, about 3-fold, about 4-fold, about 5-fold, about 6-fold, about 7-fold, about 8-fold, about 9-fold, about 10-fold less than that of the wild type CPN1.

[0065] In some embodiments, the increase in activity of a CPN1 variant of the disclosure, as compared to a wild type CPN1, comprises a decrease in EC_{50} (nM) value for cleavage of C3a and/or C5a, compared to the wild type CPN1. Generally, a decrease in EC_{50} is an increase in affinity, and therefore, in some embodiments, the increase in activity comprises a decrease in EC_{50} .

[0066] In some embodiments, the decreased EC_{50} (nM) value for cleavage of C3a and/or C5a is ranges from about 0.1 nM to about 20 nM. In some embodiments, the value is about 20 nM, or is lower than about 20 nM, e.g. about 10 nM, or about 15 nM. In some embodiments, the decreased EC_{50} value for cleavage is about 1 nM, about 0.1 nM, or less than about 0.1 nM.

[0067] In some embodiments, the improved characteristic of a CPN1 variant of the disclosure, as compared to a wild

type CPN1, is an increased half-life, wherein the increased half-life is half-life in plasma. In some embodiments, the increased half-life in plasma is greater than about 24 hours. In some embodiments, the increased half-life in plasma is about 48 hours, about 50 hours, about 60 hours, about 70 hours, about 80 hours, about 90 hours, about 100 hours, or about 150 hours. In some embodiments, the increased half-life in plasma is from about 70 hours to about 150 hours.

[0068] In some embodiments, the improved characteristic of a CPN1 variant of the disclosure, as compared to a wild type CPN1, is an increased sensitivity for a substrate, wherein the increased sensitivity comprises increased in catalytic activity upon complement activation.

[0069] The CPN1 variants can be generated by introducing one or more modifications to a wild type CPN1. In some embodiments, the CPN1 variant comprises at least one modification corresponding to a wild type non-human CPN1.

[0070] In some embodiments, a modification to the amino acid sequence as set forth in SEQ ID NO: 6 can increase affinity of a CPN1 variant for C3a and/or C5a as compared to a CPN that is not modified. In some embodiments, a modification to the amino acid sequence as set forth in SEQ ID NO: 6 includes a truncation of a loop domain.

[0071] In some embodiments, the CPN1 variant comprises the deletion of one or more amino acid residues, e.g., a truncated CPN1 variant of the disclosure may be selected from the group consisting of SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 11, and SEQ ID NO: 12. These truncated variants can further serve as base molecules for fusion constructions comprising CPN1 variants of the disclosure.

Fusion Constructs

[0072] Provided herein are fusion constructs comprising wild type CPN1 or any of the CPN1 variants described herein.

[0073] In some embodiments, provided herein are CPN1-containing fusion constructs (fusions to other proteins or components), referred to herein as “CPN1 constructs”. In some embodiments, CPN1 constructs comprise a fusion to a protein or component that is from CPN (such as CPN2). In some embodiments, CPN1 constructs comprise a fusion to a protein or component that is not from CPN. For example, such components can be, but are not limited to, other members of the carboxypeptidase family such as CPB2, or a half-life extenders (e.g. albumin).

[0074] In some embodiments, hCPN1 (1-438), SEQ ID NO: 6 can be considered a core CPN1 molecule based upon which fusion constructs are generated. In some embodiments, hCPN1 (1-398), SEQ ID NO: 7 can be considered a core CPN1 molecule based upon which fusion constructs are generated. In some embodiments, hCPN1 (1-320), SEQ ID NO: 8 can be considered a core CPN1 molecule based upon which fusion constructs are generated.

[0075] In some embodiments, the fusion construct comprises hCPN1 (1-398), SEQ ID NO: 7. In some embodiments, the fusion construct comprises hCPN1 (1-320), SEQ ID NO: 8. In some embodiments, the fusion construct comprises hCPN1 (1-396), SEQ ID NO: 12. In some embodiments, the fusion construct comprises hCPN1 (1-397), SEQ ID NO: 11. In some embodiments, the fusion construct comprises hCPN1 (1-438), SEQ ID NO: 6. In some embodiments, the fusion construct comprises hCPN2

(1-456), SEQ ID NO: 14. In some embodiments, the fusion construct comprises hCPN2 (1-524), SEQ ID NO: 13.

[0076] Table 1 provides exemplary components of certain fusion constructs of the disclosure. Table 2 lists exemplary CPN1-containing fusion constructs of the disclosure and their amino acid sequences. In some embodiments the construct comprises one or more of a signal sequence, N-terminal fusion partner 2, N-terminal linker 2, N-terminal fusion partner 1, N-terminal linker 1, a core CPN molecule (such as hCPN1 (1-398), SEQ ID NO: 7, or hCPN1 (1-320), SEQ ID NO: 8), C-terminal linker 1, C-terminal fusion partner, C-terminal linker 2, C-terminal fusion partner 2.

[0077] In some embodiments the fusion construct comprises one or more of Human Serum Albumin (HSA), the signal sequence of HSA, the signal sequence of Azurocidin, the signal sequence of interleukin 2, the signal sequence of immunoglobulin G, the activation peptide of CPB2 (the N-terminal 96 amino acids of CPB2 protease), Small Ubiquitin Modifying enzyme (SUMO), 10 histidine residues followed by SUMO, Tobacco Etch Virus (TEV) protease cleavage site, linkers consisting of amino acids (such as repeating GS) of various lengths (such as 8, 12, or 20 amino acid residues), His-tag, Fc region (the fragment crystallizable region is the tail region of an antibody that interacts with cell surface receptors called Fc receptors and some proteins of the complement system), mutations introduced in the Fc region (such as no 1st DK LALA-PG), Xa (factor Xa protease cleavage site), mammalian maltose binding protein (mMBP), CPB2, and/or CPN2. To the extent certain abbreviations are provided herein, including the drawings, reference is made to the abbreviations in Table 1.

[0078] Table 5 provides the full amino acid sequences of exemplary fusion constructs of the disclosure. Accordingly, in some embodiments, provided herein is a CPN1 fusion construct selected from any of those presented in Table 5. Also provided in Table 5 are optional signal sequence peptides that may be used for the expression of the fusion constructs presented therein.

[0079] In some embodiments the activation peptide of CPB2 (the N-terminal 96 amino acids of CPB2 protease) may be mutated to improve binding with CPN1, and is included in a fusion construct of the disclosure.

[0080] FIG. 2 depicts a schematic diagram of CPB2 (also known as carboxypeptidase U (CPU), plasma carboxypeptidase B (pCPB) or thrombin-activatable fibrinolysis inhibitor (TAFI)), another member of the carboxypeptidase M14 family, which has similar substrates as CPN1. Because CPB2 is secreted as a zymogen in circulation, its domains may be useful in the modification of CPN1 to generate CPN1 zymogen constructs. CPB2 includes an activation peptide (cleavable substrate) and is endogenously activated by thrombin-thrombomodulin or plasmin but is also sensitive to mannan-binding lectin serine protease 1 (MASP1) cleavage in vitro.

[0081] As discussed above, it should be understood that a CPN1 variant may also be fused with another component, such as a half-life extender or a portion of another carboxypeptidase providing stability in circulation, as will be further discussed herein.

[0082] The fusion constructs of the disclosure may be engineered to display an increase in sensitivity comprises an increase in the sensitivity to any one or more of: Mannan-binding lectin serine protease 1 (MASP1), Mannan-binding lectin-associated serine protease 3 (MASP3), Factor D,

methyl-accepting chemotaxis protein (mCPA3) secreted during mast cell degranulation, and cathepsin G secreted during neutrophil degranulation. In some embodiments, the decrease in sensitivity comprises decrease in thrombin-thrombomodulin.

[0083] In some embodiments, the fusion constructs provided herein comprise one or more of domains of CPN selected from: a first peptide signal, a catalytic domain (CPN1) or a portion of CPN1, an activation peptide sensitive to complement activation, and a regulatory subunit (CPN2) or a portion of CPN2. FIG. 3 depicts schematic diagrams of a number of exemplary fusion constructs, e.g. a CPN1 with its catalytic subunit alone, a CPN1 with its catalytic subunit plus its regulatory subunit. FIG. 3 also includes a schematic diagram of a CPB2 having a signal peptide (“peptide signal” in FIG. 3 schematics), an activation peptide (AP) domain, and a catalytic domain. By way of example, a fusion construct can include an activation peptide or a portion of the activation peptide from CPB2, and in some embodiments, can be expressed as a single heterodimer.

[0084] It should be understood that, while exemplary fusion constructs disclosed herein can include an activation peptide, any of the fusion constructs disclosed herein can be provided optionally without an activation peptide, or optionally with a first and a second activation peptide, or optionally with a single activation peptide.

[0085] Exemplary fusion constructs shown as schematic diagrams in FIG. 3 include:

[0086] (a) a fusion construct comprising a peptide signal, with an activation peptide from CPB2, a CPN1 domain, with a deletion of its CPN2;

[0087] (b) a fusion construct comprising a first peptide signal and CPN1 domain, and with an activation peptide from CPB2, and expressed as a heterodimer, also comprising a second peptide signal and its CPN2 domain;

[0088] (c) a fusion construct comprising a peptide signal, an activation peptide from CPB2, a CPN1 domain, a linker, and a CPN2 domain;

[0089] (d) a fusion construct comprising a peptide signal, an activation peptide from CPB2, a CPN1 domain, a linker, and a half-life extender (human serum albumin shown as an exemplary half-life extender).

[0090] Further, any of the fusion construct provided herein can be HSA-tagged, or His-tagged, as is also shown in FIG. 3 as examples.

[0091] In some embodiments, a fusion construct comprises at least one non-CPN domain or component. In such embodiments, the insertion of the at least one non-CPN domain of component can help to stabilize the CPN1 domain, such that the deletion of the CPN2 domain does not reduce the clearance of the CPN1 domain from plasma, for example.

[0092] In some embodiments, the at least one non-CPN domain or component comprises at least one domain of carboxypeptidase B2 (CPB2). In some embodiments, the CPB2 is a human CPB2, SEQ ID NO: 2. In some embodiments, the at least one domain of CPB2 comprises an activation peptide.

[0093] In some embodiments, the fusion construct comprises at least one modification corresponding to a wild type CPN1 comprising the amino acid sequence as set forth in SEQ ID NO: 3.

[0094] FIG. 4A depicts schematic diagrams of fusion constructs comprising various activation peptides to alter the sensitivity to certain complement components. In some embodiments, the activation peptide increases sensitivity of the fusion construct for any one or more of: MASP1, MASP3, Factor D, mCPA3, or cathepsin G. By way of example, an increased sensitivity for select complement components can direct activity of the fusion construct in the classical, alternate, and lectin pathways. In some embodiments, the fusion construct is not activated by thrombin-thrombomodulin. Exemplary fusion constructs comprising an activation peptide from CPB2 shown as schematic diagrams in FIG. 4A include the activation peptides for the following:

[0095] (a) an activation peptide for thrombin-thrombomodulin (T-TB), which can be used as a starting point for generation of peptide libraries that are more sensitive to MASP1, MASP3, Factor D, mCPA3, or cathepsin G, and less sensitive to T-TB;

[0096] (b) an activation peptide for MASP1, which can increase sensitivity to MASP1, within the classical pathway of the complement system;

[0097] (c) an activation peptide for MASP3, which can increase sensitivity to MASP3, within the alternate pathway of the complement system;

[0098] (d) an activation peptide for Factor D, which can increase sensitivity to Factor D, within the alternate pathway of the complement system;

[0099] (e) an activation peptide for mCPA3, which can increase sensitivity to mast cell degranulation, for mast cell diseases;

[0100] (f) an activation peptide for cathepsin G, which can increase sensitivity to neutrophil degranulation, within the classical, alternate, and/or lectin pathways of the complement system.

[0101] The fusion constructs may be designed with an activation peptide sensitive to neutrophil or mast cell activation. Neutrophil-sensitive activation peptide can allow targeting ANCA and a mast cell-activation peptide can allow targeting skin HS and a broad range of mast cell related disorders.

[0102] As noted above, also provided herein are fusion constructs comprise one or more domains of a non-CPN component, such as a non-CPN carboxypeptidase. In some embodiments, the activation peptide of a fusion construct masks the catalytic site of CPN1. An exemplary CPN fusion construct with this property is a CPN1-CPB2 chimera. FIGS. 4B-4C depict the structure of a CPN1-CPB2 chimera, showing the CPB2 activation peptide (TAFI activation peptide) masking the catalytic site of CPN1, and a schematic diagram of the fusion construct, respectively. The inclusion of a mask effectively allows the CPN1 containing fusion construct to be delivered in a zymogen format.

[0103] Another example of such a fusion construct is a fusion of CPN1 and carboxypeptidase A4 (CPA4). FIGS. 4D-4E depict the structure of a CPN1-CPA4 fusion construct, and a surface representation of the same, respectively. These figures show that the activation peptide from CPA4 masks the catalytic site of CPN1.

[0104] Another example of such a fusion construct is a fusion of CPN1 and carboxypeptidase A1 (CPA1). FIGS. 4F-4G depict the structure of a CPN1-CPA1 fusion, and a

surface representation of the same, respectively. These figures show that the activation peptide from CPA1 masks the catalytic site of CPN1.

[0105] In some embodiments, the activation peptide increases sensitivity of the fusion construct for any one or more of: mast cell degranulation, neutrophil degranulation, and inflammatory cell activation. In some embodiments, the activation peptide increases sensitivity of the fusion construct to complement activation.

[0106] In some embodiments, fusion construct is a fusion and comprises at least one CPB2 domain which comprises any one or more of: a CPB2 catalytic domain, a CPA1 catalytic domain, and a CPA4 catalytic domain.

[0107] In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (first peptide signal)-(CPB2 activation peptide)-(CPN2). In some embodiments, the fusion construct comprises a structural arrangement from C-terminus to N-terminus as (first peptide signal)-(CPB2 activation peptide)-(CPN2). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (mMBP)-(optional linker)-(TEV)-(optional linker)-(CPN1 core molecule)-(optional linker)-(HSA). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (mMBP)-(optional linker)-(TEV)-(optional linker)-(CPN1 core molecule). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (mMBP)-(optional linker)-(TEV)-(optional linker)-(core CPN1 molecule)-(optional linker)-(TEV)-(optional linker)-(HSA). The mMBP can be SEQ ID NO: 30, the optional linker can be SEQ ID NOS: 26, 32, 33, 34, 35, 36, 37, 38, 39, 41, 42, or 43. The core CPN1 molecule can be SEQ ID NOS: 1, 6, 7, 11, 12, or 8. The TEV can be SEQ ID NO: 25. The HSA can be SEQ ID NO: 17. In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (GST)-(optional linker)-(TEV)-CPN1 core molecule-(optional linker)-(HSA). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (GST)-(optional linker)-(TEV)-(optional linker)-(CPN core molecule)-(optional linker)-(TEV)-(optional linker)-(HSA). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (GST)-(optional linker)-(TEV)-(core CPN1 molecule). The GST can be SEQ ID NO: 31. In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (GHHHHHHHHHHH)-(optional linker)-(SUMO)-(CPN1 core molecule). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (GHHHHHHHHHHH)-(optional linker)-(SUMO)-(CPN1 core molecule)-(optional linker)-(HSA). The GHHHHHHHHHHH can be SEQ ID NO: 34. The SUMO can be SEQ ID NO: 24. In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (CPB2-Activation Peptide)-(optional linker)-(CPN1 core molecule). The CPB2-Activation Peptide can be SEQ ID NO: 22. In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (CPB2-Activation Peptide)-(optional linker)-(CPN1 core molecule)-(optional linker)-(HHHHHHH). The HHHHHHH can be SEQ ID NO: 40. In some embodiments, the fusion construct comprises a struc-

tural arrangement from N-terminus to C-terminus as (CPB2-Activation Peptide)-(optional linker)-(CPN1 core molecule)-(optional linker)-(HSA). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (CPB2-Activation Peptide)-(optional linker)-(CPN1 core molecule)-(optional linker)-(hCPN2 (1-339)). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (CPB2-Activation Peptide)-(optional linker)-(CPN1 core molecule)-(optional linker)-(hCPN2 (1-339))-(optional linker)-(HHHHHHH). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (CPN1 core molecule)-(optional linker)-(hCPN2 (1-339)). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (CPN1 core molecule)-(optional linker)-(hCPN2 (1-456)). The hCPN2 (1-456) can be SEQ ID NO: 10. In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (CPN1 core molecule)-(optional linker)-(HHHHHHH). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (CPN1 core molecule)-(optional linker)-(HSA). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (CPN1 core molecule)-(optional linker)-(HSA)-(optional linker)-(HHHHHHH). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (CPN1 core molecule)-(optional linker)-(TEV)-(optional linker)-(Fc (no 1st DK LALA-PG)). The Fc (no 1st DK LALA-PG) can be SEQ ID NO: 28. In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (CPN1 core molecule)-(optional linker)-(Xa)-(optional linker)-(Fc (no 1st DK LALA-PG)). The Xa can be SEQ ID NO: 29. In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (CPN1 core molecule)-(optional linker)-(Xa)-(optional linker)-(HSA). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (CPN1 core molecule)-(optional linker)-(HSA)-(optional linker)-(HHHHHHHHHHH). In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as CPN1 core molecule-(optional linker)-(HSA).

[0108] Table 2 provides exemplary fusion constructs. The column titled “Const. No.” refers to the construct number, a unique number assigned to the construct. Each component of the fusion protein is presented from the N-terminus to the C-terminus in the subsequent columns, presented from N-terminus to C-terminus. If a cell is blank that indicates that the particular construct does not contain that component.

[0109] For example, the component in the left-most side of the table, if present, is found in the far N-terminus, labeled as “N2 Term.” This may then be connected via linker (“N2 Linker”) to another component in the N Terminus, if present,

is closer to the core molecule, labeled “N Term.” The “N Term” component may then be connected to the core molecule via an optional linker labeled “N Linker.” The column entitled “core” denotes the core CPN1 or CPN2 molecule, making reference to exemplary molecules in Table 1. To the right of the core molecule is an optional “C Linker”, which if present, links the core to a component in the “C Term,” which, in present, is in turn connected to a “C2 Term” component via an optional “C2 Linker.”

[0110] In some embodiments, the fusion constructs provided herein are fused to a first non-CPN domain or component, and a second non-CPN domain or component. In some embodiments, the non-CPN domain or component is a half-life extender. In some embodiments, the half-life extender is selected from the group consisting of: PEGylation, PASylation, carbohydrates, albumin, and Fc. In some embodiments, the fusion is at the C-terminal end of the CPN1 core. In some embodiments, the fusion is at the N-terminal end of the fusion construct. In some embodiments, the fusion is at both the C-terminal and N-terminal ends of the CPN1 core. Other exemplary chimeras include fusions of CPN1 with: a portion of a redesigned CPN2, a portion of CPB2 activation peptide, or other CPB2 domains.

[0111] In some embodiments, the fusion of at least one non-CPN domain or component comprises a first non-CPN domain or component and a second non-CPN domain or component. In some embodiments, the fusion construct comprises a structural arrangement from N-terminus to C-terminus as (first peptide signal)-(first non-CPN domain or component)-(CPN2)-(second non-CPN domain or component). In some embodiments, the fusion construct comprises a structural arrangement from C-terminus to N-terminus as (first peptide signal)-(first non-CPN domain or component)-(CPN2)-(second non-CPN domain or component). In some embodiments, the first non-CPN domain or component is an activation peptide. In some embodiments, the second non-CPN domain or component is a half-life extender. It may be advantageous in some embodiments to increase the half-life of CPN1. Exemplary half-life extenders include, but are not limited to albumin, such as human serum albumin, PEG, a non-biodegradable polymer, a biodegradable polymer, and Fc. Addition of a half-life extender can also increase or alter other properties of the fusion constructs provided herein, such as, but not limited to, bioavailability, trafficking ability, and immunogenicity.

[0112] In some embodiments, the half-life extender is albumin. It is noted that as used herein, albumin refers to any albumin such as any serum albumin, or an albumin variant, or albumin derivative. In exemplary embodiments, the albumin is human serum albumin (HSA). Exemplary albumin containing CPN1 fusion constructs of the disclosure are provided in Table 2 and Table 5.

[0113] In some embodiments, the addition of a component or domain to a fusion construct of the disclosure can be directly to the CPN1 variant core molecule. In some embodiments, the addition of a component or domain to a fusion construct of the disclosure can be through a linker or multiple linkers.

[0114] As discussed above, CPN is endogenously found as a tetramer. Accordingly, in some embodiments, the CPN variants provided herein are a tetramer comprising two heterodimers. In some embodiments, the CPN variants provided herein is a single heterodimer.

[0115] In some embodiments, the CPN1 variants or fusion constructs provided herein are in a zymogen form. In such embodiments, the CPN1 variants or fusion constructs engineered as a zymogen can be activated in situ at the site of dysregulated complement. By way of example, a fusion construct can be provided fused to an activation peptide, such as an activation peptide from CPB2, which can be cleaved at the site of dysregulated complement, such as by MASP1, MASP3, Factor D, mCPA3, or cathepsin G.

[0116] In some embodiments, the fusion constructs provided herein are in an active form.

Uses

[0117] Provided herein are methods of treating a disease or condition in a subject in need thereof, comprising administering to the subject any one of the CPN1 variants or fusion constructs provided herein.

[0118] In some embodiments, the disease or condition is an acute condition. In some embodiments, the disease or condition is an acute condition, and the CPN1 variant or fusion construct of the disclosure is short-lived and highly active. Examples of acute conditions for which such fusion constructs and variants can be useful include, but are not limited to, acute respiratory distress syndrome (ARDS), COVID-19, multisystem organ failure, and sepsis. Additionally, such fusion constructs and variants can be useful for acute disease in which significant complement activation occurs during extracorporeal blood treatment by cardiopulmonary bypass surgery, ischemia/reperfusion, and dialysis. These can contribute to complications of these procedures.

[0119] In some embodiments, the disease or condition is a chronic condition. In some embodiments, the disease or condition is a chronic condition, and the fusion constructs or variants have an extended half-life, and/or with a modified catalytic activity with respect to a wild type CPN1 or wild type CPN tetramer. Examples of chronic conditions for which such fusion constructs and variants can be useful include, but are not limited to, anti-neutrophil cytoplasmic autoantibody (ANCA) vasculitis, atypical hemolytic uremic syndrome (aHUS), and IgA nephropathy.

[0120] In some embodiments, the disease or condition is selected from the group consisting of: congenital complement deficiency, control protein deficiency, secondary complement disorder, immunity related disorder, chronic renal disorder, acute inflammatory disorder, antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV), C3 glomerulopathy (C3G), lupus nephritis, skin disorder, intestinal ischemia and reperfusion (I/R) injury, and sepsis. In some embodiments, the disease or condition is a skin disorder selected from the group consisting of: hidradenitis suppurativa (HS), bullous pemphigoid (BP), and Pyoderma Gangrenosum.

Administration

[0121] The in vivo administration of the CPN1 variants and fusion constructs described herein may be carried out intravenously, intramuscularly, subcutaneously, intravitreally, topically, orally, transdermally, intraperitoneally, intraorbitally, intrathecally, intraventricularly, intranasally, transmucosally, through implantation, or through inhalation. Administration may be performed with any suitable excipients, carriers, or other agents to provide suitable or improved tolerance, transfer, delivery, and the like.

[0122] In exemplary embodiments, administration of the therapeutic variants and fusion constructs described herein is a subcutaneous administration.

[0123] In exemplary embodiments, administration of the therapeutic variants and fusion constructs described herein is an intravenous administration.

[0124] Administration may be effectuated by delivery of the variant or fusion construct itself, or a gene therapy based delivery, using, for example, a viral vector.

Production

[0125] Provided herein are nucleic acids encoding the CPN1 variants of the disclosure, and vector comprising such nucleic acids. Also provided herein are nucleic acids encoding the fusion constructs of the disclosure, and vector comprising such nucleic acids. These are useful in using recombinant systems for production. Mammalian host cells

may be used for production, including, for example human and non-human cell production systems. Exemplary mammalian cells for production include, but are not limited to CHO cells, or HEK cells. Alternatively, non-mammalian systems may be used for recombinant production of the variants and fusion constructs of the disclosure, such as bacterial host systems (e.g. *E. Coli*), yeast, and insect cell systems. It is understood that the nucleic acids encoding the variants and fusion constructs of the disclosure may be codon optimized, to maximize production. The variants and fusion constructs provided herein can be cloned or isolated using any available methods known in the art for cloning and isolating nucleic acid molecules.

TABLES

[0126] Provided here are the Tables made reference herein.

TABLE 1

Exemplary Components		
Name	Description	Amino Acid Sequence
hCPN1 (1-438) with a signal peptide	Full length wild type human carboxypeptidase N 1, with a signal peptide	MSDLLSVFLHLLLFKLIVAPVTFRHHRYDDLVRTLYKVNQECPGITRV YSIGRSVEGRHLYVLEFSDHPGIHEPLEPEVKYVGNMHNALGRELML QLSEFLCEEFRNRNQRIQLIQDTRIHLPSMNPDPGYEVAAAQGNPKPGY LVGRNNANGVDLNRNFPDLNTYIYYNEKYGGPNHHLPLPDNWKSVQEP ETRAVIRWMHSEFVLSANLHGGAVVANYPYDKSFHEHRVGVRRVAST PTPDDKLFQKLAKVYSYAHGWMFQGWNCGDYFPDGTNGASWYLSLK GMQDFNYLHTNCFEITLSECDKFPPPEELQREWLGNREALIQFLEQVHQ GIKGMVLDENYNLANAVISVSGINHDTVSGDHGDFRLLLPGIYTVSA TAPGYDPETVTVTVGPAEPTLVNFHLKRSIPQVSPVRRAPSRRHGVRAKV QPQARKKEMEMRQLQRGPA (SEQ ID NO: 1)
hCPN1 (1-438)	Full length wild type human carboxypeptidase N1	VTFRHHRYDDLVRTLYKVNQECPGITRVYSIGRSVEGRHLYVLEFSDHP GIHEPLEPEVKYVGNMHNALGRELMLQLSEFLCEEFRNRNQRIQLIQ DTRIHLPSMNPDPGYEVAAAQGNPKPGYLVGRNNANGVDLNRNFPDLN TYIYYNEKYGGPNHHLPLPDNWKSVQEPETRAVIRWMHSEFVLSANLH GGAVVANYPYDKSFHEHRVGVRRVASTPTPDDKLFQKLAKVYSYAHG WMFQGWNCGDYFPDGTNGASWYLSLKGMQDFNYLHTNCFEITLSECD DKFPPEELQREWLGNREALIQFLEQVHQGIKGMVLDENYNLANAVIS VSGINHDTVSGDHGDFRLLLPGIYTVSATAPGYDPETVTVTVGPAEPTL VNFHLKRSIPQVSPVRRAPSRRHGVRAKVQPQARKKEMEMRQLQRGPA (SEQ ID NO: 6)
hCPN1 (1-398)	human carboxypeptidase N 1, containing only the N-terminal amino acids from 1-398	VTFRHHRYDDLVRTLYKVNQECPGITRVYSIGRSVEGRHLYVLEFSDHP GIHEPLEPEVKYVGNMHNALGRELMLQLSEFLCEEFRNRNQRIQLIQ DTRIHLPSMNPDPGYEVAAAQGNPKPGYLVGRNNANGVDLNRNFPDLN TYIYYNEKYGGPNHHLPLPDNWKSVQEPETRAVIRWMHSEFVLSANLH GGAVVANYPYDKSFHEHRVGVRRVASTPTPDDKLFQKLAKVYSYAHG WMFQGWNCGDYFPDGTNGASWYLSLKGMQDFNYLHTNCFEITLSECD DKFPPEELQREWLGNREALIQFLEQVHQGIKGMVLDENYNLANAVIS VSGINHDTVSGDHGDFRLLLPGIYTVSATAPGYDPETVTVTVGPAEPTL VNFHLKRS (SEQ ID NO: 7)
hCPN1 (1-397)	human carboxypeptidase N 1, containing only the N-terminal amino acids from 1-397	VTFRHHRYDDLVRTLYKVNQECPGITRVYSIGRSVEGRHLYVLEFSDHP GIHEPLEPEVKYVGNMHNALGRELMLQLSEFLCEEFRNRNQRIQLIQ DTRIHLPSMNPDPGYEVAAAQGNPKPGYLVGRNNANGVDLNRNFPDLN TYIYYNEKYGGPNHHLPLPDNWKSVQEPETRAVIRWMHSEFVLSANLH GGAVVANYPYDKSFHEHRVGVRRVASTPTPDDKLFQKLAKVYSYAHG WMFQGWNCGDYFPDGTNGASWYLSLKGMQDFNYLHTNCFEITLSECD DKFPPEELQREWLGNREALIQFLEQVHQGIKGMVLDENYNLANAVIS VSGINHDTVSGDHGDFRLLLPGIYTVSATAPGYDPETVTVTVGPAEPTL VNFHLKR (SEQ ID NO: 11)
hCPN1 (1-396)	human carboxypeptidase N 1, containing only the N-terminal amino acids from 1-396	VTFRHHRYDDLVRTLYKVNQECPGITRVYSIGRSVEGRHLYVLEFSDHP GIHEPLEPEVKYVGNMHNALGRELMLQLSEFLCEEFRNRNQRIQLIQ DTRIHLPSMNPDPGYEVAAAQGNPKPGYLVGRNNANGVDLNRNFPDLN TYIYYNEKYGGPNHHLPLPDNWKSVQEPETRAVIRWMHSEFVLSANLH GGAVVANYPYDKSFHEHRVGVRRVASTPTPDDKLFQKLAKVYSYAHG WMFQGWNCGDYFPDGTNGASWYLSLKGMQDFNYLHTNCFEITLSECD DKFPPEELQREWLGNREALIQFLEQVHQGIKGMVLDENYNLANAVIS

TABLE 1-continued

Exemplary Components		
Name	Description	Amino Acid Sequence
		VSGINHDVTSGDHGDYFRLLLPGIYTVSATAPGYDPETVTVTVGPAEPTLVNFKLK (SEQ ID NO: 12)
hCPN1 (1-320)	human carboxypeptidase N 1 C-terminal truncation, containing only the N-terminal amino acids from 1 to 320, deletion of the transthyretin domain	VTRFHHRYDDLVRTLYKVQNECPGITRVYSIGRSVEGRHLYVLEFSDHPGIHEPLEPEVKYVGNMHGNEALGREMLQLSEFLCEEFRNRNRQIVQLIQDTRIHLPSMNPDPGYEVAAAQGPKNPGYLVGRNNANGVDLNRNFPDLNTYIYYNEKYGGPNHHLPLPDNWKSVQEPETRAVIRWMHFSFNFVLSANLHGGAVVANYPYDKSFEHRVRGVRTASTPTPDDKLFQKLAKVYSYAHGWMFQGWNCGDYFPDGTNGASWYSLSKGMQDFNYLHTNCFEITLLELSCDKFPPEEELQREWLGNREALIQFLEQVHQ (SEQ ID NO: 8)
hCPN2 (1-524) with signal peptide	Full length wild type human carboxypeptidase N 2, with a signal peptide	MGWSLILLFLVAVATRVHSCPMGCDCFVQEVFCSDEELATVPLDIPPYTKNII FVETSFTTLETRAFGSPNLT KVVF LNTQLCQFRPDAGGLPRLEDLEVTGSSFLNLSSTNIFSNLTSLGKLT LNFNMLEALPEGLFQHLAALES LHLQGNLQALPRRLFPQPLTHLKT LNLQAQLPEELFHP LTLSTLQTLKLSNNALSGLPQGVFGKLSLQELFLDSNNISELPPQVFSQLFCLERLWLQRNAITHLPLSIFASLGNLTFLSLQWNMLRVLPAGLFAHTPCLVGLSLTHNQLETV AEGTFAHLSNLRSLMLSYNATHTLPAGIFRDLEELVKLYLGSNNLTALHPALFQNL SKLELLSLSKNQLTTLPEGIFDTNYNLFNLALHGNPWQCDCHLAYFNWLQQYTDRL LNIQTYCAGPAYLKGQVVPALNEKQLVCPVTRDHLGFQVTWPDESKAGGSWDLAVQERAARSQCTYSNPEGTVVLACDQAQCRWLNVLSPQQGSLGLQYNASQEWDLRSSCGSLRLTVSIEARAAGP (SEQ ID NO: 15)
hCPN2 (1-524)	Full length wild type human carboxypeptidase N 2	CPMGCDCFVQEVFCSDEELATVPLDIPPYTKNII FVETSFTTLETRAFGSPNLT KVVF LNTQLCQFRPDAGGLPRLEDLEVTGSSFLNLSSTNIFSNLTSLGKLT LNFNMLEALPEGLFQHLAALES LHLQGNLQALPRRLFPQPLTHLKT LNLQAQLPEELFHP LTLSTLQTLKLSNNALSGLPQGVFGKLSLQELFLDSNNISELPPQVFSQLFCLERLWLQRNAITHTPLSIFASLGNLTFLSLQWNMLRVLPAGLFAHTPCLVGLSLTHNQLETV AEGTFAHLSNLRSLMLSYNATHTLPAGIFRDLEELVKLYLGSNNLTALHPALFQNL SKLELLSLSKNQLTTLPEGIFDTNYNLFNLALHGNPWQCDCHLAYFNWLQQYTDRL LNIQTYCAGPAYLKGQVVPALNEKQLVCPVTRDHLGFQVTWPDESKAGGSWDLAVQERAARSQCTYSNPEGTVVLACDQAQCRWLNVLSPQQGSLGLQYNASQEWDLRSSCGSLRLTVSIEARAAGP (SEQ ID NO: 13)
hCPN2 (1-456)	wild type human carboxypeptidase N 2 containing only the N-terminal amino acids from 1-456	MGWSLILLFLVAVATRVHSCPMGCDCFVQEVFCSDEELATVPLDIPPYTKNII FVETSFTTLETRAFGSPNLT KVVF LNTQLCQFRPDAGGLPRLEDLEVTGSSFLNLSSTNIFSNLTSLGKLT LNFNMLEALPEGLFQHLAALES LHLQGNLQALPRRLFPQPLTHLKT LNLQAQLPEELFHP LTLSTLQTLKLSNNALSGLPQGVFGKLSLQELFLDSNNISELPPQVFSQLFCLERLWLQRNAITHTPLSIFASLGNLTFLSLQWNMLRVLPAGLFAHTPCLVGLSLTHNQLETV AEGTFAHLSNLRSLMLSYNATHTLPAGIFRDLEELVKLYLGSNNLTALHPALFQNL SKLELLSLSKNQLTTLPEGIFDTNYNLFNLALHGNPWQCDCHLAYFNWLQQYTDRL LNIQTYCAGPAYLKGQVVPALNEKQLVCPVTRDHLGFQVTWPDESKAGGSWDLAVQERAA (SEQ ID NO: 10)
CPB2 with signal peptide	Wild-type human CPB2 (with signal peptide)	MKLCSLAVLVPIVLFCEQHVFQSGQVLAALPRTSRQVQVLQNLTTTYEIVLWQPVTADLIVKKKQVHFFVNASDVNDNVKAHLNVSGIPCSVLLADVEDLIQQQISNDTVSPR/ASASYVEQYHSLNEIYSWIEFITERHPDMLTKIHIGSSFEKYPLYVLKVSQKQAAKNAIWIDCGIHAREWISPAFCLWFIGHTQFYGIIGQYTNLLRLVDFYVMPVVNVVDGYDSWKKNRMRWRKNSFYANNHCI GTDLNRNFASKHWCEGASSSSCSETYCYLPESEPEVKAVASFLLRRNINQIKAYISMHSYSQHIVFPYSYTRSKSKDHEELSLVASEAVRAIEKISKNTRYTHGHGSETLYLAPGGDDWIYDLGIKYSPTIELRDTGTYGFLLEPRYIKPTCREAFAAVSKIAWHVIRNV (SEQ ID NO: 2)
	Exemplary CPN1 construct comprising a CPB2 sequence	MKLCSLAVLVPIVLFCEQHVFQSGQVLAALPRTSRQVQVLQNLTTTYEIVLWQPVTADLIVKKKQVHFFVNASDVNDNVKAHLNVSGIPCSVLLADVEDLIQQQISNDTVSPR/ASASYVTRFHHRYDDLVRTLYKVQNECPGITRVYSIGRSVEGRHLYVLEFSDHPGIHEPLEPEVKYVGNMHGNEALGREMLQLSEFLCEEFRNRNRQIVQLIQDTRIHLPSMNPDPGYEVAAAQGPKNPGYLVGRNNANGVDLNRNFPDLNTYIYYNEKYGGPNHHLPLPDNWKSVQEPETRAVIRWMHFSFNFVLSANLHGGAVVANYPYDKSFEHRVRGVR/RTASTPTPDDKLFQKLAKVYSYAHGWMFQGWNCGDYFPDGTNGASWYSLSKGMQDFNYLHTNCFEITLLELSCDKFPPEEELQREWLGNREALIQFLEQVHQQGIKGMVL DENYNNLANAVIS (SEQ ID NO: 3)

TABLE 1-continued

Exemplary Components		
Name	Description	Amino Acid Sequence
TT	transthyretin domain	GIKGMVLDENYNLANAVISVSGINHDVTSGDHGDYFRLLLPGIYTVSA TAPGYDPETVTVTVGPAEPTLVNPHLKRS (SEQ ID NO: 16)
HSA	Human serum albumin	DAHKSEVAHRFKDLGEENFKALVLI AFAQYLQQCPFEDHVKLVNEVTEF AKTCVADESAENCDSLHSLFGDKLCTVATLRETYGEMADCCAKQPEPE RNECFQLQHKDDNPRLVLRPEVDVMCTAFHDNEETFLKKYLYEIARRH PYFYAPELLFFAKRYKAAFTCCQAADKAACLLPKLDELDEGKASSAK QRLKCASLQKFGERAFKAWAVARLSQRFPPKAEFAEVSKLVDTLTKVHT ECCHGDLLECADDRADLAKYICENQDSISSKLKECCEKPLLEKSHCIAEV ENDEMPADLPSLAADFVESKDVCKNYAEAKDVFLGMFLYEYARRHPDY SVVLLRLAKTYETTLKCCAAADPHECYAKVFDEFKPLVEEPQNLIKQ NCELFEQLGEYKFQNALLVRYTKKVPQVSTPTLVEVSRNLGKVGSKCK HPEAKRMPCAEDYLSVVLNQLCVLHEKTPVSDRVTKCCTESLVNRRPCF SALEVDETYVPKEFNAETFTFHADICTLSEKERQIKKQTALVELVKKPK ATKEQLKAVMDDFAAFVEKCKADDKETCFAEEGKKLVAAASQAALGL (SEQ ID NO: 17)
signal HSA	signal sequence of HSA	MKWVTFISLLFLFSSAYS (SEQ ID NO: 18)
signal Azurocidin	signal sequence of Azurocidin	MTRLTVLALLAGLLASSRA (SEQ ID NO: 19)
signal IL2	signal sequence of interleukin 2	MYRMQLLSICIALSLALVTNS (SEQ ID NO: 20)
signal Ig	signal sequence of immunoglobulin G	MGWSLILLFLVAVATRVHS (SEQ ID NO: 21)
Activation peptide	Activation peptide of CBP2, the N-terminal 96aa of CBP2 protease	FQSGQVLAALPRTSRQVQLQNLTTTYEIVLWQPVTDLIVKKKQVHFF VNASVDNVKAHLNVSGIPCSVLLADVEDLIQQQISNDTVSPRASAS (SEQ ID NO: 22)
10His-Sumo	10 Histidine residues followed by SUMO	GHHHHHHHHHSLQDSEVNQEAKPEVKPEVKPETHINLKVSDGSSEIFF KIKKTTPLRRLMEAFKRGKEMDSLTFLYDGIETIQADQTPEDLDMEDN DIEAHREQIGG (SEQ ID NO: 23)
SUMO	Small Ubiquitin Modifying enzyme	SLQDSEVNQEAKPEVKPEVKPETHINLKVSDGSSEIFFKIKKTTPLRRLME AFKRGKEMDSLTFLYDGIETIQADQTPEDLDMEDNDIEAHREQIGG (SEQ ID NO: 24)
TEV	Tobacco Etch Virus protease cleavage site	ENLYFQG (SEQ ID NO: 25)
LL	Long Linker (20amino acid-length version of GS linker)	GSSGGSSGGSSGGSSGGSSG (SEQ ID NO: 26)
Fc	The fragment crystallizable region (Fc region) is the tail region of an antibody that interacts with cell surface receptors called Fc receptors and some proteins of the complement system.	THTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPE VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEY KCKVSNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLV KGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQ QGNVFPSCSMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 27)
(no 1st DK LALA-PG)	mutations introduced in the Fc region	THTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPE VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEY KCKVSNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLV KGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQ QGNVFPSCSMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 28)
Xa	Factor Xa protease cleavage site	AEGR (SEQ ID NO: 29)

TABLE 1-continued

Exemplary Components		
Name	Description	Amino Acid Sequence
mMBP	mammalian maltose binding protein	KTEEGKLVINGDKGYNGLAIEVGGKFEKDTGIKVTVEHPDKLEEKFPQ VAATGDGPDIIFWAHDRFGGYAQSGLLAEITPAAAFQDKLYPFTWDAVR YNGKLIAYPIAVEALSLIYNKDLLPNPPKTWEEIPALDKELKAKGKSALM FNLQEPYFTWPLIAADGGYAFKYAAGKYDIKDVGVNDAGAKAGLTFLV DLIKNKHMNADTDYSIAEHAFNHGETAMTINGPWAWSNIDTSAVNYGV TVLPTFKGQPSKPFVGVLSAGINAASPNKELAKEFLENYLLTDEGLEAVN KDKPLGAVALKSYEEELVKDPRVAATMENAQKGEIMPNI PQMSAFWYA VRTAVINAASGRQTVDAAL (SEQ ID NO: 30)
GST	Glutathione S transferase	MSPILGWYKIKGLVQPTRLLLEYLEEKYEELHYERDEGDKWRNKKFELG LEFPNLPPYYIDGDVKLTSMAIRIYIADKHNMGGCPKERAIEISMLEGAV LDIRYGVSRAYSKDFETLKVDFLSKLPPEMLKMFEDRLCHKTYLNGDHV THPDFMLYDALDVLYMDPMCLDAFPKLVCFKKRIEAI PQIDKYLKSSK YIAWPLQGWQATFGGDDHPPK (SEQ ID NO: 31)

TABLE 2

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
1					hCPN1 (1-398)				
2					hCPN1 (1-398)				
3					hCPN1 (1-398)				
4					hCPN1 (1-398)				
5					hCPN1 (1-398)		HHHHHH (SEQ ID NO: 32)		
6					hCPN1 (1-398)			GSSGG SSGGS SG (SEQ ID NO: 33)	HSA
7					hCPN1 (1-398)	GSSG GSSG GSSG (SEQ ID NO: 33)	hCPN2 (1-339)		
8					hCPN1 (1-398)	GSSG GSSG GSSG (SEQ ID NO: 33)	hCPN2 (1-339)		HHHH HH (SEQ ID NO: 32)
9			CPB2- Activa- tion peptide		hCPN1 (1-398)				

TABLE 2-continued

Exemplary Fusion Constructs								
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker C2Term
10			CPB2- Activa- tion peptide		hCPN1 (1-398)		HHHH HH (SEQ ID NO: 32)	
11			CPB2- Activa- tion peptide		hCPN1 (1-398)	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
12			CPB2- Activa- tion peptide		hCPN1 (1-398)	GSSG GSSG GSSG (SEQ ID NO: 33)	hCPN2 (1-339)	
13			CPB2- Activa- tion peptide		hCPN1 (1-398)	GSSG GSSG GSSG (SEQ ID NO: 33)	hCPN2 (1-339)	HHHH HH (SEQ ID NO: 32)
14	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-398)			
15	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-398)	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
16	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-320); A (321-398)			
17	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-320); A (321-398)	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
18					hCPN1 (1-320); A (321-398)			
19					hCPN1 (1-320); A	GSSG GSSG GSSG	HSA	

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
					(321-398)		(SEQ ID NO: 33)		
20					hCPN1 (1-398)		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
21					hCPN1 (1-398)		TEV	GSSG GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
22					hCPN1 (1-398)		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	Fc (no 1st DKLA LA-PG)
23					hCPN1 (1-398) ; R37G		Xa	GSSG GSSG GSSG (SEQ ID NO: 33)	Fc (no 1st DKLA LA-PG)
24					hCPN1 (1-398) ; R37G		Xa	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
25	mMBP	AAAQ TNAA AASS ASLE (SEQ ID NO : 35)	TEV		hCPN1 (1-398)	GSSG GSSG (SEQ ID NO : 33)	HSA		
26	mMBP	AAAQ TNAA AASS ASLE (SEQ ID NO : 35)	TEV		hCPN1 (1-398)				
27	GST	AAAQ TNAA AASS ASLE (SEQ ID NO : 35)	TEV		hCPN1 (1-398)	GSSG GSSG (SEQ ID NO : 33)	HSA		

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
28					hCPN1 (1-398)	GSSG GSSG GSSG GSSG (SEQ ID NO: 26)	hCPN2 (1-456)		
29					hCPN2 (1-456)				
30					hCPN2 (1-524)				
31					hCPN1 (1-320); L55N; H161R; N188D; V190K; K229E; L230W; N281D; C282S; F283Y; R308K; Q320N				
32					hCPN1 (1-320); L55N; H161S; N188D; V190R; K229E; L230W; C282S; F283Y; R308K; Q320N				
33					hCPN1 (1-320); L55N; H161K; N188D; V190K; K229E; L230W; C282S; F283Y; R308K; Q320N				
34					hCPN1 (1-320); L55N; H161S; N188D; V190R; K229E; L230K; K233D; C282S; F283Y; R308K; Q320K				

TABLE 2-continued

Exemplary Fusion Constructs								
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker
35					hCPN1 (1-320); L55N; H161R; N188D; V190K; K229E; L230W; K233H; N281D; C282S; F283Y; R308K; Q320N			
36					hCPN1 (1-320); L55N; H161R; N188D; V190K; K229E; L230K; N281D; C282S; F283Y; R308K; Q320N			
37					hCPN1 (1-438)			
38					hCPN1 (1-438)	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
39					hCPN1 (1-438)	GSSG GSSG GSSG GSSG (SEQ ID NO: 26)	CPN2 (1-456)	
40					hCPN1 (1-438)	GSSG GSSG GSSG GSSG (SEQ ID NO: 26)	CPN2 (1-524)	
41					hCPN1 (1-397)	GGSS GGSS GGSS GGSS GGSS GGSS (SEQ ID NO: 36)	CPN2 (1-425)	

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLink-er	Core	CLinker	CTerm	C2Link-er	C2Term
42					hCPN1 (1-397)	GGSS GGSS GGSS GGSS GGSS GGSS (SEQ ID NO: 36)	CPN2 (1-370)		CD180 (45 residue capping motif)
43					hCPN1 (1-397)	GGSS GGSS GGSS GGSS GGSS GGSS (SEQ ID NO: 36)	CPN2 (1-367)		LRIG1 (59 residue capping motif)
44					hCPN1 (1-396)	GGSS GGSS GGSS GGSS GGSS GGSS (SEQ ID NO: 36)	CPN2 (1-425)		
45					hCPN1 (1-396)	GGSS GGSS GGSS GGSS GGSS GGSS (SEQ ID NO: 36)	CPN2 (1-425)		CD180 (45 residue capping motif)
46					hCPN1 (1-396)	GGSS GGSS GGSS GGSS GGSS GGSS (SEQ ID NO: 36)	CPN2 (1-425)		LRIG1 (59 residue capping motif)
47					hCPN1 (1-397)	GGSS GGSS GGSS GGSS GGSS GG	CPN2 (1-425)		

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
48					hCPN1 (1-397)	(SEQ ID NO: 37) GGSS GGSS GGSS GG SSGG SSGG SSGG SSSS GG (SEQ ID NO: 38)	CPN2 (1-425)		
49					hCPN1 (1-396)	GGSS GGSS GGSS GGSS GGSS GG (SEQ ID NO: 37)	CPN2 (1-370)		CD180 (45 residue capping motif)
50					hCPN1 (1-396)	GGSS GGSS GGSS GGSS GGSS GGSS GGSS SSGG (SEQ ID NO: 38)	CPN2 (1-370)		CD180 (45 residue capping motif)
51	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-397)	GGSS GGSS GGSS GGSS GGSS GGSS GGSS GGSS GG (SEQ ID NO: 36)	CPN2 (1-425)		
52	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-397)	GGSS GGSS GGSS GGSS GGSS GGSS GGSS GGSS GG (SEQ ID NO: 36)	CPN2 (1-370)		CD180 (45 residue capping motif)

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
53	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-397)	GGSS GGSS GGSS GGSS GGSS GGSS GGSS (SEQ ID NO: 36)	CPN2 (1-367)		LRIG1 (59 residue capping motif)
54	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-396)	GGSS GGSS GGSS GGSS GGSS GGSS GGSS GGSS (SEQ ID NO: 36)	CPN2 (1-425)		
55	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-396)	GGSS GGSS GGSS GGSS GGSS GGSS GGSS GGSS (SEQ ID NO: 36)	CPN2 (1-425)		CD180 (45 residue capping motif)
56	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-396)	GGSS GGSS GGSS GGSS GGSS GGSS GGSS GGSS (SEQ ID NO: 36)	CPN2 (1-425)		LRIG 1 (59 residue capping motif)
57	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-397)	GGSS GGSS GGSS GGSS GGSS GGSS GG (SEQ ID NO: 37)	CPN2 (1-425)		
58	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-397)	GGSS GGSS GGSS GGSS GGSS GGSS	CPN2 (1-425)		

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
						GGSS GGSS SSGG (SEQ ID NO: 38)			
59	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-396)	GGSS GGSS GGSS GGSS GGSS GG (SEQ ID NO: 37)	CPN2 (1-370)		CD180 (45 residue capping motif)
60	GHHH HHHH HHH (SEQ ID NO: 34)	G	SUMO		hCPN1 (1-396)	GGSS GGSS GGSS GGSS GGSS GGSS GGSS SSGG (SEQ ID NO: 38)	CPN2 (1-370)		CD180 (45 residue capping motif)
61					hCPN1 (1-398); K60F; V62S; V180Y; S192A; N276T; E284I			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
62					hCPN1 (1-398); K60F; V180F; S192A; N276V; E284I			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
63					hCPN1 (1-398); K60F; V62M; V180A; S192A; M272F; N276V; E284I			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
64					hCPN1 (1-398); K60A; V62L; M184L; S192M; M272L; N276I; E284V			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
65					hCPN1 (1-398); A193W; A202V; Y238F; I285L; L287I; L289Y; E314A			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
66					hCPN1 (1-398); A202V; L287I; L289T			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
67					hCPN1 (1-398); A202I			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
68					hCPN1 (1-398); A235L			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
69					hCPN1 (1-398); V1I; F3W; E74L; Q78W			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
70					hCPN1 (1-398); V1I; E74L; Q78W			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
71					hCPN1 (1-398); K60F; V62S; V180Y; S192A; N276T; E284I	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA		
72					hCPN1 (1-398); K60F; V180F; S192A; N276V; E284I	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA		
73					hCPN1 (1-398); K60F;	GSSG GSSG	HSA		

TABLE 2-continued

Exemplary Fusion Constructs								
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker
				er				er
								C2Term
					V62M; V180A; S192A; M272F; N276V; E284I	(SEQ ID NO: 33)		
74					hCPN1 (1-398); K60A; V62L; M184L; S192M; M272L; N276I; E284V	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
75					hCPN1 (1-398); A193W; A202V; Y238F; I285L; L287I; L289Y; E314A	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
76					hCPN1 (1-398); A202V; L287I; L289T	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
77					hCPN1 (1-398); A202I	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
78					hCPN1 (1-398); A235L	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
79					hCPN1 (1-398); V1I; F3W; E74L; Q78W	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
80					hCPN1 (1-398); V1I; E74L; Q78W	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
81					hCPN1 (1-398); A70V	GSSG GSSG GSSG	HSA	

TABLE 2-continued

Exemplary Fusion Constructs								
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	C2Linker	C2Term
						(SEQ ID NO: 33)		
82					hCPN1 (1-398); N68D; P125S	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
83					hCPN1 (1-398); N68D; A70V; P125S	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
84					hCPN1 (1-398); E288Q	TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
85					hCPN1 (1-398); Y266F	TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
86					hCPN1 (1-398); Y266F; E288Q	TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
87					hCPN1 (1-398); A199S	TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
88					hCPN1 (1-398); K124R; P125S	TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
89					hCPN1 (1-398); G121T; P122Y; K124Q; P125D	TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs								
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker C2Term
90					hCPN1 (1-398); N123Y; K124R		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
91					hCPN1 (1-398); G121D; P122D; N123T; K124I		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
92					hCPN1 (1-398); G121D; P122H; K124I		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
93					hCPN1 (1-398); G121R; P122E; N123Y; K124I		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
94					hCPN1 (1-398); G121D; P122H; K124Q		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
95					hCPN1 (1-398); G121I; N123D		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
96					hCPN1 (1-398); P122W; N123R; K124Y		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
97					hCPN1 (1-398); P122Y; N123R; K124Y		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
98					hCPN1 (1-398); K124Q		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLink-er	Core	CLinker	CTerm	C2Link-er	C2Term
99					hCPN1 (1-398); G121T; P122Y; K124Q; P125D; E297P; E298Q		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
100					hCPN1 (1-398); G121T; P122Y; K124Q; P125D; E297P		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
101					hCPN1 (1-398); D292I; K293I; F294W		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
102					hCPN1 (1-398); K293L; E297P; E298Q		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
103					hCPN1 (1-398); D292L; K293I; P295L		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
104					hCPN1 (1-398); D292L; K293I		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
105					hCPN1 (1-398); D292N; K293A; P295T; E298S		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
106					hCPN1 (1-398); D292N; K293A; E298S		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
107					hCPN1 (1-398); D292I; K293A; E294W		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
108					hCPN1 (1-398); D292L; K293A; E294W		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
109					hCPN1 (1-398); D292L; K293V; E295L		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
110					hCPN1 (1-398); D292L; K293V; E298D		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
111					hCPN1 (1-398); K293N		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
112					hCPN1 (1-398); K293L; F294W; E298L		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
113					hCPN1 (1-398); D292N; K293L; F294W		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
114					hCPN1 (1-398); N123W; K124R; P125A; Y127D; D292I; K293I; F294Q		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
115					hCPN1 (1-398); N123Y; K124R; P127E; D292I; K293V; F294M		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
116					hCPN1 (1-398); G121T;		TEV	GSSG GSSG GSSG	HSA

TABLE 2-continued

Exemplary Fusion Constructs								
Const. No.	N2 Term	N2 Linker	N Term	NLink-er	Core	CLinker	CTerm	C2Link-er C2Term
					P122Y; K124Q; P125D; Y127W; D292I; K293L; E297P			(SEQ ID NO: 33)
117					hCPN1 (1-398); N123Y; K124R; Y127D; K293I		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
118					hCPN1 (1-398); G121D; P122S; N123T; K124I; D292N; K293A; E298S		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
119					hCPN1 (1-398); G121Q; P122W; N123D; Y127I; D292L; K293A; E294I		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
120					hCPN1 (1-398); G121D; P122W; N123D; Y127I; K293A; E294W		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
121					hCPN1 (1-398); P122W; N123R; K124H; Y127S; D292L; K293V; E298W		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
122					hCPN1 (1-398); P122W; N123R; K124Y; D292L; K293V; E298D		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
123					hCPN1 (1-398); K124Q; K293L; F294W; E298L		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
124					hCPN1 (1-398); K124D; K293L; F294W; E297D		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
125					hCPN1 (1-398); P125H		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
126					hCPN1 (1-398); P125L		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
127					hCPN1 (1-398); P122A; N123D; K124I; P125S		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
128					hCPN1 (1-398); P122A; N123D; K124S; P125S		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
129					hCPN1 (1-398); G121E; P122R; N123D; K124I; P125S		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
130					hCPN1 (1-398); G121D; P122R; N123D; K124T; P125S; G126Q		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
131					hCPN1 (1-398); N123D; K124D; P125S; G126Q		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
132					hCPN1 (1-398); N68D		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
133					hCPN1 (1-398); N68K; K293N		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
134					hCPN1 (1-398); V201H		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
135					hCPN1 (1-398); N203D		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
136					hCPN1 (1-398); G198D		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
137					hCPN1 (1-398); G198D; V201Y		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
138					hCPN1 (1-398); A70V		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
139					hCPN1 (1-398); N68D; P125S		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
140					hCPN1 (1-398); N68D; A70V; P125S		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
141					hCPN1 (1-398); E288Q			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
142					hCPN1 (1-398); Y266F			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
143					hCPN1 (1-398); Y266F; E288Q			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
144					hCPN1 (1-398); A199S			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
145					hCPN1 (1-398); K124R; P125S			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
146					hCPN1 (1-398); G121T; P122Y; K124Q; P125D			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
147					hCPN1 (1-398); N123Y; K124R			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
148					hCPN1 (1-398); G121D; P122D; N123T; K124I			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
149					hCPN1 (1-398); G121D; P122H; K124I			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
150					hCPN1 (1-398); G121R; P122E; N123Y; K124I			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
151					hCPN1 (1-398); G121D; P122H; K124Q			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
152					hCPN1 (1-398); G121I; N123D			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
153					hCPN1 (1-320); L55N; H161R; N188D; V190K; K229E; L230W; N281D; C282S; F283Y; R308K; Q320N		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
154					hCPN1 (1-320); L55N; H161S; N188D; V190R; K229E; L230W; C282S; F283Y; R308K; Q320N		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
155					hCPN1 (1-320); L55N; H161K; N188D; V190K; K229E; L230W; C282S; F283Y; R308K; Q320N		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
156					hCPN1 (1-398); P122W; N123R; K124Y			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
157					hCPN1 (1-398); P122Y; N123R; K124Y			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs								
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker C2Term
158					hCPN1 (1-398); K124Q			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
159					hCPN1 (1-398); G121D;			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
160					hCPN1 (1-398); N123D;			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
161					hCPN1 (1-398); G121D; N123D;			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
162					hCPN1 (1-398); P122A;			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
163					hCPN1 (1-398); P122R;			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
164					hCPN1 (1-398); K124I;			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
165					hCPN1 (1-398); G121D; N123D; K124I;			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
166					hCPN1 (1-398); P125S;			GSSG GSSG GSSG (SEQ ID NO: 33) HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
167					hCPN1 (1-398); P122G;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
168					hCPN1 (1-398); K124S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
169					hCPN1 (1-398); K124G;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
170					hCPN1 (1-398); P125N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
171					hCPN1 (1-398); P125G;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
172					hCPN1 (1-398); E297Q;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
173					hCPN1 (1-398); V180F;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
174					hCPN1 (1-398); S192A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
175					hCPN1 (1-398); N276V;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs								
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker C2Term
176					hCPN1 (1-398); E284I;			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
177					hCPN1 (1-398); K60F;			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
178	mMBP	AAAQ TNAA AASS ASLE (SEQ ID NO: 35)	TEV		hCPN1 (1-398); Y266F; E288Q			
179	mMBP	AAAQ TNAA AASS ASLE (SEQ ID NO: 35)	TEV		hCPN1 (1-398); Y266F; E288Q		TEV	GSSG GSSG GSSG (SEQ ID NO: 33) HSA
180	mMBP	AAAQ TNAA AASS ASLE (SEQ ID NO: 35)	TEV		hCPN1 (1-438)			
181	mMBP	AAAQ TNAA AASS ASLE (SEQ ID NO: 35)	TEV		hCPN1 (1-438)		TEV	GSSG GSSG GSSG (SEQ ID NO: 33) HSA
182	mMBP	AAAQ TNAA AASS ASLE (SEQ ID NO: 35)	TEV		hCPN1 (1-438); Y266F; E288Q			
183	mMBP	AAAQ TNAA AASS ASLE (SEQ ID NO: 35)	TEV		hCPN1 (1-438); Y266F; E288Q		TEV	GSSG GSSG GSSG (SEQ ID NO: 33) HSA

TABLE 2-continued

Exemplary Fusion Constructs								
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker C2Term
184	GST	AAAO	TEV		hCPN1 (1-398); Y266F; E288Q		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
185	GST	AAAO	TEV		hCPN1 (1-398)			
186	GST	AAAO	TEV		hCPN1 (1-398); Y266F; E288Q			
187					hCPN1 (1-438); Y266F; E288Q; 398DDDD K399		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
188					hCPN1 (1-438); Y266F; E288Q		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)
189					hCPN1 (1-438); Y266F; E288Q			GSSG GSSG GSSG (SEQ ID NO: 33)
190					hCPN1 (1-320); L55N; H161R; N188D; V190K; K229E; L230W; N281D; C282S; F283Y; R308K; Q320N			GSSG GSSG GSSG (SEQ ID NO: 33)
191					hCPN1 (1-320); L55N; H161S; N188D; V190R; K229E;			GSSG GSSG GSSG (SEQ ID NO: 33)

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
					L230W; C282S; F283Y; R308K; Q320N				
192					hCPN1 (1-320); L55N; H161K; N188D; V190K; K229E; L230W; C282S; F283Y; R308K; Q320N			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
193					hCPN1 (1-320); L55N; H161S; N188D; V190R; K229E; L230K; K233D; C282S; F283Y; R308K; Q320K			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
194					hCPN1 (1-320); L55N; H161R; N188D; V190K; K229E; L230W; K233H; N281D; C282S; F283Y; R308K; Q320N			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
195					hCPN1 (1-320); L55N; H161R; N188D; V190K; K229E; L230K; N281D; C282S; F283Y; R308K; Q320N			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
196					hCPN1 (1-398); A199D;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
197					hCPN1 (1-398); G121D; A199D;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
198					hCPN1 (1-398); N123D; A199D;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
199					hCPN1 (1-398); G121D; N123D; A199D;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
200					hCPN1 (1-398); G121D; N123D; K124I; A199D;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
201					hCPN1 (1-398); H66K;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
202					hCPN1 (1-398); Q78E;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
203					hCPN1 (1-398); P125E;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
204					hCPN1 (1-398); P125D;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
205					hCPN1 (1-398); D138K;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
206					hCPN1 (1-398); P296E;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
207					hCPN1 (1-398); P296G;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
208					hCPN1 (1-398); E298G;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
209					hCPN1 (1-398); L300E;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
210					hCPN1 (1-398); Q301R;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
211					hCPN1 (1-398); Q301E;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
212					hCPN1 (1-398); R302E;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
213					hCPN1 (1-398); D292N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
214					hCPN1 (1-398); F189C;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs								
Const. No.	N2 Term	N2 Linker	N Term	NLink-er	Core	CLinker	CTerm	C2Link-er C2Term
215					hCPN1 (1-398)	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	GSSG HHHH (SEQ ID HH NO: 39) (SEQ ID NO: 40)
216					aCPN1			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
217					bCPN1			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
218					mCPN1			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
219					vCPN1			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
220					uCPN1			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
221					hCPN1 (1-398); R308K; E329Q;			GSSG GSSG GSSG (SEQ ID NO: 33) HSA
222					hCPN1 (1-398);			HSA
223					hCPN1 (1-398);			GSSG (SEQ ID NO: 41) HSA
224					hCPN1 (1-398);			GSSG GSSG (SEQ ID NO: 42) HSA

Exemplary Fusion Constructs

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
225					hCPN1 (1-398);			GSSG GSSG GSSG (SEQ ID NO: 43)	HSA
226					hCPN1 (1-398);		TEV		HSA
227					hCPN1 (1-398);		TEV	GSSG (SEQ ID NO: 41)	HSA
228					hCPN1 (1-398);		TEV	GSSG GSSG (SEQ ID NO: 42)	HSA
229					hCPN1 (1-398);		TEV	GSSG GSSG GSSG (SEQ ID NO: 43)	HSA
230					hCPN1 (1-398); A70V; P125S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
231					hCPN1 (1-398); P125S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
232					hCPN1 (1-398); A70V; P125N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
233					hCPN1 (1-398); A70V; P125I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
234					hCPN1 (1-398); A70V; P125F;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLink-er	Core	CLinker	CTerm	C2Link-er	C2Term
235					hCPN1 (1-398); N68D; A70V;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
236					hCPN1 (1-398); N68D; A70V; P125Y;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
237					hCPN1 (1-398); N68D; P125Y;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
238					hCPN1 (1-398); N123D; K124S; P125N; V129T;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
239					hCPN1 (1-398); K124T; P125S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
240					hCPN1 (1-398); G121D; P122R; N123D; K124I; P125S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
241					hCPN1 (1-398); N123D; K124I; P125S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
242					hCPN1 (1-398); N123D; K124S; P125I; V129T;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
243					hCPN1 (1-398); N123D; K124A; P125S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	C Term	C2Linker	C2Term
244					hCPN1 (1-398); K124A; P125S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
245					hCPN1 (1-398); N123D; K124A; P125N; V129T;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
246					hCPN1 (1-398); G121E; P122R; N123D; K124V; P125S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
247					hCPN1 (1-398); N123S; K124T; P125S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
248					hCPN1 (1-398); K60F; V62I; V180I; M184L; S192A; M272L; N276I; E284V;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
249					hCPN1 (1-398); K60F; V62A; V180F; M184L; S192A; M272A; N276V; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
250					hCPN1 (1-398); K60F; V62T; V180F; S192A; N276V; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
251					hCPN1 (1-398); K60G; V62I; V180I; S192M; M272L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLink-er	Core	CLinker	CTerm	C2Link-er	C2Term
					N276I; E284I;				
252					hCPN1 (1-398); K60M; V62I; V180I; M184F; S192A; M272L; N276L; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
253					hCPN1 (1-398); K60F; V180Y; S192A; N276T; E284L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
254					hCPN1 (1-398); K60F; V62A; V180Y; M184L; S192A; M272A; N276T; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
255					hCPN1 (1-398); K60F; V180Y; S192A; N276T; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
256					hCPN1 (1-398); K60Y; V180Y; M184L; S192A; M272I; N276T; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
257					hCPN1 (1-398); K60F; V180F; S192A; N276V; C282T; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
258					hCPN1 (1-398); K60A; V62L; M184L; S192M; M272L; N276I; C282T; E284V;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
259					hCPN1 (1-398); K60F; V62I; V180I; M184L; S192A; M272L; N276I; C282T; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
260					hCPN1 (1-398); K60G; V180F; S192M; N276M; C282T; E284V;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
261					hCPN1 (1-398); K60G; V62I; V180I; M184Q; S192Q; N276I; C282S; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
262					hCPN1 (1-398); K60F; S192A; N276W; E284V;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
263					hCPN1 (1-398); K60G; S192M; N276W; E284V;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
264					hCPN1 (1-398); K60A; S192M; N276I; E284V;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
265					hCPN1 (1-398); K60F; S192A; N276W; C282T; E284V;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
266					hCPN1 (1-398); K60G; S192A; N276W; C282T; E284V;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
267					hCPN1 (1-398); K60F; S192A; N276W; C282L; E284V;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
268					hCPN1 (1-398); Y266F; E288Q;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
269					hCPN1 (1-398); Y266F; E288Q;		TEV	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
270					hCPN1 (1-398); W244A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
271					hCPN1 (1-398); Q247A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
272					hCPN1 (1-398); W249A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
273					hCPN1 (1-398); D253A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
274					hCPN1 (1-398); Y254A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
275					hCPN1 (1-398); D292A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
276					hCPN1 (1-398); E298A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
277					hCPN1 (1-398); E299A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
278					hCPN1 (1-398); R302A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
279					hCPN1 (1-398); K293A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
280					hCPN1 (1-398); P296A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
281					hCPN1 (1-398); G126A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
282					hCPN1 (1-398); Y127A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
283					hCPN1 (1-398); L128A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
284					hCPN1 (1-398); Q247L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLink-er	Core	CLinker	CTerm	C2Link-er	C2Term
285					hCPN1 (1-398); W244K; Q247L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
286					hCPN1 (1-398); W244L; Q247M; W249Y;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
287					hCPN1 (1-398); W244L; Q247L; W249Y;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
288					hCPN1 (1-398); W244F; Q247T;			GSSG GSSG GSSG (SEQ ID NO: 33	HSA
289					hCPN1 (1-398); Q247M;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
290					hCPN1 (1-398); W249Y;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
291					hCPN1 (1-398); W244D; Q247L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
292					hCPN1 (1-398); Y254R;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
293					hCPN1 (1-398); Q247L; Y254R;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
294					hCPN1 (1-398); W244K; Q247L; D253T;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
295					hCPN1 (1-398); Q247L; D253T;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
296					hCPN1 (1-398); D292L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
297					hCPN1 (1-398); R302M;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
298					hCPN1 (1-398); R302W;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
299					hCPN1 (1-398); R.302K;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
300					hCPN1 (1-398); R302L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
301					hCPN1 (1-398); R302Q;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
302					hCPN1 (1-398); K293P; R302M;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
303					hCPN1 (1-398); D292L; K293A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
304					hCPN1 (1-398); K293Y;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
305					hCPN1 (1-398); K293Y; P296D;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
306					hCPN1 (1-398); K293P; R302I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
307					hCPN1 (1-398); K293M; P296D;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
308					hCPN1 (1-398); Y127L; K293P;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
309					hCPN1 (1-398); Y127A; K293Y;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
310					hCPN1 (1-398); D292L; K293A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
311					hCPN1 (1-398); Y127A; K293M;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
312					hCPN1 (1-398); Y127K; K293P;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
313					hCPN1 (1-398); D292V; K293A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
314					hCPN1 (1-398); K293P;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
315					hCPN1 (1-398); Y127K; L128D; K293Y;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
316					hCPN1 (1-398); Y127L; L128H; K293P;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
317					hCPN1 (1-398); Y127K; L128H; K293P;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
318					hCPN1 (1-398); L128D;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
319					hCPN1 (1-398); C251S; C291S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
320					hCPN1 (1-398); T223C; A263C;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
321					hCPN1 (1-398); S209C; A263C;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
322					hCPN1 (1-398); S209C; S267C;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
323					hCPN1 (1-398); S267C; S269C;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
324					hCPN1 (1-398); K60F; V180F; S192A; T223C; A263C; N276V; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
325					hCPN1 (1-398); K60F; V180F; S192A; S209C; A263C; N276V; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
326					hCPN1 (1-398); K60F; V180F; S192A; S209C; S267C; N276V; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
327					hCPN1 (1-398); K60F; V180F; S192A; S267C; S269C; N276V; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
328					hCPN1 (1-398); G121A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
329					hCPN1 (1-398); P122A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
330					hCPN1 (1-398); N123A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
331					hCPN1 (1-398); K124A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
332					hCPN1 (1-398); P125A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
333					hCPN1 (1-398); D138A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
334					hCPN1 (1-398); V180A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
335					hCPN1 (1-398); F189A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
336					hCPN1 (1-398); A199A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
337					hCPN1 (1-398); N276A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
338					hCPN1 (1-398); E284A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
339					hCPN1 (1-398); E297A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
340					hCPN1 (1-398); L300A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
341					hCPN1 (1-398); Q301A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
342					hCPN1 (1-398); K60A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
343					hCPN1 (1-398); H66A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
344					hCPN1 (1-398); Q78A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
345					hCPN1 (1-398); K60F; V180F; S192A; N276V; E284I;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
346					hCPN1 (1-398); K60F; A70V; V180F;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
					S192A; N276V; E284I				
347					hCPN1 (1-398); K60F; N68D; P125S; V180F; S192A; N276V; E284I			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
348					hCPN1 (1-398); K60F; N123D; K124D; P125S; G126Q; V180F; S192A; N276V; E284I			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
349					hCPN1 (1-398); K60F; G121T; P122Y; K124Q; P125D; V180F; S192A; N276V; E284I			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
350					hCPN1 (1-398); K60F; N68D; A70V; P125S; V180F; S192A; N276V; E284I			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
351					hCPN1 (1-398)			GSSG GSSG (SEQ ID NO: 42)	HSA
352					hCPN1 (1-398); W244Y; Q247S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
353					hCPN1 (1-398); W244Y; Q247L; W249L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLink-er	Core	CLinker	CTerm	C2Link-er	C2Term
354					hCPN1 (1-398); W244Y; Q247L; W249M;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
355					hCPN1 (1-398); Q247Y; W249M;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
356					hCPN1 (1-398); Q247Y;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
357					hCPN1 (1-398); W244Y; Q247L; W249F;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
358					hCPN1 (1-398); W244Y; Q247L; W249Y;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
359					hCPN1 (1-398); Q247Y; W249L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
360					hCPN1 (1-398); W244Y; Q247M; W249L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
361					hCPN1 (1-398); W244D; Q247F; W249H;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
362					hCPN1 (1-398); W244Y; Q247N; W249F;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
363					hCPN1 (1-398); G243E; W244Y; F246Y; Q247L; W249Y; N250S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
364					hCPN1 (1-398); G243A; W244Y; F246W; Q247S; D253L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
365					hCPN1 (1-398); G243Y; W244M; F246W; Q247S; D253L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
366					hCPN1 (1-398); G243A; W244Y; F246W; Q247A; D253L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
367					hCPN1 (1-398); G243W; F246Y; Q247L; N250S; D253N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
368					hCPN1 (1-398); G243E; W244Y; F246A; W249Y; N250S;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
369					hCPN1 (1-398); G243Y; W244I; F246W; Q247S; D253L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
370					hCPN1 (1-398); G243W; Q247Y; N250S; D253N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
371					hCPN1 (1-398); G243W; F246W; Q247L; N250S; D253N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	C Term	C2Linker	C2Term
372					hCPN1 (1-398); G243W; F246L; Q247Y; W249L; N250S; D253N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
373					hCPN1 (1-398); G243W; W244D; F246R; Q247F; W249L; N250S; D253N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
374					hCPN1 (1-398); G243W; Q247Y; W249L; N250S; D253N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
375					hCPN1 (1-398); G243W; F246R; Q247F; W249L; N250S; D253N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
376					hCPN1 (1-398); G243W; W244D; F246W; Q247Y; W249F; N250S; D253N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
377					hCPN1 (1-398); R302E; G306A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
378					hCPN1 (1-398); G306Q;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
379					hCPN1 (1-398); R302E; G306T;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
380					hCPN1 (1-398); D292I; R302L; G306H;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
381					hCPN1 (1-398); G306N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
382					hCPN1 (1-398); D292I; R302Q; G306D;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
383					hCPN1 (1-398); R302E; G306N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
384					hCPN1 (1-398); D292I; G306N;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
385					hCPN1 (1-398); G306F;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
386					hCPN1 (1-398); D292I; R302L; G306W;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
387					hCPN1 (1-398); D292I; R302D; G306H;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
388					hCPN1 (1-398); D292I; R302E; G306H;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLink-er	Core	CLinker	CTerm	C2Link-er	C2Term
389					hCPN1 (1-398); Y127E; L128T;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
390					hCPN1 (1-398); Y127L; L128T;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
391					hCPN1 (1-398); Y127T; L128E;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
392					hCPN1 (1-398); Y127T; L128T;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
393					hCPN1 (1-398); Y127R; L128T;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
394					hCPN1 (1-398); Y127L; L128Y;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
395					hCPN1 (1-398); Y127L;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
396					hCPN1 (1-398); Y127D; L128Y;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
397					hCPN1 (1-398); G243A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
398					hCPN1 (1-398); F246A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
399					hCPN1 (1-398); N250A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
400					hCPN1 (1-398); G306A;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
401					hCPN1 (1-398); Y240E;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
402					hCPN1 (1-398); K124E;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
403					hCPN1 (1-398); K293E;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
404					hCPN1 (1-398); K124E; K293E;			GSSG GSSG GSSG (SEQ ID NO: 33)	HSA
405					hCPN1 (1-398); A193W; A202V; Y238F; I285L; L287I; L289Y; F314A				
406					hCPN1 (1-398); D292I; K293A; F294W				

TABLE 2-continued

Exemplary Fusion Constructs								
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker
				er				C2Term
407					hCPN1 (1-398); D292L; K293V; P295L			
408					hCPN1 (1-398); N123Y; K124R; Y127E; D292I; K293V; F294M			
409					hCPN1 (1-398); G121Q; P122W; N123D; Y127I; D292L; K293A; F294I			
410					hCPN1 (1-398); G121D; P122W; N123D; Y127I; D292I; K293A; F294W			
411					hCPN1 (1-398); K60F; V62S; V180Y; S192A; N276T; E284I			
412					hCPN1 (1-398); K60F; V180F; S192A; N276V; E284I			
413					hCPN1 (1-398); K60F; V62M; V180A; S192A; M272F; N276V; E284I			
414					hCPN1 (1-320); L55N; H161R; N188D; V190K; K229E; L230W; K233H; N281D;			

TABLE 2-continued

Exemplary Fusion Constructs								
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker
				er				er
								C2Term
415					C282S; F283Y; R308K; Q320N hCPN1 (1-398); S209G; ΔF210- T220; A221R;	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
416					hCPN1 (1-398); S209G; ΔF210- T220; A221H;	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
417					hCPN1 (1-398); S209G; ΔF210- R219; T220P; A221S;	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
418					hCPN1 (1-398); S209G; ΔF210- R219; T220E; A221D;	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
419					hCPN1 (1-398); S209G; ΔF210- R219; T220D; A221D;	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
420					hCPN1 (1-398); S209N; F210W; E211G; ΔH212- T220; A221G;	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
421					hCPN1 (1-398); S209A; F210E; E211D; ΔH212- R219; T220N; A221D;	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	
422					hCPN1 (1-398); F210D; ΔE211- R218; R219K; T220N; A221D;	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA	

TABLE 2-continued

Exemplary Fusion Constructs									
Const. No.	N2 Term	N2 Linker	N Term	NLinker	Core	CLinker	CTerm	C2Linker	C2Term
423					hCPN1 (1-398); F210S; ΔE211- R219; A221S;	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA		
424					hCPN1 (1-398); F210D; ΔE211- R219; T220G; A221D;	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA		
425					hCPN1 (1-398); ΔF210- R218; R219E; A221G;	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA		
426					hCPN1 (1-398); F210D; ΔE211- R219; T220S; A221G;	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA		
427					hCPN1 (1-398); F210E; E211D; ΔH212- T220; A221S; S222E;	GSSG GSSG GSSG (SEQ ID NO: 33)	HSA		

TABLE 3A

Exemplary Modification Strings of hCPN1 (1-398)	
R37G	G121D; P122H; K124Q
K60F; V62S; V180Y; S192A; N276T; E284I	G121I; N123D
K60F; V180F; S192A; N276V; E284I	P122W; N123R; K124Y
K60F; V62M; V180A; S192A; M272F; N276V; E284I	P122Y; N123R; K124Y
K60A; V62L; M184L; S192M; M272L; N276I; E284V	K124Q
A193W; A202V; Y238F; I285L; L287I; L289Y; E314A	G121T; P122Y; K124Q; P125D; E297P; E298Q
A202V; L287I; L289T	G121T; P122Y; K124Q; P125D; E297P
A202I	D292I; K293I; F294W
A235L	K293L; E297P; E298Q
V1I; F3W; E74L; Q78W	D292L; K293I; P295L
V1I; E74L; Q78W	D292L; K293I
K60F; V62S; V180Y; S192A; N276T; E284I	D292N; K293A; P295T; E298S
K60F; V180F; S192A; N276V; E284I	D292N; K293A; E298S
K60F; V62M; V180A; S192A; M272F; N276V; E284I	D292I; K293A; E294W
K60A; V62L; M184L; S192M; M272L; N276I; E284V	D292L; K293A; E294W
A193W; A202V; Y238F; I285L; L287I; L289Y; E314A	D292L; K293V; E295L
A202V; L287I; L289T	D292L; K293V; E298D
A202I	K293N
A235L	K293L; F294W; E298L
V1I; F3W; E74L; Q78W	D292N; K293L; F294W
V1I; E74L; Q78W	N123W; K124R; P125A; Y127D; D292I; K293I;
A70V	F294Q

TABLE 3A-continued

Exemplary Modification Strings of hCPN1 (1-398)	
N68D; P125S	N123Y; K124R; P127E; D292I; K293V; F294M
N68D; A70V; P125S	G121T; P122Y; K124Q; P125D; Y127W; D292I;
E288Q	K293L; E297P
Y266F	N123Y; K124R; Y127D; K293I
Y266F; E288Q	G121D; P122S; N123T; K124I; D292N; K293A; E298S
A199S	G121Q; P122W; N123D; Y127I; D292L; K293A;
K124R; P125S	E294I
G121T; P122Y; K124Q; P125D	G121D; P122W; N123D; Y127I; K293A; E294W
N123Y; K124R	P122W; N123R; K124H; Y127S; D292L; K293V;
G121D; P122D; N123T; K124I	E298W
G121D; P122H; K124I	P122W; N123R; K124Y; D292L; K293V; E298D
G121R; P122E; N123Y; K124I	K124Q; K293L; F294W; E298L
K124D; K293L; F294W; E297D	P122R;
P125H	K124I;
P125L	G121D; N123D; K124I;
P122A; N123D; K124I; P125S	P125S;
P122A; N123D; K124S; P125S	P122G;
G121E; P122R; N123D; K124I; P125S	K124S;
G121D; P122R; N123D; K124T; P125S; G126Q	K124G;
N123D; K124D; P125S; G126Q	P125N;
N68D	P125G;
N68K; K293N	E297Q;
V201H	V180F;
N203D	S192A;
G198D	N276V;
G198D; V201Y	E284I;
A70V	K60F;
N68D; P125S	Y266F; E288Q
N68D; A70V; P125S	Y266F; E288Q
E288Q	Y266F; E288Q
Y266F	CPN1 (1-398)
Y266F; E288Q	Y266F; E288Q
A199S	A199D;
K124R; P125S	G121D; A199D;
G121T; P122Y; K124Q; P125D	N123D; A199D;
N123Y; K124R	G121D; N123D; A199D;
G121D; P122D; N123T; K124I	G121D; N123D; K124I; A199D;
G121D; P122H; K124I	H66K;
G121R; P122E; N123Y; K124I	Q78E;
G121D; P122H; K124Q	P125E;
G121I; N123D	P125D;
P122W; N123R; K124Y	D138K;
P122Y; N123R; K124Y	P296E;
K124Q	P296G;
G121D;	E298G;
N123D;	L300E;
G121D; N123D;	Q301R;
P122A;	Q301E;
R302E;	K60A; V62L; M184L; S192M; M272L; N276I; C282T;
D292N;	E284V;
F189C;	K60F; V62I; V180I; M184L; S192A; M272L; N276I;
R308K; E329Q;	C282T; E284I;
A70V; P125S;	K60G; V180F; S192M; N276M; C282T; E284V;
P125S;	K60G; V62I; V180I; M184Q; S192Q; N276I; C282S;
A70V; P125N;	E284I;
A70V; P125I;	K60F; S192A; N276W; E284V;
A70V; P125F;	K60G; S192M; N276W; E284V;
N68D; A70V;	K60A; S192M; N276I; E284V;
N68D; A70V; P125Y;	K60F; S192A; N276W; C282T; E284V;
N68D; P125Y;	K60G; S192A; N276W; C282T; E284V;
N123D; K124S; P125N; V129T;	K60F; S192A; N276W; C282L; E284V;
K124T; P125S;	Y266F; E288Q;
G121D; P122R; N123D; K124I; P125S;	Y266F; E288Q;
N123D; K124I; P125S;	W244A;
N123D; K124S; P125I; V129T;	Q247A;
N123D; K124A; P125S;	W249A;
K124A; P125S;	D253A;
N123D; K124A; P125N; V129T;	Y254A;
G121E; P122R; N123D; K124V; P125S;	D292A;
N123S; K124T; P125S;	E298A;
K60F; V62I; V180I; M184L; S192A; M272L; N276I;	E299A;
E284V;	R302A;
K60F; V62A; V180F; M184L; S192A; M272A;	K293A;
N276V; E284I;	P296A;

TABLE 3A-continued

Exemplary Modification Strings of hCPN1 (1-398)	
K60F; V62T; V180F; S192A; N276V; E284I;	G126A;
K60G; V62I; V180I; S192M; M272L; N276I; E284I;	Y127A;
K60M; V62I; V180I; M184F; S192A; M272L; N276L;	L128A;
E284I;	Q247L;
K60F; V180Y; S192A; N276T; E284L;	W244K; Q247L;
K60F; V62A; V180Y; M184L; S192A; M272A;	W244L; Q247M; W249Y;
N276T; E284I;	W244L; Q247L; W249Y;
K60F; V180Y; S192A; N276T; E284I;	W244F; Q247T;
K60Y; V180Y; M184L; S192A; M272I; N276T; E284I;	Q247M;
K60F; V180F; S192A; N276V; C282T; E284I;	W249Y;
W244D; Q247L;	K60F; V180F; S192A; S267C; S269C; N276V; E284I;
Y254R;	G121A;
Q247L; Y254R;	P122A;
W244K; Q247L; D253T;	N123A;
Q247L; D253T;	K124A;
D292L;	P125A;
R302M;	D138A;
R302W;	V180A;
R302K;	F189A;
R302L;	A199A;
R302Q;	N276A;
K293P; R302M;	E284A;
D292L; K293A;	E297A;
K293Y;	L300A;
K293Y; P296D;	Q301A;
K293P; R302I;	K60A;
K293M; P296D;	H66A;
Y127L; K293P;	Q78A;
Y127A; K293Y;	K60F; V180F; S192A; N276V; E284I;
D292L; K293A;	K60F; A70V; V180F; S192A; N276V; E284I
Y127A; K293M;	K60F; N68D; P125S; V180F; S192A; N276V; E284I
Y127K; K293P;	K60F; N123D; K124D; P125S; G126Q; V180F;
D292V; K293A;	S192A; N276V; E284I
K293P;	K60F; G121T; P122Y; K124Q; P125D; V180F; S192A;
Y127K; L128D; K293Y;	N276V; E284I
Y127L; L128H; K293P;	K60F; N68D; A70V; P125S; V180F; S192A; N276V;
Y127K; L128H; K293P;	E284I
L128D;	W244Y; Q247S;
C251S; C291S;	W244Y; Q247L; W249L;
T223C; A263C;	W244Y; Q247L; W249M;
S209C; A263C;	Q247Y; W249M;
S209C; S267C;	Q247Y;
S267C; S269C;	W244Y; Q247L; W249F;
K60F; V180F; S192A; T223C; A263C; N276V; E284I;	W244Y; Q247L; W249Y;
K60F; V180F; S192A; S209C; A263C; N276V; E284I;	Q247Y; W249L;
K60F; V180F; S192A; S209C; S267C; N276V; E284I;	W244Y; Q247M; W249L;

TABLE 3B

Exemplary Modification Strings of hCPN1 (1-320)
Δ(321-398)
L55N; H161R; N188D; V190K; K229E; L230W; N281D; C282S; F283Y; R308K; Q320N
L55N; H161S; N188D; V190R; K229E; L230W; C282S; F283Y; R308K; Q320N
L55N; H161K; N188D; V190K; K229E; L230W; C282S; F283Y; R308K; Q320N
L55N; H161S; N188D; V190R; K229E; L230K; K233D; C282S; F283Y; R308K; Q320K
L55N; H161R; N188D; V190K; K229E; L230W; K233H; N281D; C282S; F283Y; R308K; Q320N
L55N; H161R; N188D; V190K; K229E; L230K; N281D; C282S; F283Y; R308K; Q320N
L55N; H161S; N188D; V190R; K229E; L230W; N281D; C282S; F283Y; R308K; Q320N
L55N; H161S; N188D; V190R; K229E; L230W; C282S; F283Y; R308K; Q320N
L55N; H161K; N188D; V190K; K229E; L230W; C282S; F283Y; R308K; Q320N
L55N; H161R; N188D; V190K; K229E; L230W; N281D; C282S; F283Y; R308K; Q320N
L55N; H161S; N188D; V190R; K229E; L230W; C282S; F283Y; R308K; Q320N
L55N; H161K; N188D; V190K; K229E; L230W; C282S; F283Y; R308K; Q320N
L55N; H161S; N188D; V190R; K229E; L230K; K233D; C282S; F283Y; R308K; Q320K
L55N; H161R; N188D; V190K; K229E; L230W; K233H; N281D; C282S; F283Y; R308K; Q320N
L55N; H161R; N188D; V190K; K229E; L230K; N281D; C282S; F283Y; R308K; Q320N
L55N; H161S; N188D; V190R; K229E; L230W; K233H; N281D; C282S; F283Y; R308K; Q320N

TABLE 3C

Exemplary Modification Strings of hCPN1 (1-438)
Y266F; E288Q
Y266F; E288Q; 398DDDDK399

TABLE 4A

Exemplary Modifications of hCPN1 (1-398)						
V1I	V62L	Q78W	P122Y	N123T	K124H	P125A
F3W	V62I	Q78E	P122D	N123D	K124D	P125H
R37G	V62A	Q78A	P122H	N123R	K124S	P125L
K60F	V62T	G121T	P122E	N123W	K124T	P125N
K60A	H66K	G121D	P122W	N123S	K124G	P125G
K60G	H66A	G121R	P122S	N123A	K124A	P125E
K60M	N68D	G121I	P122A	K124R	K124V	P125I
K60Y	N68K	G121Q	P122R	K124Q	K124E	P125F
V62S	A70V	G121E	P122G	K124I	P125S	P125Y
V62M	E74L	G121A	N123Y	K124Y	P125D	G126Q
G126A	S192M	F246L	N276T	K293Y	Q301E	A221S
Y127D	S192Q	F246R	N276V	K293M	Q301A	T220E
P127E	A193W	Q247A	N276I	K293E	R302E	A221D
Y127W	G198D	Q247L	N276L	F294W	R302A	T220D
Y127I	A199S	Q247M	N276M	E294W	R302M	S209N
Y127S	A199D	Q247S	N276W	F294Q	R302W	F210W
Y127A	A199A	Q247Y	N276A	F294M	R302K	E211G
Y127L	V201H	Q247F	C282T	E294I	R302L	ΔH212-
Y127K	V201Y	Q247N	C282S	F294I	R302Q	T220
Y127E	A202V	W249A	C282L	P295L	R302I	A221G
Y127T	A202I	W249L	E284I	P295T	R302D	S209A
Y127R	N203D	W249M	E284V	E295L	G306A	F210E
L128A	S209C	W249F	E284L	P296E	G306Q	E211D
L128D	T223C	W249Y	E284A	P296G	G306T	ΔH212-
L128H	A235L	W249H	I285L	P296A	G306H	R219
L128T	Y238F	N250S	L287I	P296D	G306N	T220N
L128E	Y240E	N250A	E288Q	E297P	G306D	F210D
L128Y	G243E	C251S	L289Y	E297D	G306F	ΔE211-
V129T	G243A	D253A	L289T	E297Q	G306W	R218
D138K	G243Y	D253T	C291S	E297A	R308K	R219K
D138A	G243W	D253L	D292I	E298Q	E314A	F210S
V180Y	W244A	D253N	D292L	E298S	F314A	ΔE211-
V180F	W244K	Y254A	D292N	E298D	E329Q	R219
V180A	W244L	A263C	D292A	E298L	S209G	T220G
V180I	W244Y	Y266F	D292V	E298W	ΔF210-	ΔF210-
M184L	W244D	S267C	K293I	E298G	T220	R218
M184F	W244M	S269C	K293L	E298A	A221R	R219E
M184Q	W244I	M272F	K293A	E299A	A221H	T220S
F189C	F246Y	M272L	K293V	L300E	ΔF210-	S222E
F189A	F246W	M272A	K293N	L300A	R219	
S192A	F246A	M272I	K293P	Q301R	T220P	

TABLE 4B

Exemplary Modifications of hCPN1 (1-320)						
L55N	H161S	V190R	L230W	N281D	R308K	Δ(321-398)
H161K	N188D	K229E	K233D	C282S	Q320K	
H161R	V190K	L230K	K233H	F283Y	Q320N	

TABLE 4C

Exemplary Modifications of hCPN1 (1-438)						
Y266F						
E288Q						
398DDDDK399 (SEQ ID NO: 44)						

TABLE 5

Amino Acid Sequences of Exemplary Fusion Constructs			
Const. No.	Signal Sequence (Optional)	Signal Sequence ID (Optional)	Full Protein SEQ ID
1	MKWVTFISLLFLFSSAYS	487	73
2	MTRLTVLALLAGLLASSRA	19	74
3	MYRMQLLSICIALSLALVTNS	20	75
4	MGWSLILLFLVAVATRVHS	21	76
5	MGWSLILLFLVAVATRVHS	21	77
6	MGWSLILLFLVAVATRVHS	21	78
7	MGWSLILLFLVAVATRVHS	21	79
8	MGWSLILLFLVAVATRVHS	21	80
9	MGWSLILLFLVAVATRVHS	21	81
10	MGWSLILLFLVAVATRVHS	21	82
11	MGWSLILLFLVAVATRVHS	21	83
12	MGWSLILLFLVAVATRVHS	21	84
13	MGWSLILLFLVAVATRVHS	21	85
14	MGWSLILLFLVAVATRVHS	21	86
15	MGWSLILLFLVAVATRVHS	21	87
16	MGWSLILLFLVAVATRVHS	21	88
17	MGWSLILLFLVAVATRVHS	21	89
18	MGWSLILLFLVAVATRVHS	21	90
19	MGWSLILLFLVAVATRVHS	21	91
20	MGWSLILLFLVAVATRVHS	21	92
21	MGWSLILLFLVAVATRVHS	21	93
22	MGWSLILLFLVAVATRVHS	21	94
23	MGWSLILLFLVAVATRVHS	21	95
24	MGWSLILLFLVAVATRVHS	21	96
25	MGWSLILLFLVAVATRVHS	21	97
26	MGWSLILLFLVAVATRVHS	21	98
27	MGWSLILLFLVAVATRVHS	21	99
28	MGWSLILLFLVAVATRVHS	21	100
29	MGWSLILLFLVAVATRVHS	21	101
30	MGWSLILLFLVAVATRVHS	21	102
31	MGWSLILLFLVAVATRVHS	21	103
32	MGWSLILLFLVAVATRVHS	21	104
33	MGWSLILLFLVAVATRVHS	21	105
34	MGWSLILLFLVAVATRVHS	21	106

TABLE 5-continued

Amino Acid Sequences of Exemplary Fusion Constructs			
Const. No.	Signal Sequence (Optional)	Signal Sequence SEQ ID (Optional)	Full Protein SEQ ID
35	MGWSLILLFLVAVATRVHS	21	107
36	MGWSLILLFLVAVATRVHS	21	108
37	no signal seq		109
38	MGWSLILLFLVAVATRVHS	21	110
39	MGWSLILLFLVAVATRVHS	21	111
40	MGWSLILLFLVAVATRVHS	21	112
41	TMGWSLILLFLVAVATRVHS	488	113
42	TMGWSLILLFLVAVATRVHS	488	114
43	TMGWSLILLFLVAVATRVHS	488	115
44	TMGWSLILLFLVAVATRVHS	488	116
45	TMGWSLILLFLVAVATRVHS	488	117
46	TMGWSLILLFLVAVATRVHS	488	118
47	TMGWSLILLFLVAVATRVHS	488	119
48	TMGWSLILLFLVAVATRVHS	488	120
49	TMGWSLILLFLVAVATRVHS	488	121
50	TMGWSLILLFLVAVATRVHS	488	122
51	TMGWSLILLFLVAVATRVHS	488	123
52	TMGWSLILLFLVAVATRVHS	488	124
53	TMGWSLILLFLVAVATRVHS	488	125
54	TMGWSLILLFLVAVATRVHS	488	126
55	TMGWSLILLFLVAVATRVHS	488	127
56	TMGWSLILLFLVAVATRVHS	488	128
57	TMGWSLILLFLVAVATRVHS	488	129
58	TMGWSLILLFLVAVATRVHS	488	130
59	TMGWSLILLFLVAVATRVHS	488	131
60	TMGWSLILLFLVAVATRVHS	488	132
61	TMGWSLILLFLVAVATRVHS	488	133
62	TMGWSLILLFLVAVATRVHS	488	134
63	TMGWSLILLFLVAVATRVHS	488	135
64	TMGWSLILLFLVAVATRVHS	488	136
65	TMGWSLILLFLVAVATRVHS	488	137
66	TMGWSLILLFLVAVATRVHS	488	138
67	TMGWSLILLFLVAVATRVHS	488	139
68	TMGWSLILLFLVAVATRVHS	488	140

TABLE 5-continued

Amino Acid Sequences of Exemplary Fusion Constructs			
Const. No.	Signal Sequence (Optional)	Signal Sequence SEQ ID (Optional)	Full Protein SEQ ID
69	TMGWSLILLFLVAVATRVHS	488	141
70	TMGWSLILLFLVAVATRVHS	488	142
71	TMGWSLILLFLVAVATRVHS	488	143
72	TMGWSLILLFLVAVATRVHS	488	144
73	TMGWSLILLFLVAVATRVHS	488	145
74	TMGWSLILLFLVAVATRVHS	488	146
75	TMGWSLILLFLVAVATRVHS	488	147
76	TMGWSLILLFLVAVATRVHS	488	148
77	TMGWSLILLFLVAVATRVHS	488	149
78	TMGWSLILLFLVAVATRVHS	488	150
79	TMGWSLILLFLVAVATRVHS	488	151
80	TMGWSLILLFLVAVATRVHS	488	152
81	TMGWSLILLFLVAVATRVHS	488	153
82	TMGWSLILLFLVAVATRVHS	488	154
83	TMGWSLILLFLVAVATRVHS	488	155
84	TMGWSLILLFLVAVATRVHS	488	156
85	TMGWSLILLFLVAVATRVHS	488	157
86	TMGWSLILLFLVAVATRVHS	488	158
87	TTTMGWSLILLFLVAVATRVHS	489	159
88	TTTMGWSLILLFLVAVATRVHS	489	160
89	TTTMGWSLILLFLVAVATRVHS	489	161
90	TTTMGWSLILLFLVAVATRVHS	489	162
91	TTTMGWSLILLFLVAVATRVHS	489	163
92	TTTMGWSLILLFLVAVATRVHS	489	164
93	TTTMGWSLILLFLVAVATRVHS	489	165
94	TTTMGWSLILLFLVAVATRVHS	489	166
95	TTTMGWSLILLFLVAVATRVHS	489	167
96	TTTMGWSLILLFLVAVATRVHS	489	168
97	TTTMGWSLILLFLVAVATRVHS	489	169
98	TTTMGWSLILLFLVAVATRVHS	489	170
99	TTTMGWSLILLFLVAVATRVHS	489	171
100	TTTMGWSLILLFLVAVATRVHS	489	172
101	TTTMGWSLILLFLVAVATRVHS	489	173
102	TTTMGWSLILLFLVAVATRVHS	489	174

TABLE 5-continued

Amino Acid Sequences of Exemplary Fusion Constructs			
Const. No.	Signal Sequence (Optional)	Signal Sequence SEQ ID (Optional)	Full Protein SEQ ID
103	TTTMGWSLILLFLVAVATRVHS	489	175
104	TTTMGWSLILLFLVAVATRVHS	489	176
105	TTTMGWSLILLFLVAVATRVHS	489	177
106	TTTMGWSLILLFLVAVATRVHS	489	178
107	TTTMGWSLILLFLVAVATRVHS	489	179
108	TTTMGWSLILLFLVAVATRVHS	489	180
109	TTTMGWSLILLFLVAVATRVHS	489	181
110	TTTMGWSLILLFLVAVATRVHS	489	182
111	TTTMGWSLILLFLVAVATRVHS	489	183
112	TTTMGWSLILLFLVAVATRVHS	489	184
113	TTTMGWSLILLFLVAVATRVHS	489	185
114	TTTMGWSLILLFLVAVATRVHS	489	186
115	TTTMGWSLILLFLVAVATRVHS	489	187
116	TTTMGWSLILLFLVAVATRVHS	489	188
117	TTTMGWSLILLFLVAVATRVHS	489	189
118	TTTMGWSLILLFLVAVATRVHS	489	190
119	TTTMGWSLILLFLVAVATRVHS	489	191
120	TTTMGWSLILLFLVAVATRVHS	489	192
121	TTTMGWSLILLFLVAVATRVHS	489	193
122	TTTMGWSLILLFLVAVATRVHS	489	194
123	TTTMGWSLILLFLVAVATRVHS	489	195
124	TTTMGWSLILLFLVAVATRVHS	489	196
125	TTTMGWSLILLFLVAVATRVHS	489	197
126	TTTMGWSLILLFLVAVATRVHS	489	198
127	TTTMGWSLILLFLVAVATRVHS	489	199
128	TTTMGWSLILLFLVAVATRVHS	489	200
129	TTTMGWSLILLFLVAVATRVHS	489	201
130	TTTMGWSLILLFLVAVATRVHS	489	202
131	TTTMGWSLILLFLVAVATRVHS	489	203
132	TTTMGWSLILLFLVAVATRVHS	489	204
133	TTTMGWSLILLFLVAVATRVHS	489	205
134	TTTMGWSLILLFLVAVATRVHS	489	206
135	TTTMGWSLILLFLVAVATRVHS	489	207
136	TTTMGWSLILLFLVAVATRVHS	489	208

TABLE 5-continued

Amino Acid Sequences of Exemplary Fusion Constructs			
Const. No.	Signal Sequence (Optional)	Signal Sequence SEQ ID (Optional)	Full Protein SEQ ID
137	TTTMGWSLILLFLVAVATRVHS	489	209
138	TMGWSLILLFLVAVATRVHS	488	210
139	TMGWSLILLFLVAVATRVHS	488	211
140	TMGWSLILLFLVAVATRVHS	488	212
141	TMGWSLILLFLVAVATRVHS	488	213
142	TMGWSLILLFLVAVATRVHS	488	214
143	TMGWSLILLFLVAVATRVHS	488	215
144	TTTMGWSLILLFLVAVATRVHS	489	216
145	TTTMGWSLILLFLVAVATRVHS	489	217
146	TTTMGWSLILLFLVAVATRVHS	489	218
147	TTTMGWSLILLFLVAVATRVHS	489	219
148	TTTMGWSLILLFLVAVATRVHS	489	220
149	TTTMGWSLILLFLVAVATRVHS	489	221
150	TTTMGWSLILLFLVAVATRVHS	489	222
151	TTTMGWSLILLFLVAVATRVHS	489	223
152	TTTMGWSLILLFLVAVATRVHS	489	224
153	TTTMGWSLILLFLVAVATRVHS	489	225
154	TTTMGWSLILLFLVAVATRVHS	489	226
155	TTTMGWSLILLFLVAVATRVHS	489	227
156	TTTMGWSLILLFLVAVATRVHS	489	228
157	TTTMGWSLILLFLVAVATRVHS	489	229
158	TTTMGWSLILLFLVAVATRVHS	489	230
159	AATMGWSLILLFLVAVATRVHS	486	231
160	AATMGWSLILLFLVAVATRVHS	486	232
161	AATMGWSLILLFLVAVATRVHS	486	233
162	AATMGWSLILLFLVAVATRVHS	486	234
163	AATMGWSLILLFLVAVATRVHS	486	235
164	AATMGWSLILLFLVAVATRVHS	486	236
165	AATMGWSLILLFLVAVATRVHS	486	237
166	AATMGWSLILLFLVAVATRVHS	486	238
167	AATMGWSLILLFLVAVATRVHS	486	239
168	AATMGWSLILLFLVAVATRVHS	486	240
169	AATMGWSLILLFLVAVATRVHS	486	241
170	AATMGWSLILLFLVAVATRVHS	486	242

TABLE 5-continued

Amino Acid Sequences of Exemplary Fusion Constructs			
Const. No.	Signal Sequence (Optional)	Signal Sequence SEQ ID (Optional)	Full Protein SEQ ID
171	AATMGWSLILLFLVAVATRVHS	486	243
172	AATMGWSLILLFLVAVATRVHS	486	244
173	AATMGWSLILLFLVAVATRVHS	486	245
174	AATMGWSLILLFLVAVATRVHS	486	246
175	AATMGWSLILLFLVAVATRVHS	486	247
176	AATMGWSLILLFLVAVATRVHS	486	248
177	AATMGWSLILLFLVAVATRVHS	486	249
178	AATMGWSLILLFLVAVATRVHS	486	250
179	AATMGWSLILLFLVAVATRVHS	486	251
180	AATMGWSLILLFLVAVATRVHS	486	252
181	AATMGWSLILLFLVAVATRVHS	486	253
182	AATMGWSLILLFLVAVATRVHS	486	254
183	AATMGWSLILLFLVAVATRVHS	486	255
184	AATMGWSLILLFLVAVATRVHS	486	256
185	AATMGWSLILLFLVAVATRVHS	486	257
186	AATMGWSLILLFLVAVATRVHS	486	258
187	AATMGWSLILLFLVAVATRVHS	486	259
188	AATMGWSLILLFLVAVATRVHS	486	260
189	AATMGWSLILLFLVAVATRVHS	486	261
190	AATMGWSLILLFLVAVATRVHS	486	262
191	AATMGWSLILLFLVAVATRVHS	486	263
192	AATMGWSLILLFLVAVATRVHS	486	264
193	AATMGWSLILLFLVAVATRVHS	486	265
194	AATMGWSLILLFLVAVATRVHS	486	266
195	AATMGWSLILLFLVAVATRVHS	486	267
196	AATMGWSLILLFLVAVATRVHS	486	268
197	AATMGWSLILLFLVAVATRVHS	486	269
198	AATMGWSLILLFLVAVATRVHS	486	270
199	AATMGWSLILLFLVAVATRVHS	486	271
200	AATMGWSLILLFLVAVATRVHS	486	272
201	AATMGWSLILLFLVAVATRVHS	486	273
202	AATMGWSLILLFLVAVATRVHS	486	274
203	AATMGWSLILLFLVAVATRVHS	486	275
204	AATMGWSLILLFLVAVATRVHS	486	276

TABLE 5-continued

Amino Acid Sequences of Exemplary Fusion Constructs			
Const. No.	Signal Sequence (Optional)	Signal Sequence SEQ ID (Optional)	Full Protein SEQ ID
205	AATMGWSLILLFLVAVATRVHS	486	277
206	AATMGWSLILLFLVAVATRVHS	486	278
207	AATMGWSLILLFLVAVATRVHS	486	279
208	AATMGWSLILLFLVAVATRVHS	486	280
209	AATMGWSLILLFLVAVATRVHS	486	281
210	AATMGWSLILLFLVAVATRVHS	486	282
211	AATMGWSLILLFLVAVATRVHS	486	283
212	AATMGWSLILLFLVAVATRVHS	486	284
213	AATMGWSLILLFLVAVATRVHS	486	285
214	AATMGWSLILLFLVAVATRVHS	486	286
215	GSATMETDTLLLWVLLWVPG-STG	490	287
216	AATMGWSLILLFLVAVATRVHS	486	288
217	AATMGWSLILLFLVAVATRVHS	486	289
218	AATMGWSLILLFLVAVATRVHS	486	290
219	AATMGWSLILLFLVAVATRVHS	486	291
220	AATMGWSLILLFLVAVATRVHS	486	292
22	AATMGWSLILLFLVAVATRVHS	486	293
222	AATMGWSLILLFLVAVATRVHS	486	294
223	AATMGWSLILLFLVAVATRVHS	486	295
224	AATMGWSLILLFLVAVATRVHS	486	296
225	AATMGWSLILLFLVAVATRVHS	486	297
226	AATMGWSLILLFLVAVATRVHS	486	298
227	AATMGWSLILLFLVAVATRVHS	486	299
228	AATMGWSLILLFLVAVATRVHS	486	300
229	AATMGWSLILLFLVAVATRVHS	486	301
230	AATMGWSLILLFLVAVATRVHS	486	302
231	AATMGWSLILLFLVAVATRVHS	486	303
232	AATMGWSLILLFLVAVATRVHS	486	304
233	AATMGWSLILLFLVAVATRVHS	486	305
234	AATMGWSLILLFLVAVATRVHS	486	306
235	AATMGWSLILLFLVAVATRVHS	486	307
236	AATMGWSLILLFLVAVATRVHS	486	308
237	AATMGWSLILLFLVAVATRVHS	486	309
238	AATMGWSLILLFLVAVATRVHS	486	310

TABLE 5-continued

Amino Acid Sequences of Exemplary Fusion Constructs			
Const. No.	Signal Sequence (Optional)	Signal Sequence SEQ ID (Optional)	Full Protein SEQ ID
239	AATMGWSLILLFLVAVATRVHS	486	311
240	AATMGWSLILLFLVAVATRVHS	486	312
241	AATMGWSLILLFLVAVATRVHS	486	313
242	AATMGWSLILLFLVAVATRVHS	486	314
243	AATMGWSLILLFLVAVATRVHS	486	315
244	AATMGWSLILLFLVAVATRVHS	486	316
245	AATMGWSLILLFLVAVATRVHS	486	317
246	AATMGWSLILLFLVAVATRVHS	486	318
247	AATMGWSLILLFLVAVATRVHS	486	319
248	AATMGWSLILLFLVAVATRVHS	486	320
249	AATMGWSLILLFLVAVATRVHS	486	321
250	AATMGWSLILLFLVAVATRVHS	486	322
251	AATMGWSLILLFLVAVATRVHS	486	323
252	AATMGWSLILLFLVAVATRVHS	486	324
253	AATMGWSLILLFLVAVATRVHS	486	325
254	AATMGWSLILLFLVAVATRVHS	486	326
255	AATMGWSLILLFLVAVATRVHS	486	327
256	AATMGWSLILLFLVAVATRVHS	486	328
257	AATMGWSLILLFLVAVATRVHS	486	329
258	AATMGWSLILLFLVAVATRVHS	486	330
259	AATMGWSLILLFLVAVATRVHS	486	331
260	AATMGWSLILLFLVAVATRVHS	486	332
261	AATMGWSLILLFLVAVATRVHS	486	333
262	AATMGWSLILLFLVAVATRVHS	486	334
263	AATMGWSLILLFLVAVATRVHS	486	335
264	AATMGWSLILLFLVAVATRVHS	486	336
265	AATMGWSLILLFLVAVATRVHS	486	337
266	AATMGWSLILLFLVAVATRVHS	486	338
267	AATMGWSLILLFLVAVATRVHS	486	339
268	AATMGWSLILLFLVAVATRVHS	486	340
269	AATMGWSLILLFLVAVATRVHS	486	341
270	AATMGWSLILLFLVAVATRVHS	486	342
271	AATMGWSLILLFLVAVATRVHS	486	343
272	AATMGWSLILLFLVAVATRVHS	486	344

TABLE 5-continued

Amino Acid Sequences of Exemplary Fusion Constructs			
Const. No.	Signal Sequence (Optional)	Signal Sequence SEQ ID (Optional)	Full Protein SEQ ID
273	AATMGWSLILLFLVAVATRVHS	486	345
274	AATMGWSLILLFLVAVATRVHS	486	346
275	AATMGWSLILLFLVAVATRVHS	486	347
276	AATMGWSLILLFLVAVATRVHS	486	348
277	AATMGWSLILLFLVAVATRVHS	486	349
278	AATMGWSLILLFLVAVATRVHS	486	350
279	AATMGWSLILLFLVAVATRVHS	486	351
280	AATMGWSLILLFLVAVATRVHS	486	352
281	AATMGWSLILLFLVAVATRVHS	486	353
282	AATMGWSLILLFLVAVATRVHS	486	354
283	AATMGWSLILLFLVAVATRVHS	486	355
284	AATMGWSLILLFLVAVATRVHS	486	356
285	AATMGWSLILLFLVAVATRVHS	486	357
286	AATMGWSLILLFLVAVATRVHS	486	358
287	AATMGWSLILLFLVAVATRVHS	486	359
288	AATMGWSLILLFLVAVATRVHS	486	360
289	AATMGWSLILLFLVAVATRVHS	486	361
290	AATMGWSLILLFLVAVATRVHS	486	362
291	AATMGWSLILLFLVAVATRVHS	486	363
292	AATMGWSLILLFLVAVATRVHS	486	364
293	AATMGWSLILLFLVAVATRVHS	486	365
294	AATMGWSLILLFLVAVATRVHS	486	366
295	AATMGWSLILLFLVAVATRVHS	486	367
296	AATMGWSLILLFLVAVATRVHS	486	368
297	AATMGWSLILLFLVAVATRVHS	486	369
298	AATMGWSLILLFLVAVATRVHS	486	370
299	AATMGWSLILLFLVAVATRVHS	486	371
300	AATMGWSLILLFLVAVATRVHS	486	372
301	AATMGWSLILLFLVAVATRVHS	486	373
302	AATMGWSLILLFLVAVATRVHS	486	374
303	AATMGWSLILLFLVAVATRVHS	486	375
304	AATMGWSLILLFLVAVATRVHS	486	376
305	AATMGWSLILLFLVAVATRVHS	486	377
306	AATMGWSLILLFLVAVATRVHS	486	378

TABLE 5-continued

Amino Acid Sequences of Exemplary Fusion Constructs			
Const. No.	Signal Sequence (Optional)	Signal Sequence SEQ ID (Optional)	Full Protein SEQ ID
307	AATMGWSLILLFLVAVATRVHS	486	379
308	AATMGWSLILLFLVAVATRVHS	486	380
309	AATMGWSLILLFLVAVATRVHS	486	381
310	AATMGWSLILLFLVAVATRVHS	486	382
311	AATMGWSLILLFLVAVATRVHS	486	383
312	AATMGWSLILLFLVAVATRVHS	486	384
313	AATMGWSLILLFLVAVATRVHS	486	385
314	AATMGWSLILLFLVAVATRVHS	486	386
315	AATMGWSLILLFLVAVATRVHS	486	387
316	AATMGWSLILLFLVAVATRVHS	486	388
317	AATMGWSLILLFLVAVATRVHS	486	389
318	AATMGWSLILLFLVAVATRVHS	486	390
319	AATMGWSLILLFLVAVATRVHS	486	391
320	AATMGWSLILLFLVAVATRVHS	486	392
321	AATMGWSLILLFLVAVATRVHS	486	393
322	AATMGWSLILLFLVAVATRVHS	486	394
323	AATMGWSLILLFLVAVATRVHS	486	395
324	AATMGWSLILLFLVAVATRVHS	486	396
325	AATMGWSLILLFLVAVATRVHS	486	397
326	AATMGWSLILLFLVAVATRVHS	486	398
327	AATMGWSLILLFLVAVATRVHS	486	399
328	AATMGWSLILLFLVAVATRVHS	486	400
329	AATMGWSLILLFLVAVATRVHS	486	401
330	AATMGWSLILLFLVAVATRVHS	486	402
331	AATMGWSLILLFLVAVATRVHS	486	403
332	AATMGWSLILLFLVAVATRVHS	486	404
333	AATMGWSLILLFLVAVATRVHS	486	405
334	AATMGWSLILLFLVAVATRVHS	486	406
335	AATMGWSLILLFLVAVATRVHS	486	407
336	AATMGWSLILLFLVAVATRVHS	486	408
337	AATMGWSLILLFLVAVATRVHS	486	409
338	AATMGWSLILLFLVAVATRVHS	486	410
339	AATMGWSLILLFLVAVATRVHS	486	411
340	AATMGWSLILLFLVAVATRVHS	486	412

TABLE 5-continued

Amino Acid Sequences of Exemplary Fusion Constructs			
Const. No.	Signal Sequence (Optional)	Signal Sequence SEQ ID (Optional)	Full Protein SEQ ID
341	AATMGWSLILLFLVAVATRVHS	486	413
342	AATMGWSLILLFLVAVATRVHS	486	414
343	AATMGWSLILLFLVAVATRVHS	486	415
344	AATMGWSLILLFLVAVATRVHS	486	416
345	AATMGWSLILLFLVAVATRVHS	486	417
346	AATMGWSLILLFLVAVATRVHS	486	418
347	AATMGWSLILLFLVAVATRVHS	486	419
348	AATMGWSLILLFLVAVATRVHS	486	420
349	AATMGWSLILLFLVAVATRVHS	486	421
350	AATMGWSLILLFLVAVATRVHS	486	422
351	AATMGWSLILLFLVAVATRVHS	486	423
352	AATMGWSLILLFLVAVATRVHS	486	424
353	AATMGWSLILLFLVAVATRVHS	486	425
354	AATMGWSLILLFLVAVATRVHS	486	426
355	AATMGWSLILLFLVAVATRVHS	486	427
356	AATMGWSLILLFLVAVATRVHS	486	428
357	AATMGWSLILLFLVAVATRVHS	486	429
358	AATMGWSLILLFLVAVATRVHS	486	430
359	AATMGWSLILLFLVAVATRVHS	486	431
360	AATMGWSLILLFLVAVATRVHS	486	432
361	AATMGWSLILLFLVAVATRVHS	486	433
362	AATMGWSLILLFLVAVATRVHS	486	434
363	AATMGWSLILLFLVAVATRVHS	486	435
364	AATMGWSLILLFLVAVATRVHS	486	436
365	AATMGWSLILLFLVAVATRVHS	486	437
366	AATMGWSLILLFLVAVATRVHS	486	438
367	AATMGWSLILLFLVAVATRVHS	486	439
368	AATMGWSLILLFLVAVATRVHS	486	440
369	AATMGWSLILLFLVAVATRVHS	486	441
370	AATMGWSLILLFLVAVATRVHS	486	442
371	AATMGWSLILLFLVAVATRVHS	486	443
372	AATMGWSLILLFLVAVATRVHS	486	444
373	AATMGWSLILLFLVAVATRVHS	486	445
374	AATMGWSLILLFLVAVATRVHS	486	446

TABLE 5-continued

Amino Acid Sequences of Exemplary Fusion Constructs			
Const. No.	Signal Sequence (Optional)	Signal Sequence SEQ ID (Optional)	Full Protein SEQ ID
375	AATMGWSLILLFLVAVATRVHS	486	447
376	AATMGWSLILLFLVAVATRVHS	486	448
377	AATMGWSLILLFLVAVATRVHS	486	449
378	AATMGWSLILLFLVAVATRVHS	486	450
379	AATMGWSLILLFLVAVATRVHS	486	451
380	AATMGWSLILLFLVAVATRVHS	486	452
381	AATMGWSLILLFLVAVATRVHS	486	453
382	AATMGWSLILLFLVAVATRVHS	486	454
383	AATMGWSLILLFLVAVATRVHS	486	455
384	AATMGWSLILLFLVAVATRVHS	486	456
385	AATMGWSLILLFLVAVATRVHS	486	457
386	AATMGWSLILLFLVAVATRVHS	486	458
387	AATMGWSLILLFLVAVATRVHS	486	459
388	AATMGWSLILLFLVAVATRVHS	486	460
389	AATMGWSLILLFLVAVATRVHS	486	461
390	AATMGWSLILLFLVAVATRVHS	486	462
391	AATMGWSLILLFLVAVATRVHS	486	463
392	AATMGWSLILLFLVAVATRVHS	486	464
393	AATMGWSLILLFLVAVATRVHS	486	465
394	AATMGWSLILLFLVAVATRVHS	486	466
395	AATMGWSLILLFLVAVATRVHS	486	467
396	AATMGWSLILLFLVAVATRVHS	486	468
397	AATMGWSLILLFLVAVATRVHS	486	469
398	AATMGWSLILLFLVAVATRVHS	486	470
399	AATMGWSLILLFLVAVATRVHS	486	471
400	AATMGWSLILLFLVAVATRVHS	486	472
415	AATMGWSLILLFLVAVATRVHS	486	473
416	AATMGWSLILLFLVAVATRVHS	486	474
417	AATMGWSLILLFLVAVATRVHS	486	475
418	AATMGWSLILLFLVAVATRVHS	486	476
419	AATMGWSLILLFLVAVATRVHS	486	477
420	AATMGWSLILLFLVAVATRVHS	486	478
421	AATMGWSLILLFLVAVATRVHS	486	479
422	AATMGWSLILLFLVAVATRVHS	486	480

TABLE 5-continued

Amino Acid Sequences of Exemplary Fusion Constructs			
Const. No.	Signal Sequence (Optional)	Signal Sequence SEQ ID (Optional)	Full Protein SEQ ID
423	AATMGWSLILLFLVAVATRVHS	486	481
424	AATMGWSLILLFLVAVATRVHS	486	482
425	AATMGWSLILLFLVAVATRVHS	486	483
426	AATMGWSLILLFLVAVATRVHS	486	484
427	AATMGWSLILLFLVAVATRVHS	486	485

[0127] The present description sets forth numerous exemplary configurations, methods, parameters, and the like. It should be recognized, however, that such description is not intended as a limitation on the scope of the present disclosure, but is instead provided as a description of exemplary embodiments. Embodiments of the present subject matter described above may be beneficial alone or in combination, with one or more other aspects or embodiments. Without limiting the foregoing description, certain non-limiting embodiments of the disclosure are provided below. As will be apparent to those of skill in the art upon reading this disclosure, each of the individually numbered embodiments may be used or combined with any of the preceding or following individually numbered embodiments. This is intended to provide support for all such combinations of embodiments and is not limited to combinations of embodiments explicitly provided below.

[0128] The following Enumerated Embodiments and Examples are merely illustrative and are not meant to limit any aspects of the present disclosure in any way.

ENUMERATED EMBODIMENTS

Set I

[0129] Embodiment I-1. A variant of a carboxypeptidase N (CPN) comprising at least one modification with respect to a wild type CPN, wherein the variant has at least one improved characteristic as compared to the wild type CPN.

[0130] Embodiment I-2. The variant of embodiment I-1, wherein the improved characteristic is selected from an increase or a decrease in any one or more of: half-life, activity, potency, affinity for one or more substrates, sensitivity, cofactor affinity, catalytic capability, and selectivity.

[0131] Embodiment I-3. The variant of embodiment I-2, wherein the at least one improved characteristic comprises an increase in affinity for one or more substrates, and wherein at least one substrate is C3a.

[0132] Embodiment I-4. The variant of any one of embodiments I-2 to I-3, wherein the at least one improved characteristic comprises an increase in affinity for one or more substrates, and wherein at least one substrate is C5a.

- [0133] Embodiment I-5. The variant of any one of embodiments I-2 to I-4, wherein an increase in activity comprises an increase in the cleavage of C3a and/or C5a.
- [0134] Embodiment I-6. The variant of any one of embodiments I-2 to I-5, wherein the increase in sensitivity comprises an increase in the sensitivity to any one or more of: MASP1, MASP3, Factor D, mast cell degranulation, mCPA3, and neutrophil degranulation cathepsin G.
- [0135] Embodiment I-7. The variant of any one of embodiments I-2 to I-6, wherein the increase in activity comprises an increased k_{cat}/K_M ($M^{-1} s^{-1}$) for cleavage of C3a and/or C5a.
- [0136] Embodiment I-8. The variant of embodiment I-7, wherein the increased k_{cat}/K_M ($M^{-1} s^{-1}$) for cleavage of C3a and/or C5a is about 2-fold, about 3-fold, about 4-fold, about 5-fold, about 6-fold, about 7-fold, about 8-fold, about 9-fold, about 10-fold, about 20-fold, about 30-fold, about 40-fold, about 50-fold, about 60-fold, about 70-fold, about 80-fold, about 90-fold, or about 100-fold greater than that of the wild type CPN.
- [0137] Embodiment I-9. The variant of embodiment I-7, wherein the increase in activity comprises an increase in k_{cat} with a decrease in K_M .
- [0138] Embodiment I-10. The variant of any one of embodiments I-1 to I-9, wherein the increased activity comprises a decreased K_D (nM) value for cleavage of C3a and/or C5a.
- [0139] Embodiment I-11. The variant of embodiment I-10, wherein the decreased K_D (nM) value is about 2-fold, about 3-fold, about 4-fold, about 5-fold, about 6-fold, about 7-fold, about 8-fold, about 9-fold, about 10-fold less than that of the wild type CPN.
- [0140] Embodiment I-12. The variant of any one of embodiments I-2 to I-11, wherein the increased activity comprises a decreased EC_{50} (nM) value for cleavage of C3a and/or C5a, compared to the wild type CPN.
- [0141] Embodiment I-13. The variant of embodiment I-12, wherein the decreased EC_{50} (nM) value for cleavage of C3a and/or C5a is about 10, about 15, about 20, or less than about 20.
- [0142] Embodiment I-14. The variant of any one of embodiments I-2 to I-13, wherein the increased half-life is half-life in plasma.
- [0143] Embodiment I-15. The variant of embodiment I-14, wherein the increased half-life in plasma is greater than about 24 hours.
- [0144] Embodiment I-16. The variant of any one of embodiments I-2 to I-15, wherein the increased half-life in plasma is from about 70 hours to about 150 hours.
- [0145] Embodiment I-17. The variant of any one of embodiments I-2 to I-16, wherein the increased sensitivity comprises increased sensitivity for complement activation.
- [0146] Embodiment I-18. The variant of any one of embodiments I-1 to I-17, wherein the variant comprises at least one modification corresponding to a wild type non-human CPN.
- [0147] Embodiment I-19. The variant of any one of embodiments I-1 to I-17, wherein the variant comprises at least one modification corresponding to a wild type human CPN.
- [0148] Embodiment I-20. The variant of embodiment I-19, wherein the CPN variant comprises at least one modification corresponding to a wild type CPN comprising the amino acid sequence as set forth in SEQ ID NO: 1.
- [0149] Embodiment I-21. The variant of any one of embodiments I-1 to I-20, comprising one or more of domains of a wild type CPN selected from: a first peptide signal, a catalytic domain (CPN1), a second peptide signal, and a regulatory subunit (CPN2).
- [0150] Embodiment I-22. The variant of any one of embodiments I-1 to I-21, wherein the modification with respect to a wild type CPN comprises any one or more of: a deletion of one or more amino acid residues, a deletion of one or more CPN domains, a substitution of one or more amino acid residues, an insertion of one or more amino acid residues, an insertion of one or more CPN domains, a swapping of one or more CPN domains, and an insertion of at least one non-CPN domain or component.
- [0151] Embodiment I-23. The variant of embodiment I-22, wherein the at least one non-CPN domain or component comprises at least one domain of any one or more of: carboxypeptidase B2 (CPB2), carboxypeptidase A4 (CPA4), and carboxypeptidase A1 (CPA1).
- [0152] Embodiment I-24. The variant of embodiment I-23, wherein the at least one non-CPN domain is at least one domain of CPB2 comprising an activation peptide.
- [0153] Embodiment I-25. The variant of embodiment I-23, wherein the at least one non-CPN domain is at least one domain of CPA4 comprising an activation peptide.
- [0154] Embodiment I-26. The variant of embodiment I-23, wherein the at least one non-CPN domain is at least one domain of CPA1 comprising an activation peptide.
- [0155] Embodiment I-27. The variant of any one of embodiments I-24 to I-26, wherein the activation peptide increases sensitivity of the CPN variant for any one or more of: thrombin-thrombomodulin, MASP1, MASP3, Factor D, mCPA3, cathepsin G.
- [0156] Embodiment I-28. The variant of any one of embodiments I-24 to I-27, wherein the activation peptide increases sensitivity of the CPN variant for any one or more of: mast cell degranulation, neutrophil degranulation, and inflammatory cell activation.
- [0157] Embodiment I-29. The variant of any one of embodiments I-24 to I-28, wherein the activation peptide increases sensitivity of the CPN variant to complement activation.
- [0158] Embodiment I-30. The variant of any one of embodiments I-1 to I-29, wherein the deletion of one or more CPN domains comprises deletion of a CPN2 domain.
- [0159] Embodiment I-31. The variant of any one of embodiments I-23 to I-29, wherein the at least one CPB2 domain comprises a CPB2 catalytic domain.
- [0160] Embodiment I-32. The variant of any one of embodiments I-23 to I-31, wherein the CPN variant

- comprises a structural arrangement from N-terminus to C-terminus as (first peptide signal)-(CPB2 activation peptide)-(CPN2).
- [0161] Embodiment I-33. The variant of any one of embodiments I-1 to I-32, wherein the CPN variant is further fused to a second component.
- [0162] Embodiment I-34. The variant of embodiment I-33, wherein second component comprises a half-life extender.
- [0163] Embodiment I-35. The variant of embodiment I-34, wherein the half-life extender is selected from the group consisting of: PEGylation, PASylation, carbohydrates, albumin, and Fc.
- [0164] Embodiment I-36. The variant of any one of embodiments I-22 to I-35, wherein the insertion of at least one non-CPN domain or component comprises a first non-CPN domain or component a second non-CPN domain or component.
- [0165] Embodiment I-37. The variant of embodiment I-36, wherein the CPN variant comprises a structural arrangement from N-terminus to C-terminus as (first peptide signal)-(first non-CPN domain or component)-(CPN2)-(second non-CPN domain or component).
- [0166] Embodiment I-38. The variant of embodiment I-37, wherein the first non-CPN domain or component is an activation peptide or a half-life extender.
- [0167] Embodiment I-39. The variant of any one of embodiments I-37 to I-38, wherein the second non-CPN domain or component is a half-life extender or activation peptide.
- [0168] Embodiment I-40. The variant of any one of embodiments I-1 to I-39, wherein the variant is a tetramer comprising two heterodimers.
- [0169] Embodiment I-41. The variant of any one of embodiments I-1 to I-39, wherein the variant is a single heterodimer.
- [0170] Embodiment I-42. The variant of any one of embodiments I-1 to I-41, wherein the variant is non-immunogenic.
- [0171] Embodiment I-43. The variant of any one of embodiments I-1 to I-42, wherein the variant is in a zymogen form.
- [0172] Embodiment I-44. The variant of any one of embodiments I-1 to I-42, wherein the variant is in an active form.
- [0173] Embodiment I-45. A method of treating a disease or condition in a subject in need thereof, comprising administering to the subject any one of the variants of embodiments I-1 to I-44.
- [0174] Embodiment I-46. The method of embodiment I-45, wherein the disease or condition is selected from the group consisting of: congenital complement deficiency, control protein deficiency, secondary complement disorder, immunity related disorder, chronic renal disorder, acute inflammatory disorder, antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV), C3 glomerulopathy (C3G), lupus nephritis, skin disorder, intestinal ischemia and reperfusion (I/R) injury, sepsis, mast cell related disorders, and solid tumors refractory to immunotherapy agents such as pembrolizumab and ipilimumab.
- [0175] Embodiment I-47. The method of embodiment I-45, wherein the disease or condition is an acute condition selected from the group consisting of: acute respiratory distress syndrome (ARDS), COVID-19, multisystem organ failure, and sepsis.
- [0176] Embodiment I-48. The method of embodiment I-45, wherein the disease or condition is a chronic condition selected from the group consisting of: anti-neutrophil cytoplasmic autoantibody (ANCA) vasculitis, atypical hemolytic uremic syndrome (aHUS), and IgA nephropathy.
- [0177] Embodiment I-49. The method of embodiment I-46, wherein the disease or condition is a skin disorder selected from the group consisting of: hidradenitis suppurativa (HS), bullous pemphigoid (BP), and Pyoderma Gangrenosum.
- [0178] Embodiment I-50. The method of any one of embodiments I-45 to I-49, wherein the variant is administered in a zymogen form.
- [0179] Embodiment I-51. The method of any one of embodiments I-45 to I-49, wherein the variant is administered in an active form.
- [0180] Embodiment I-52. The method of any one of embodiments I-45 to I-49, wherein the administration is a subcutaneous administration.
- [0181] Embodiment I-53. The method of any one of embodiments I-45 to I-49, wherein the administration is an intravenous administration.
- [0182] Embodiment I-54. The method of any one of embodiments I-45 to I-49, wherein the administration comprises administering a vector comprising a nucleic acid encoding any one of the variants of embodiments I-1 to I-44.
- [0183] Embodiment I-55. A nucleic acid encoding any one of the variants of embodiments I-1 to I-44.
- [0184] Embodiment I-56. A pharmaceutical composition comprising any one of the variants of embodiments I-1 to I-44, and optionally a pharmaceutically acceptable carrier.
- [0185] Embodiment I-57. A variant of a carboxypeptidase N1 (CPN1) comprising at least one modification with respect to a wild type CPN1, wherein the variant has at least one improved characteristic as compared to the wild type CPN1.
- [0186] Embodiment I-58. A CPN1 variant selected from FIG. 11 of U.S. Ser. No. 63/222,929.
- [0187] Embodiment I-59. A CPN1 construct selected from FIG. 12 of U.S. Ser. No. 63/222,929.
- [0188] Embodiment I-60. A CPN1 construct comprising a CPN1 variant of FIG. 11 of U.S. Ser. No. 63/222,929.

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- [0189] Embodiment II-1. A variant of a carboxypeptidase N catalytic subunit (CPN1) comprising at least one modification with respect to a wild type CPN1, wherein the variant has at least one improved characteristic as compared to the wild type CPN1.
- [0190] Embodiment II-2. The variant of embodiment II-1, wherein the modification with respect to a wild type CPN1 comprises any one or more of: a substitution of one or more amino acid residues, a deletion of one or more amino acid residues, an insertion of one or more amino acid residues, an insertion of one or more CPN domains, and an insertion of one or more non-CPN domains or components.
- [0191] Embodiment II-3. The variant of any one of embodiments II-1 to II-2, wherein the improved char-

acteristic is selected from an increase or a decrease in any one or more of: half-life, activity, potency, substrate affinity, substrate specificity, substrate selectivity, proteolytic sensitivity, cofactor affinity, and catalytic capability.

- [0192] Embodiment II-4. The variant of embodiment II-3, wherein the at least one improved characteristic comprises an increase in affinity for one or more substrates, and wherein at least one substrate is C3a.
- [0193] Embodiment II-5. The variant of any one of embodiments II-3 to II-4, wherein the at least one improved characteristic comprises an increase in affinity for one or more substrates, and wherein at least one substrate is C5a.
- [0194] Embodiment II-6. The variant of any one of embodiments II-3 to II-5, wherein an increase in activity comprises an increase in the cleavage of C3a and/or C5a.
- [0195] Embodiment II-7. The variant of any one of embodiments II-3 to II-6, wherein the increase in activity comprises an increased k_{cat}/K_M ($M^{-1} s^{-1}$) for cleavage of C3a and/or C5a.
- [0196] Embodiment II-8. The variant of embodiment II-7, wherein the increased k_{cat}/K_M ($M^{-1} s^{-1}$) for cleavage of C3a and/or C5a is about 2-fold, about 3-fold, about 4-fold, about 5-fold, about 6-fold, about 7-fold, about 8-fold, about 9-fold, about 10-fold, about 20-fold, about 30-fold, about 40-fold, about 50-fold, about 60-fold, about 70-fold, about 80-fold, about 90-fold, or about 100-fold greater than that of the wild type CPN.
- [0197] Embodiment II-9. The variant of embodiment II-7, wherein the increase in activity comprises an increase in k_{cat} with a decrease in K_M .
- [0198] Embodiment II-10. The variant of any one of embodiments II-1 to II-9, wherein the increased activity comprises a decreased K_D (nM) value for cleavage of C3a and/or C5a.
- [0199] Embodiment II-11. The variant of embodiment II-10, wherein the decreased K_D (nM) value is about 2-fold, about 3-fold, about 4-fold, about 5-fold, about 6-fold, about 7-fold, about 8-fold, about 9-fold, about 10-fold less than that of the wild type CPN.
- [0200] Embodiment II-12. The variant of any one of embodiments II-3 to II-11, wherein the increased activity comprises a decreased EC_{50} (nM) value for cleavage of C3a and/or C5a, compared to the wild type CPN.
- [0201] Embodiment II-13. The variant of embodiment II-12, wherein the decreased EC_{50} (nM) value for cleavage of C3a and/or C5a is about 10, about 15, about 20, or less than about 20.
- [0202] Embodiment II-14. The variant of any one of embodiments II-3 to II-13, wherein the increased half-life is observed in plasma.
- [0203] Embodiment II-15. The variant of embodiment II-14, wherein the increased half-life in plasma is greater than about 24 hours.
- [0204] Embodiment II-16. The variant of any one of embodiments II-3 to II-15, wherein the increased half-life in plasma is from about 70 hours to about 150 hours.

- [0205] Embodiment II-17. The variant of any one of embodiments II-1 to II-16, wherein the variant comprises at least one modification corresponding to a wild type non-human CPN1.
- [0206] Embodiment II-18. The variant of any one of embodiments II-1 to II-16, wherein the variant comprises at least one modification corresponding to a wild type human CPN1.
- [0207] Embodiment II-19. The variant of embodiment II-18, wherein the CPN1 variant comprises at least one modification corresponding to a wild type CPN1 comprising the amino acid sequence as set forth in SEQ ID NO: 6.
- [0208] Embodiment II-20. The variant of any one of embodiments II-1 to II-19, wherein the CPN1 variant comprises an amino acid having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity to SEQ ID NO: 6.
- [0209] Embodiment II-21. The variant of any one of embodiments II-1 to II-19, wherein the CPN1 variant comprises the amino acid sequence as set forth in SEQ ID NO: 7, or at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity thereto.
- [0210] Embodiment II-22. The variant of any one of embodiments II-1 to II-19, wherein the CPN1 variant comprises the amino acid sequence as set forth in SEQ ID NO: 8, or at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity thereto.
- [0211] Embodiment II-23. The variant of any one of embodiments II-1 to II-19, wherein the CPN1 variant comprises the amino acid sequence as set forth in SEQ ID NO: 11, or at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity thereto.
- [0212] Embodiment II-24. The variant of any one of embodiments II-1 to II-19, wherein the CPN1 variant comprises the amino acid sequence as set forth in SEQ ID NO: 12, or at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity thereto.
- [0213] Embodiment II-25. The variant of any one of embodiments II-1 to II-19, wherein the CPN1 variant comprises SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 11, or SEQ ID NO: 12, comprising one of the modification strings selected from the group consisting of the modification strings provided in Table 4A, Table 4B, and Table 4C.
- [0214] Embodiment II-26. The variant of any one of embodiments II-1 to II-19, wherein the CPN1 variant comprises SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 11, or SEQ ID NO: 12, comprising one of the modification strings selected from the group

consisting of the modification strings provided in Table 3A, Table 3B, and Table 3C.

- [0215] Embodiment II-27. A fusion construct comprising a carboxypeptidase N catalytic subunit (CPN1) or variant thereof.
- [0216] Embodiment II-28. The fusion construct of embodiment II-27, selected from the group consisting of SEQ ID NO: 73-SEQ ID NO: 485.
- [0217] Embodiment II-29. The fusion construct of embodiment II-27, wherein the construct comprises a CPN1 of SEQ ID NO: 6.
- [0218] Embodiment II-30. The fusion construct of embodiment II-27, wherein the CPN1 variant comprises an amino acid having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity to SEQ ID NO: 6.
- [0219] Embodiment II-31. The fusion construct of embodiment II-27, wherein the CPN1 variant is selected from any one of the variants of embodiments II-1 to II-26.
- [0220] Embodiment II-32. The fusion construct of embodiment II-27, wherein the CPN1 variant comprises at least one modification corresponding to a wild type CPN1 comprising the amino acid sequence as set forth in SEQ ID NO: 6.
- [0221] Embodiment II-33. The fusion construct of embodiment II-27, wherein the CPN1 variant comprises the amino acid sequence as set forth in SEQ ID NO: 7, or at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity thereto.
- [0222] Embodiment II-34. The fusion construct of embodiment II-27, wherein the CPN1 variant comprises the amino acid sequence as set forth in SEQ ID NO: 8, or at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity thereto.
- [0223] Embodiment II-35. The fusion construct of embodiment II-27, wherein the CPN1 variant comprises the amino acid sequence as set forth in SEQ ID NO: 11, or at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity thereto.
- [0224] Embodiment II-36. The fusion construct of embodiment II-27, wherein the CPN1 variant comprises the amino acid sequence as set forth in SEQ ID NO: 12, or at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or even at least 99% sequence identity thereto.
- [0225] Embodiment II-37. The fusion construct of embodiment II-27, wherein the CPN1 variant comprises SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 11, or SEQ ID NO: 12, comprising one or

more modifications selected from the group consisting of the modifications provided in Table 4A, Table 4B, and Table 4C.

- [0226] Embodiment II-38. The fusion construct of embodiment II-27, wherein the CPN1 variant comprises SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 11, or SEQ ID NO: 12, comprising one of the modification strings selected from the group consisting of the modification strings provided in Table 3A, Table 3B, and Table 3C.
- [0227] Embodiment II-39. The fusion construct of any one of embodiments II-27 to II-38, wherein the fusion construct comprises a Glutathione S transferase (GST) amino acid sequence.
- [0228] Embodiment II-40. The fusion construct of embodiment II-39, wherein the GST amino acid sequence comprises SEQ ID NO: 31.
- [0229] Embodiment II-41. The fusion construct of any one of embodiments II-27 to II-40, wherein the fusion construct comprises a mammalian maltose binding protein (mMBP) amino acid sequence.
- [0230] Embodiment II-42. The fusion construct of embodiment II-41, wherein the mMBP amino acid sequence comprises SEQ ID NO: 30.
- [0231] Embodiment II-43. The fusion construct of any one of embodiments II-27 to II-42, wherein the fusion construct comprises a small ubiquitin modifying enzyme (SUMO) amino acid sequence.
- [0232] Embodiment II-44. The fusion construct of embodiment II-43, wherein the SUMO amino acid sequence comprises SEQ ID NO: 23 or SEQ ID NO: 24.
- [0233] Embodiment II-45. The fusion construct of any one of embodiments II-27 to II-44, wherein the fusion construct comprises a Tobacco Etch Virus protease cleavage site (TEV) amino acid sequence.
- [0234] Embodiment II-46. The fusion construct of embodiment II-45, wherein the TEV amino acid sequence comprises SEQ ID NO: 25.
- [0235] Embodiment II-47. The fusion construct of any one of embodiments II-27 to II-46, wherein the fusion construct comprises an activation peptide of CBP2 (the N-terminal 96aa of a CBP2 protease) amino acid sequence.
- [0236] Embodiment II-48. The fusion construct of embodiment II-47, wherein the amino acid sequence of the activation peptide of CBP2 comprises SEQ ID NO: 22.
- [0237] Embodiment II-49. The fusion construct of any one of embodiments II-27 to II-48, wherein the fusion construct comprises an Factor Xa protease cleavage site (Xa) amino acid sequence.
- [0238] Embodiment II-50. The fusion construct of embodiment II-49, wherein the Factor Xa protease cleavage site (Xa) comprises the amino acid sequence of SEQ ID NO: 29.
- [0239] Embodiment II-51. The fusion construct of any one of embodiments II-27 to II-50, wherein the fusion construct comprises a portion of a regulatory CPN2 subunit amino acid sequence.
- [0240] Embodiment II-52. The fusion construct of embodiment II-51, wherein the CPN2 amino acid sequence is selected from the group consisting of:

- CPN2 (1-367), CPN2 (1-370), CPN2 (1-425), CPN2 (1-456), and CPN2 (1-524).
- [0241] Embodiment II-53. The fusion construct of any one of embodiments II-27 to II-52, wherein the fusion construct comprises an CD180 amino acid sequence.
- [0242] Embodiment II-54. The fusion construct of any one of embodiments II-27 to II-53, wherein the fusion construct comprises an CD180 amino acid sequence.
- [0243] Embodiment II-55. The fusion construct of any one of embodiments II-27 to II-54, wherein the fusion construct comprises LR1G1 amino acid sequence.
- [0244] Embodiment II-56. The fusion construct of any one of embodiments II-27 to II-38, wherein the fusion construct comprises at least one non-CPN1 or non-CPN2 domain or component.
- [0245] Embodiment II-57. The fusion construct of embodiment II-56, wherein the at least one non-CPN1 or non-CPN2 domain or component comprises at least one domain of any one or more of: carboxypeptidase B2 (CPB2), carboxypeptidase A4 (CPA4), and carboxypeptidase A1 (CPA1).
- [0246] Embodiment II-58. The fusion construct of any one of embodiments II-27 to II-57, wherein the fusion construct comprises an activation peptide that increases sensitivity of the fusion construct.
- [0247] Embodiment II-59. The fusion construct of embodiment II-58, wherein the construct comprises an activation peptide that increases sensitivity of the fusion construct for any one or more of: mast cell degranulation, neutrophil degranulation, and inflammatory cell activation.
- [0248] Embodiment II-60. The fusion construct of embodiment II-58, wherein the activation peptide increases sensitivity of the fusion construct for any one or more of: thrombin-thrombomodulin, MASP1, MASP3, Factor D, mCPA3, complement activation, and cathepsin G.
- [0249] Embodiment II-61. The fusion construct of any one of embodiments II-27 to II-60, wherein the fusion construct comprises a half-life extender.
- [0250] Embodiment II-62. The fusion construct of embodiment II-61, wherein the half-life extender is selected from the group consisting of: PEG, PAS, carbohydrates, albumin, and Fc.
- [0251] Embodiment II-63. The fusion construct of embodiment II-62, wherein the albumin comprises human serum albumin.
- [0252] Embodiment II-64. The fusion construct of any one of embodiments II-27 to II-63, wherein the fusion construct is non-immunogenic.
- [0253] Embodiment II-65. The fusion construct of any one of embodiments II-27 to II-64, wherein the fusion construct is in a zymogen form.
- [0254] Embodiment II-66. The fusion construct of any one of embodiments II-27 to II-65, wherein the fusion construct is in an active form.
- [0255] Embodiment II-67. A method of treating a disease or condition in a subject in need thereof, comprising administering to the subject any one of the CPN1 variants or fusion constructs of embodiments II-1 to II-66.
- [0256] Embodiment II-68. The method of embodiment II-67, wherein the disease or condition is selected from the group consisting of: congenital complement deficiency, control protein deficiency, secondary complement disorder, immunity related disorder, chronic renal disorder, acute inflammatory disorder, antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV), C3 glomerulopathy (C3G), lupus nephritis, skin disorder, intestinal ischemia and reperfusion (I/R) injury, sepsis, mast cell related disorders, and solid tumors refractory to immunotherapy agents such as pembrolizumab and ipilimumab.
- [0257] Embodiment II-69. The method of embodiment II-67, wherein the disease or condition is an acute condition selected from the group consisting of: acute respiratory distress syndrome (ARDS), COVID-19, multisystem organ failure, and sepsis.
- [0258] Embodiment II-70. The method of embodiment II-67, wherein the disease or condition is a chronic condition selected from the group consisting of: anti-neutrophil cytoplasmic autoantibody (ANCA) vasculitis, atypical hemolytic uremic syndrome (aHUS), and IgA nephropathy.
- [0259] Embodiment II-71. The method of embodiment II-68, wherein the disease or condition is a skin disorder selected from the group consisting of: hidradenitis suppurativa (HS), bullous pemphigoid (BP), and Pyoderma Gangrenosum.
- [0260] Embodiment II-72. The method of any one of embodiments II-67 to II-71, wherein the fusion construct as administered is in a zymogen form.
- [0261] Embodiment II-73. The method of any one of embodiments II-67 to II-71, wherein the CPN1 variant fusion construct as administered is in an active form.
- [0262] Embodiment II-74. The method of any one of embodiments II-67 to II-71, wherein the administration of the CPN1 variant or fusion construct is a subcutaneous administration.
- [0263] Embodiment II-75. The method of any one of embodiments II-67 to II-71, wherein the administration of the CPN1 variant or fusion construct is an intravenous administration.
- [0264] Embodiment II-76. The method of any one of embodiments II-67 to II-71, wherein the administration comprises administering a vector comprising a nucleic acid encoding any one of the CPN1 variants or fusion constructs of embodiments II-1 to II-66.
- [0265] Embodiment II-77. A nucleic acid encoding any one of the CPN1 variants or fusion constructs of embodiments II-1 to II-66.
- [0266] Embodiment II-78. A pharmaceutical composition comprising any one of the CPN1 variants or fusion constructs of embodiments II-1 to II-66, and optionally a pharmaceutically acceptable carrier.

EXAMPLES

Example 1: Activation Peptide Screening for Fusion With CPN

[0267] A peptide library can be used for the selection of effective activation peptide sequences, which can be expressed as an 8-mer, 10-mer, or a 12-mer, and so on. Purified MASP1, MASP3, Factor D, mCPA3, and cathepsin G can be used to screen for the activation peptide sequences. A mass spectrometry-based method can be used for each protease to evaluate the effectiveness of each activation peptide sequence tested.

[0268] In order to identify sequences efficiently cleaved by a selected enzyme, the positional scanning approach is used. The linear 8-mer (i.e., 8-amino acid long) peptide libraries covering the P4 to P4' positions are generated for each enzyme based on the sequence of the specific substrates described in the literature. For example, the library for MASP-1 will be based on the cleavage sequence of protease-activated receptor 4 (YPGK↓F, SEQ ID NO: 67)⁵, for MASP-3 on sequence derived of the combinatorial peptide library (GGK↓IFGG, SEQ ID NO: 68)⁶, for Factor D on the cleavage sequence of Factor B (QQKR↓KIVL, SEQ ID NO: 69)⁷, and for cathepsin G on the cleavage sequence of interleukin-33 (VECF↓AFGI, SEQ ID NO: 70)⁸. The selected sequence will be incorporated into the thrombin cleavage site from CPB2 activation peptide (i.e., VSPR↓A-SAS, SEQ ID NO: 71). For each library, the screening for the best substitutions (residues with the highest affinity towards the selected enzyme and selectivity over others) will be performed as follows: each of the selected position (with the exception of the positions indicated in red which are necessary for enzyme recognition) will be replaced, one by one, with natural amino acid residues (except for Cys) to generate 18 individual peptides per one position screened. For instance, the library for MASP-3 will consist of 7 sub-libraries for the following positions: P4, P3, P2, P1, P1', P2', P3', P4', wherein the positions P are counted from the point of cleavage, which is between P1 and P', each containing 18 individual peptides (total of 126 peptides). Subsequently, the cleavage studies are performed by incubating the obtained analogs with the selected enzymes at different time points (at least 6 time points). FIG. 5 depicts a general schematic diagram of the screening process, and examples of libraries for the activation peptide screening.

Methodology

[0269] Peptides are prepared by manual and automated solid-phase peptide synthesis based on standard Fmoc strategy protocols using 2-chlorotriyl chloride resin. After completion of the synthesis, peptides are cleaved using trifluoroacetic acid (TFA) and required scavengers (selected based on the peptide sequence). The obtained crudes are purified using high performance liquid chromatography (HPLC) and their purity (>95%) will be characterized by HPLC. Enzymes will be purchased from commercial suppliers (i.e., R&D Systems, Enzo Life Science). The conditions (e.g., enzyme and peptide concentrations, pH) for the cleavage studies will be optimized for each enzyme using commercially available fluorogenic substrates. The cleavage experiments will be performed by incubating each peptide with the selected enzyme at different time points (0, 5, 15, 30, 60, 120 and 240 min). The reactions will be quenched with formic acid. The samples will be centrifuged, supernatant will be collected recovered, and stored in -80° C. Cleavage analysis will be done by HPLC and mass spectrometry analysis.

Example 2: Screening for Activated CPN Variants

[0270] Assays are performed to evaluate the activation of a CPN zymogen, followed by C3a and/or C5a digestion, for selection of effective CPN variants. The assays are used to determine the k_{cat}/K_M of the activated CPN. A first assay is used to screen the CPN variants' k_{cat}/K_M of towards C3a

versus C5a, and a second assay is used to screen the zymogen forms of CPN chimeras for their activation by complement enzymes.

[0271] Briefly, CPB2, CPN wild type and CPN chimeras are incubated with their respective complement activation enzyme and the negative control T-TM complex for a period of time before the reaction is terminated by the addition of PPACK. Kinetic analysis of hydrolysis of 8-10-mer C3a and C5a peptides by CPB2 and CPN is performed, and Michaelis-Menten kinetics is used to determine the k_{cat}/K_M for the hydrolysis by CPB2 or CPN of the 10-mer peptide. The portion of the C3a and C5a peptides used for the assay are indicated in bold in Table 2 below.

TABLE 2.1	
C3a and C5a Peptide Substrate Sequences	
C3a	SVQLTEKRM D KVGKYPKELR K CCEDGMRENPMRFS CQ RRTRFISLGEACKKVELDCCNYITELRRQHARASHL G LAR (SEQ ID NO: 4)
C5a	TLQKKIEETIAAKYKHSVVKCCYD G ACVNNDETCEQR AARISLGPRCIKAPTECCVVASQLRANISHKDMQL GR (SEQ ID NO: 5)

[0272] The cleaved peptide is resolved by HPLC, and the nmol of the peptide generated is determined from the peak area of cleaved peptide. The values for K_M and k_{cat} are then determined by plotting the initial velocities of cleavage against the different substrate concentrations tested, and fitting to the Michaelis-Menten equation by non-linear regression, as is known in the art.

Example: 3: Receptor Activation Assay

[0273] A beta-arrestin GPCR assay can be used to measure the activity of the CPN variants for their substrates, which can be C3a and/or C5a. C3a and/or C5a receptor activation upon binding of C3a and/or C5a is measured by the amount of luminescence generated, indicative of recruitment of beta-arrestin, which is directly related to receptor activation, thus providing a determination of the activity level of the C3a and/or C5a. Briefly, CPN variants are incubated with C3a and/or C5a and the digestion is stopped with a CPN inhibitor. Digests are then incubated with cells and detecting reagents are added. Luminescence is read to measure the amount of residual C3a or C5a resulting from the CPN variant digestion.

Example 4: Functional Assay—Calcium Signaling

[0274] A functional assay measuring calcium signaling can be performed to assess the level of C3a or C5a inhibition by CPN variants. Imaging can be done to show the calcium signaling taking place. Briefly, cells transfected with C3aR or C5aR are incubated with the CPN variants to be tested, to determine whether the CPN variants are capable of directly activating C3aR or C5aR, and whether the CPN variants are capable of interfering with C3a or C5a to thus avoid C3aR or C5aR activation. The main readout for cell line activation will be the monitoring of intracellular calcium levels (increased after C3aR or C5aR activation) by real-time fluorescence microscopy in a temperature-controlled atmosphere. Following treatments, intracellular calcium will be visualized with a time-lapse fluorescent microscope. Levels

of calcium signaling shown through fluorescent imaging will determine whether the CPN variants are effectively interfering with C3a or C5a to thus prevent or reduce C3aR or C5aR activation. HEK cells expressing or not expressing the receptors C3a and C5a are loaded with Fluo-4, a calcium sensitive probe, then imaged under an Apotome microscope wide field, 20× objective with a temperature-controlled atmosphere. Quantification of fluorescent signals are performed using the software ImageJ, and global increase in mean fluorescence intensity (MFI) is evaluated.

Example 5: In Vivo Models of Disease

[0275] Animal models can be used for further study of diseases related to dysregulation of complement. Chronic models include: lupus nephritis using MRL/lpr and NZB/W F1 mice, C3G using rats, ANCA AAV using mice, bullous pemphigoid using mice. Acute models include: cecal ligation and puncture using rats, peritonitis using mice, LPS lung injury using mice, and kidney transplant I/R injury using mice.

Example 6: Expression of CPN1 Variants

[0276] In general, the Expi293™ Expression System Kit (A14635 From Thermo Fisher Scientific) was used to transfect and transiently express CPN1 and its variants. Briefly, cells were diluted in pre-warmed media to indicated desired starting density in an appropriate vessel size based on volume. Optionally, cells were pre-diluted to desired transfection density for multiple transfections and desired total mL column used. DNA/optimem and Expifectamine/optimem were prepared as two separate mixtures, inverted, and incubated for 5 mins at RT. Optionally, expifectamine/optimem master mix with ~3% extra volume for multiple transfections can be made. Expifectamine mix was added to DNA mix, mixed by inversion, and allowed to complex for 10 mins. Complexes were added dropwise to cells while swirling. Cells were placed in an incubator at an appropriate shake speed for vessel size.

[0277] FIGS. 6A-6C depict Coomassie staining from SDS-PAGE analysis showing various CPN chimeras expressed in Expi293 cells. Generally, the expression of CPN1-HSA and CPN1-CPB2-HSA were shown to have better expression in Expi293 cells than the other tested constructs.

[0278] Thirteen CPN1 constructs were transiently transfected in Expi293 cells. After 2 or 3 days of transient expression, cell supernatants were collected for SDS PAGE gel analysis. Cell supernatants were mixed with NuPage loading dye containing reducing agent and samples heated 95° C. for 5 minutes. Samples were loaded in Tris-Bis gel 4-12% and run at 150V for 1 hour. Gels were incubated with SimplyBlue Safe stain to detect proteins. Among the 13 constructs tested CPN1-SP-Ig-HSA tag and CPN1-CPB2-SP-Ig-HSA tag expressed best.

[0279] FIG. 6D shows a Coomassie staining from SDS-PAGE analysis of CPN1 construct expression, using a similar process as depicted in FIGS. 6A-6C, with five CPN1 constructs. These five CPN1 constructs were also transiently transfected in Expi293 cells. After 3 days of transient expression, cell supernatants were collected for SDS-PAGE gel analysis. Cell supernatants were mixed with NuPage loading dye with (R) or without (NR) reducing agent and samples heated 95° C. for 5 minutes. Samples were loaded

in Tris-Bis gel 4-12% and run at 150V for 1 hour. Gels were incubated with SimplyBlue Safe stain to detect proteins. Overall, among these five constructs, CPN1-SP-Ig-HSA tag expressed best.

[0280] FIGS. 6E-6F show eight exemplary CPN1 constructs and their corresponding SDS-PAGE analysis of cell culture supernatants after 4 days of transient expression in Expi293 cells. Generally, the constructs with HSA fusion showed the best expression. Similarly, FIG. 6G-6H shows the construct His-SUMO-CPN1-HSA and its SDS-PAGE analysis. The results show that His-SUMO-CPN1-HSA can be expressed and purified by His-Trap and SUMO cleavage.

[0281] FIGS. 6I-6J shows exemplary CPN1 constructs with a TEV protease cleavage site (CPN1-TEV-HSA and CPN1-TEV-LL-HSA) and without a TEV protease cleavage site (CPN1-HSA). High expression of all three constructs was seen in SDS-PAGE analysis of cell culture supernatants after 4 days of transient expression in Expi293 (FIG. 6K). Further, FIG. 6L shows SDS-PAGE analysis of purified CPN1-TEV-HSA and CPN1-TEV-LL-HSA proteins treated with or without TEV protease. The cleavage of the constructs into CPN1 and TEV-HSA components in samples exposed to TEV protease is clearly seen. Further, to test the CPN1 catalytic activity of CPN1-TEV-HSA and CPN1-TEV-LL-HSA after HSA cleavage by TEV protease, a peptide based TAFI activity assay (Pefakit®) TAFI, Catalogue Number: 800186) was carried out. CPN1-HSA, which has no TEV cleavage site, served as control. Results of the TAFI activity assay are shown in FIG. 6M, which show that CPN1 preserves its catalytic activity after HSA cleavage in both CPN1-TEV-HSA and CPN1-TEV-LL-HSA constructs. Further, FIGS. 6N-6O show additional five exemplary CPN1 constructs and their corresponding SDS-PAGE expression analysis.

[0282] FIGS. 6P-6W show the SDS-PAGE expression analysis of further exemplary CPN1 constructs. Samples for the gels were generally prepared with 10 µL 4xLDS/DTT and 30 µL sample. The samples were heated to 90° C. for 5 minutes. 8 µL ladder loaded and 15 µL sample were loaded in relevant lanes. 4-12% Bis-Tris Gels were used and run at 150 V for 50 min. SimplyBlue Safe Stain was used to detect proteins.

Example 7A: Purification of CPN1 Variants by a 2-Step Purification Process

[0283] FIGS. 7A-7C depict various chromatography and SDS-PAGE results from purification of CPN variants. The CPN variants provided herein can be purified by a generic 2-step purification process negating the need for an affinity purification step with expensive resins. Briefly, clarified culture supernatant (with or without flocculant pretreatment) is diluted 1/10 in Buffer A (50 mM Tris-HCl, 500 mM NaCl pH 7.5) and applied to a Benzamidine-Sepharose column. The column is washed with 4CV of Buffer A to baseline and then eluted with a 20 CV linear gradient from Buffer A to 100% Buffer B (50 mM Tris-HCl, 500 mM NaCl, 1 M Arginine pH 7.5). Peak fractions were pooled corresponding to relative purity on a SDS-PAGE gel (FIG. 7A). Purity was assessed by absolute size exclusion chromatography (aSEC) with major peak purity of 82%, HMWS 5% and LMWS 13%. Pooled fractions were desalted to AEX Buffer A (25 mM Tris-HCl pH 7.5 and loaded onto a AEX column. The column was washed with 4CV of AEX Buffer A to baseline and then eluted with a 30 CV linear gradient. From AEX

Buffer A to 100% AEX Buffer B (25 mM Tris-HCl, 1 M NaCl, pH 7.5). Fractions were pooled based on relative purity from a SDS-PAGE gel (FIG. 7B). An increase in purity was increased to 94% main peak fraction and 6% HMWS (FIG. 7C).

Example 7B: Purification of CPN1 Variants by Affinity Purification

[0284] CPN1 constructs may also be purified using affinity purification-based methods. For example, CPN1-HSA (SEQ ID NO: 9) purification and high molecular weight removal was carried out as described below. CPN1-HSA was expressed using the Expi293 expression system, according to manufacturer's recommendations. After expression, cell culture supernatant was harvested and clarified by centrifugation at 3000×g for 30 min followed by vacuum filtration through a 0.22 µm filter.

[0285] CaptureSelect human albumin affinity matrix resin (ThermoFisher Scientific, Catalog No 191297050) was equilibrated by incubating the resin slurry with 20 mM Tris, pH 7.4 in a conical bottom centrifuge tube. The mixture was centrifuged at 3000×g for 25 min at room temperature and the supernatant was decanted. This process was repeated 3 times. After equilibration, the resin was loaded with cell culture supernatant, incubated for 30 mins, and centrifuged at 3000×g for 30 mins at room temperature. The supernatant flow-thru was decanted. The resin was then washed 3 times by incubating with 20 mM Tris, pH 7.4, centrifuging at 3000×g for 25 mins and pouring off the supernatant. Finally, the target protein was eluted by incubating the resin with 20 mM Tris, 1 M NaCl, 0.5 M Arg HCl, pH 7.4, spinning down at 3000×g for 10 min at room temperature, and collecting the supernatant. An exemplary SDS-PAGE gel stained with Coomassie showing load, flow-through, and 3 elutions from this procedure using a 500 mL cell culture supernatant load with 25 mL resin slurry is shown in FIG. 7D.

[0286] Elution fractions were pooled and dialyzed in 1×PBS and subsequently concentrated to >5 mg/mL using Amicon 10 K 15 mL filters (Millipore Sigma, Catalog No 901024). After concentration, preparative SEC was performed to remove high molecular weight (HMW) species. To perform the preparative SEC, A HiPrep 26/60 Sepharacyl S-200 High Resolution column (Cytiva, Catalog No 17119501) was first equilibrated with 4 column volumes (CV) of 1×PBS. Then, 3 mL concentrated protein sample was loaded via a sample loop onto the column. Separation was performed by running 4 CV 1×PBS over the loaded column. An exemplary chromatogram in FIG. 7E shows a clear separation between HMW (rt~97 min) and main peak (rt~118 min). The main peak fractions (rt 113 min-149 min) were pooled and concentrated for endotoxin removal and finishing.

[0287] CPN1-TEV-HSA was similarly purified using the CaptureSelect spin column method though at a smaller scale. FIG. 7F shows SDS-PAGE gels stained with Coomassie showing load, flow-through, and 3 elutions at different load and resin volumes.

Example 8A: C3a and C5a Activity Assay

[0288] FIGS. 8A-8B depict the results of an activity assay measuring the activation of C3aR and C5aR, respectively. The activity assay is used to examine the activation of C3aR and C5aR by C3a and C5a, and can also be used to screen for CPN activity on C3a and C5a. Briefly, target cells are infected with Retroparticles containing β-Arrestin Enzyme Acceptor ("EA"). Next, the PathHunter β-Arrestin parental cell line is transfected with the GPCR-PK plasmid (containing the GPCR of interest and the β-gal ProLink peptide tag). Next, the ligand to be tested, C3a or C5a, is added. Then, the substrate for active β-Arrestin EA is added. Luminescence is read to measure the amount of residual C3a or C5a and EC₅₀ calculated.

Example 8B: Activity Characterization Assay

[0289] Cell culture supernatants (CPN1-HSA and TAFI activation peptide-CPN1-HSA) or purified protein (CPN1-HSA) were evaluated for carboxypeptidase activity against anaphylatoxin substrates C3a or C5a. CPN1-HSA and TAFI activation peptide-CPN1-HSA amino acid sequences are shown in Table 1.

[0290] Carboxypeptidase enzymatic cleavage of terminal arginine from specific substrates (C3a or C5a, Complement Technology, Texas, USA) was assessed by mass spectrometric detection of released arginine. Following in vitro cleavage at 37° C., reactions were stopped via addition of acid (0.4 M perchloric acid, Sigma, Missouri, USA) or specific inhibitors (250 nM 1,10-phenanthroline or EDTA, Sigma, Missouri, USA). Reactions were derivatized (SymDAQ, Charles Rivers Laboratories, California, USA), internal standard (13C6-Arginine, Sigma, Missouri, USA) spiked into reactions and products separated via liquid chromatography-mass spectrometry (LC-MS/MS). Released arginine and spiked calibrator were then measured by mass transition. Released arginine was quantified by linear regression against the spiked in calibrator. Reactant and buffer alone controls were included for background correction.

[0291] To further characterize the activity of purified carboxypeptidases cleavage of C3a or C5a anaphylatoxin substrate was assessed using the PathHunter® β-Arrestin assay (Eurofins DiscoverX, California, USA) for GPCR C3aR1 or C5aR1 cell lines. To provide an approximation for percent cleavage, the cleavage reaction mixtures were run with a full titration curve and compared to native ligand (C3a or C5a) and fully cleaved native ligand (C3a-desArg and C5a-desArg) on their respective receptors. Substrate concentration was varied to generate response curves used to calculate EC₅₀ by nonlinear regression (Prism 9, log (agonist) vs. response-4 parameter variable slope model). Using the calculated EC₅₀'s, percent cleavage was estimated by the equation 100%-(native substrate EC₅₀/test article EC₅₀×100). Percent cleavage reported was normalized by subtracting C3a-desArg or C5a-desArg alone background signaling. Native anaphylatoxins and their native-desArg versions, buffer and vehicle alone controls were included for background evaluation.

TABLE 8.1

In vitro carboxypeptidase activity of CPN1-HSA variants for the anaphylotoxin substrate C3a.				
Variant	Enzyme	Assay	Metric	Measurement
CPN1-HSA	Purified	Arg LC-MS/MS	K_{cat}/K_M	$8 \times 10^5 \text{ M}^{-1} \text{ s}^{-1}$
CPN1-HSA	Purified	Arg LC-MS/MS	EC_{50}	$4.42 \times 10^{-9} \text{ M}$
CPN1-HSA	Supernatant	Arg LC-MS/MS	% Arg released	100%
TAFI activation peptide-CPN1-HSA	Supernatant	Arg LC-MS/MS	% Arg released	72%
CPN1-HSA	Purified	GPCR	% Cleavage	100%
Variant	Enzyme	Assay	Metric	Measurement
CPN1-HSA	Supernatant	GPCR	% Cleavage	100%

TABLE 4

In vitro carboxypeptidase activity of CPN1-HSA variants for the anaphylotoxin substrate C5a				
Variant	Enzyme	Assay	Metric	Measurement
CPN1-HSA	Purified	Arg LC-MS/MS	K_{cat}/K_M	ND
CPN1-HSA	Purified	Arg LC-MS/MS	EC_{50}	$3.53 \times 10^{-8} \text{ M}$
CPN1-HSA	Supernatant	Arg LC-MS/MS	% Arg released	20%
TAFI activation peptide-CPN1-HSA	Supernatant	Arg LC-MS/MS	% Arg released	14%
CPN1-HSA	Purified	GPCR	% Cleavage	82%
CPN1-HSA	Supernatant	GPCR	% Cleavage	97%

ND = not determined

Example 8C: Dansyl-Ala-Arg Activity Assay

[0292] Assay principle: CPN1 cleaves C-terminal arginine from dansyl-ala-arg (CAS #: 87687-46-5) substrate. Activity was measured in an end-point assay format. The product dansyl-ala-OH was separated from dansyl-ala-arg after acidification and extraction with chloroform. The dansyl-ala-OH product (CAS #: 53332-27-7) was measured via absorbance at $\lambda=340 \text{ nm}$. Fluorescence of the chloroform-extracted dansyl-ala-OH product can also be measured at $\lambda_{ex}/\lambda_{em} 340/495 \text{ nm}$.

[0293] Stock materials included 10 mM dansyl-ala-arg dissolved in DMSO, 10 mM dansyl-ala-OH dissolved in DMSO for calibration curve, Activity assay buffer: 100 mM Tris, pH 7.4, 50 mM NaCl, 0.01% Tween 80 Quartz cuvette (10 cm path length).

[0294] Each reaction mixture (200 μL) was prepared in 1.5 mL Eppendorf tubes and included 400 μM dansyl-ala-arg, activity assay buffer, and 0.35 μM of relevant CPN1 enzyme. The reaction was allowed to proceed at 37° C. for 2 h. To terminate and acidify the reaction, 20 μL of 1 M HCl was added to the reaction tubes. 1 mL of chloroform was then added to each reaction mixture and vortexed for 15 s to extract the product (dansyl-ala-OH) into the chloroform layer. 0.8 mL of chloroform (the bottom phase) was pipetted into the quartz cuvette and absorbance at 340 nm was measured (or scanned for better visualization). The cuvette was rinsed twice with 0.8 mL chloroform between sample measurements. A calibration curve with dansyl-ala-OH absorbance at 340 nm was used to quantify amount of product. The substrate dansyl-ala-arg did not extract into chloroform, even at high (2.5 mM) concentrations; in contrast, dansyl-ala-OH provided large signal ($\lambda_{max}=340 \text{ nm}$) after extraction. The calibration curve of dansyl-ala-OH was observed to be linear ($\sim 100 \mu\text{M}$ -1.5 mM).

[0295] CPB1 (positive control) and CPN1-HSA were tested using the above protocol. Both CPB1 and CPN1-HSA showed activity as seen in FIG. 8C and FIG. 8D respectively. The corresponding tables in FIG. 8C (for CPB1 positive control) and FIG. 8D (for CPN1-HSA) depict the signal levels and concentrations for CPB1 and CPN1-HSA, respectively.

Example 8D: Hippuryl-Arg Activity Assay

[0296] Assay principle: CPN1 cleaves C-terminal arginine. Both hippuryl-arg (CAS: 744-46-7) and hippuric acid (CAS: 495-69-2) absorb at 254 nm, but hippuric acid has a slightly higher ϵ . The amount of product produced is quantified by ΔOD_{254} between sample and [S]₀.

[0297] Stocks/Materials included 10 mM hippuryl-arg dissolved in water, 10 mM hippuric acid dissolved in water for calibration curve, activity assay buffer: 100 mM Tris, pH 7.4, 50 mM NaCl, 0.01% Tween 80, and UV-detectable 96-well plates.

[0298] 2 $\times$ enzyme solution in a non-binding plate (i.e. 0.7 μM [E]) was prepared. Serial dilutions in this plate were performed when multiple [E] were tested. 50 μL of buffer (for substrate and product standards) or 2 $\times$ enzyme solution to a UV-detectable plate were added. 2 $\times$ solutions of substrate and product (i.e. 2 mM hippuryl-arg and 2 mM hippuric acid, respectively) were prepared. 50 μL of substrate and product standard wells were added, followed by 50 μL of substrate to wells with enzyme to start the reaction. The plates were loaded in the plate reader as soon as possible, typically within 20 s of starting the reactions. Absorbance at 254 nm was read in 20 s intervals for 1 h. For analysis the extent of reaction and amount of product produced at each time point was quantified. Mixing effects were seen up to 5 min, so productivity curves (integrated

rate equations) were used to fit kinetic parameters. CPB1 was found to generate signal from hippuryl-arg substrate in a dose-dependent manner in above assay protocol.

[0299] Activities of CPN1-HSA and TAFI activation peptide-CPN1-HSA were tested using the above assay protocol at various concentrations ranging from 0.044 uM to 0.35 uM, as shown in FIGS. 8E-8F, respectively. In FIG. 8E each concentration of CPN1-HSA was tested in duplicate. Similarly, in FIG. 8F, each concentration of TAFI activation peptide-CPN1-HSA was tested in duplicate.

Example 9: CPN1-HSA Efficacy after Intravenous Administration in a Rodent Model of Complement Activation

[0300] Examination of in vivo activity of CPN1-HSA in Acute Respiratory Distress Syndrome and evaluation of the therapeutic effects of CPN1-HSA and anti-C5 in an LPS-induced acute respiratory distress syndrome (ARDS) mouse model was carried out. The purpose of this study was to assess the efficacy of CPN1-HSA and anti-murine C5 antibody to limit complement mediated acute pulmonary inflammation in a mouse model of ARDS induced by a single administration of lipopolysaccharide (LPS).

[0301] Method description: A mouse model of aseptic ARDS was used to study complement involvement following an intratracheal instillation (IT) of LPS. Male C57BL/6 mice (Charles River Laboratories) weighing 20 to 25 g at enrolment were anesthetized under isoflurane and intratracheally instilled with 50 µg LPS (1 mg/mL LPS isolated from *E. coli* 0111: B4 in 0.9% saline solution, Sigma).

[0302] Immediately prior to IT LPS administration, animals received an IV injection of 5 mg/kg CPN1-HSA (as shown in FIG. 12, SEQ ID NO: 9, n=8) or control article (PBST with 0.01% P80; n=8) at a dosing volume of 5 mL/kg. Positive control animals received 1 mg/animal anti-mouse C5 recombinant antibody (Clone BB5.1, Creative Biolabs, n=8) intraperitoneally at a dose volume of 1 mL. This antibody is highly specific for mouse C5 and, like the FDA-approved anti-human C5 monoclonal antibody, eculizumab, BB5.1 binding to C5 efficiently inhibits cleavage of C5 to C5a and C5b. All animals were sacrificed 24 hours post-LPS IT.

[0303] Whole body plethysmography was performed prior to LPS instillation, as well as 6- and 24-hours post-instillation. Bronchoalveolar lavage fluid (BALF) was harvested in three 300 µL perfusions of the right lung with cold PBS 1x containing Protease Inhibitor 1x (SigmaFAST®). Evaluation of complement component fragments by mass spectrometry (MS) was performed on K₂-EDTA plasma samples collected at baseline (>5 days prior to study start) and sacrifice as well as lung tissue and BALF samples collected at sacrifice. Cytokine and chemokine levels (Mouse 31 plex Multiplex Immunoassay analyzed with a BioPlex 200 Cytokine Array, Assay Kit Millipore MILLIPLEX, performed by Eve Technologies, Calgary, Canada) were assessed in K₂-EDTA plasma, BALF, and lung tissue (homogenized in PBS 1x+0.1% Triton X-100 with protease cocktail inhibitors) collected at sacrifice and baseline (plasma only). A cell count differential was performed on BALF samples to assess leukocyte recruitment to the lung. Myeloperoxidase (MPO) and histamine levels were assessed in BALF samples by commercially available ELISA kits (Abcam ab155458 and Abcam ab213975).

[0304] Conclusions: CPN1-HSA administered prophylactically at the time of LPS intratracheal instillation significantly protected against the development of ARDS-like phenotype in mice. PenH, or enhanced respiratory pause, an index of pulmonary congestion associated with ARDS disease severity, increased on average 726% from healthy baseline levels in untreated animals (FIG. 9A). In animals receiving CPN1-HSA this was significantly reduced to an average of 337% greater than baseline (FIG. 9A). A similar but non-significant trend was observed in anti-C5 antibody treated animals. Both treatments demonstrated trends toward protection at 6 hours. The less prominent protection of anti-C5 antibody after 24 hours compared to CPN1-HSA suggests the protease had a longer pharmacodynamic window, possibly due to a longer circulating half-life or increased substrate processing.

[0305] Consistent with the physiologically protective role of CPN1-HSA, administration was associated with significant reductions in pro-inflammatory chemokines in key body compartments at 24 hours (FIGS. 10A-F). Specifically, circulating macrophage inflammatory protein-1 alpha (MIP-1α) was significantly reduced in CPN1-HSA treated animal plasma with slight reductions also observed in the lung but not BALF (FIG. 10A-C). Levels of RANTES (Regulated upon Activation, Normal T Cell Expressed and Presumably Secreted), a proinflammatory multi-functional chemokine secreted by lung epithelial cells, were significantly lower in the lung tissue in CPN1-HSA treated animals compared to untreated controls with a similar non-significant trend observed in anti-C5 antibody treated animals (FIG. 10F). These effects were not observed in plasma levels, consistent with the epithelial surface role of this chemokine in lung injury (FIG. 10D). Mice engineered to constitutively over-express RANTES demonstrate increased neutrophil migration to the airways. In mice, deletion of CCR1, a receptor that detects MIP-1α and RANTES, is associated with protection from inflammatory lung injury secondary to pancreatitis. The ability to modulate this signaling pathway speaks to the potential role of CPN1-HSA to similarly regulate other inflammatory pathologies due to complement dysregulation.

[0306] The findings demonstrated in this model confirm that CPN1-HSA administration is as efficacious, if not more, than anti-C5 antibody treatment to protect against LPS-induced development of ARDS in mice. The number of molecules of CPN1-HSA administered required to produce a comparable or superior effect was over 7-fold less compared to anti-C5 antibody dosing (Table 5, columns C and D). The assumptions for this calculation are described in Table 5 columns A and B below.

TABLE 9.1

In vivo study parameters				
Treatment	A	B	C	D
	Molecular Weight (g/mol)			Dose Level Fold-Difference (Anti-C5 Ab:CPN1-HSA)
CPN1-HSA	112,700	5.0	0.04	7.25
Anti-C5 Antibody	150,000 ^A	44.05 ^B	0.29	

^AMolecular weight estimated based on standard murine IgG;
^BDose level calculated using the mean body weight of anti-C5 treated animals at time of dosing (22.7 g)

Example 10: In Vivo Pharmacokinetics Assays in Rat Model

[0307] Examination of in vivo pharmacokinetics of CPN1-HSA was carried out in rats. The pharmacokinetics profile of pooled CPN1-HSA construct numbers 351 and 6 (the only difference being the presence of an 8 or 12 linker-length, respectively) was assessed following a single intravenous injection (IV) in adult male and female Sprague Dawley rats, four each for a total of eight animals. Animals were dosed at 2 mg/kg intravenously via tail vein injection. Blood samples were collected into EDTA pre-dose and at 0.5 h, 1 h, 3 h, 6 h, 12 h, 24 h, 48 h, 72 h, 120 h, 168 h, 240 h, and 336 h followed by plasma separation for PK bioassay analysis.

[0308] CPN1-HSA concentration in plasma was assessed via a quantitative sandwich enzyme electrochemiluminescence (ECL) antigen assay for detection of construct numbers 351 and 6 in rat EDTA plasma. For quantification, Meso Scale Discovery (MSD, Rockville, MA) assay plates were prepared by coating with 1 µg/ml of immunogen purified rabbit polyclonal capture antibody specific to the antigen (MyBioSource, San Diego, CA). Standards, QCs, and plasma samples were added to the plate wells and incubated to facilitate capture. Plates were washed followed by addition of 1 µg/ml of a biotinylated human HSA monoclonal antibody (Invitrogen, Waltham, MA) and 0.3 µg/ml SULFO-TAG Streptavidin (MSD, Rockville, MD) for detection. The plates were once again washed and MSD Read Buffer T (2×) added. Plates were read using a MSD Sector S 600 reporting ECL relative light units (RLU).

[0309] CPN1-HSA plasma concentrations were interpolated via standard curve and pharmacokinetic (PK) parameters derived from a noncompartmental analysis performed in Excel based on the determined concentrations. Area Under the Curve (AUC) was calculated using the linear trapezoidal method, AUC 0-inf was calculated from the AUC 0-t (determined up to the last measurable concentration) and then extrapolated to infinity using the estimated half-life. Mean Residence Time (MRT) was calculated using the moment theory method (Area Under the Moment Curve [AUMC 0-inf] divided by AUC 0-inf). Sufficient sampling was determined by confirming that the extrapolated AUC did not exceed 20% of total measurable AUC.

[0310] For PK parameter estimates, three subjects were included in the analysis. Of the eight total animals, one was excluded due to inconsistently high baseline detection compared to the other animals and four animals did not meet model criteria due to too few datapoints above assay Lower Limit Of Quantitation (LLOQ). Of note, a sample collected immediately post-dose was not obtained. Given the immediate and complete absorption of an IV dose into circulation, the C_{max} is likely under-estimated and the T_{max} over-estimated for this report. PK parameter estimates from the three animals assessed are summarized in FIG. 11A. The PK profiles of the 3 animals analyzed are depicted in FIG. 11B, and the representative PK profile is depicted in FIG. 11C.

[0311] The pharmacokinetic profile of CPN1-HSA is determined following a single subcutaneous injection in Sprague Dawley rats. Adult male and female rats weighing between 200 g and 250 g at the time are used and pair-housed during the acclimation period and experimental phase of the study. Prior to the study, a blood sample is taken for pre-dose measurement a minimum of one day prior to dosing.

[0312] Animals are injected subcutaneously with test compounds. Experimental groups are dosed at 6 mg/kg of CPN1-HSA subcutaneously. Subcutaneous doses are administered via bolus injection between the skin and underlying layers of tissue in the scapular region on the back of each animal.

Example 11: In Vivo Pharmacokinetics Assays in Primate Model

[0313] The pharmacokinetics profile of CPN1-HSA are determined following a single intravenous injection or single subcutaneous injection in cynomolgus macaques. Adult male and female macaques age 2-4 years at study start and weighing between 2-4 kg at study start are used.

[0314] On study day 0, the test article is delivered by a single intravenous bolus or a single subcutaneous injection to alert, chair-restrained adult male and female macaques. Experimental groups are dosed at either 2 mg/kg of CPN1-HSA intravenously or 6 mg/kg of CPN1-HSA subcutaneously. FIG. 12, Table 1 summarizes the dose levels and regimen of the test articles and vehicles.

[0315] Test articles and vehicles are administered by single intravenous or subcutaneous injection. For intravenous doses, 2 mg/kg of CPN1-HSA are administered as a bolus via catheter followed by approximately 5 mL sterile saline flush. The dose time is recorded at the time the saline flush is completed. For subcutaneous doses, 6 mg/kg of CPN1-HSA are administered to alert, chair-restrained animals by injection between shoulder blades.

[0316] For both intravenous and subcutaneous administration, baseline blood is collected immediately prior to dosing. Blood collections are performed in alert, chair-restrained animals. For animals dosed intravenously, blood is collected from the leg opposite the intravenous dosing site. FIG. 12, Table 2 summarizes the blood collection schedule.

[0317] Samples for plasma isolation are collected into tubes containing K₂-EDTA until processing. Samples are centrifuged at 2000×g for 10 minutes at 4° C. within 1 hour of collection. The presence of hemolysis is documented following centrifugation, and the maximum amount of plasma is recovered and frozen prior to analysis.

[0318] Samples for serum isolation are collected into serum separator tubes, stored at ambient temperature for at least 15 minutes or until blood is clotted, and then processed. Samples are centrifuged at 2000×g for 10 minutes at 4° C. within 1 hour of collection. The presence of hemolysis is documented following centrifugation, and the maximum amount of serum is recovered and frozen prior to analysis.

Example 12: Serum Stability Assays

[0319] Serum stability was measured for CPN1-HSA constructs at 37° C. for extended times. To quantify protein levels, semi-quantitative Western blots were performed with CPN1 primary antibodies. Constructs were dosed into serum and diluted 1:50, diluted in LDS, reduced with a reducing agent, and boiled for 5 minutes.

[0320] For the Western blot, 15 μ L of samples were loaded into each well, and the gel run at 100V for 10 minutes, then 150V for 90 minutes. Afterwards, the gel was transferred to a nitrocellulose membrane and blocked for 1 hour in 5% nfd milk at room temperature. Then, the blot was incubated with CPN1 primary antibodies at 1:1000 dilution in 5% nfd milk overnight at 4° C. The following day, the blot was subjected to three 10-minute TBST washes, then incubated with secondary antibody at 1:20000 dilution in 5% nfd milk for 1 hour at room temperature. Afterwards, the blot was again washed three times for 10 minutes with TBST. Lastly, the blot was developed and imaged with SuperSignal™ West Pico PLUS Chemiluminescent Substrate. FIGS. 13A and 13B depict Western blots for two exemplary CPN1-HSA constructs.

[0321] Using the semi-quantitative Western blot, a half-life curve was generated for CPN1-HSA constructs. FIGS. 13C and 13D depict half-life curves for exemplary CPN1-HSA constructs in serum.

[0322] Western blot analysis was also performed on plasma isolated from rats following intravenous or subcutaneous injection of CPN1-HSA constructs. FIGS. 14A and 14B depict two different rats' plasma samples and the presence of two exemplary CPN1-HSA constructs. Western blot analysis was then performed on plasma isolated from rats following subcutaneous injection of CPN1-HSA constructs, and is depicted in FIG. 14C. Lastly, Western blot analysis was performed on plasma derived from rats subjected to intravenous injection of human plasma purified CPN1, and is depicted in FIG. 14D.

Example 13: Protease Activity Assay

[0323] The relative activity of CPN constructs was measured via modified use of the Pefakit TAFI (thrombin activatable fibrinolysis inhibitor) assay, which determines the activity of CPN proteins on a synthetic substrate relative to a TAFI-containing plasma calibration curve. The assay provides no information on activity or enzyme kinetics of CPN proteins on native substrates, but serves as a high-throughput method for determining relative activities of CPN proteins.

[0324] Enzymatic hydrolysis of the synthetic substrate CPN protein generates colorimetric 5-mercapto-2-nitro-benzoic acid product. The rate of increase in absorbance is interpolated against a plasma calibration curve to yield the % TAFIa (thrombin activatable fibrinolysis inhibitor activated form) activity.

[0325] The activity was calculated in two ways. First, the product generation rate was interpolated from the plasma calibration curve and dilution corrected to yield a volumetric

activity in units of % TAFIa activity. The volumetric % TAFIa activity indicates the amount of calibrated TAFI-containing plasma required to achieve activity equal to the sample. This is useful for comparing samples with equal concentrations. The second activity was calculated by dividing the % TAFIa activity of the samples by the sample enzyme concentration to yield a specific activity in units of % TAFIa activity per μ M enzyme (μ M). The specific % TAFIa activity indicates the intrinsic activity of the CP protein in the sample regardless of the concentration. For example, if multiple samples have equally active CP protein at different concentrations, the volumetric activities will differ while the specific activities would not.

[0326] The specific activity of CPN constructs was measured and compiled into a graph depicted in FIG. 15. The bars marked with asterisks indicate improved performance over Const. No. 6.

Example 14: In Silico Design of CPN Constructs

[0327] CPN1 was reengineered based on structural understanding of the enzyme from crystallographic data of human CPN1 (PDB ID: 2 nsm) and homologous carboxypeptidases. A number of computational tools were used in the development of designs aimed at improving and modulating a number of CPN1 properties including stability, solubility, activity, and substrate selectivity. The primary tool used in these designs was the Rosetta suite of programs in addition to the docking program PIPER and molecular dynamics simulations via OpenMM.

[0328] A computational site-saturation mutagenesis experiment was run to identify positions in the native CPN1 structure that could be modified in order to improve stability. Clusters of amino acids around the identified hot spots were mutated for improved stability using the FastDesign algorithm of Rosetta. Molecular dynamics simulations of select mutations revealed further sites of instability which were rectified via the introduction of strategic disulfide bonds. Additionally, a workflow involving docking, molecular dynamics simulations, and homology modeling was used to determine the binding interaction between CPN1 and the complements C3a and C5a. These complexes served as the basis for CPN1 mutations that aimed at improving selectivity for either C3a or C5a. Further docking experiments with endogenous CPN1 substrates was used to identify potential prodomains that could be fused to the N-terminus of CPN1 with a cleavable linker to create a zymogen out of the enzyme, using the same cleavage site (SRVRIAT; SEQ ID NO: 72) found in thrombin activatable fibrinolysis inhibitor (TAFI).

SEQUENCE LISTING

The patent application contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US20250277200A1>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

1. A variant of a carboxypeptidase N catalytic subunit 1 (CPN1 variant) comprising at least one modification with respect to a wild type CPN1, wherein the CPN1 variant has at least one improved characteristic as compared to the wild type CPN1.

2. The CPN1 variant of claim 1, wherein the modification with respect to a wild type CPN1 comprises any one or more of: a substitution of one or more amino acid residues, a deletion of one or more amino acid residues, an insertion of one or more amino acid residues, an insertion of one or more CPN domains, and an insertion of one or more non-CPN domains or components.

3. The CPN1 variant of claim 1, wherein the at least one improved characteristic is selected from an increase or a decrease in any one or more of: half-life, activity, potency, substrate affinity, substrate specificity, substrate selectivity, proteolytic sensitivity, cofactor affinity, and catalytic capability.

4. The CPN1 variant of claim 3, wherein the at least one improved characteristic comprises an increase in affinity for one or more substrates, and wherein at least one substrate is C3a.

5. The CPN1 variant of claim 1, wherein the at least one improved characteristic comprises:

- (a) an increase in affinity for one or more substrates, and wherein at least one substrate is C5a;
- (b) an increase in the cleavage of C3a and/or C5a;
- (c) an increased k_{cat}/K_M ($M^{-1} s^{-1}$) for cleavage of C3a and/or C5a;
- (d) a decreased K_D (nM) value for cleavage of C3a and/or C5a;
- (e) a decreased EC_{50} (nM) value for cleavage of C3a and/or C5a, compared to the wild type CPN1;
- (f) increased half-life observed in plasma; or
- (g) a combination of (a)-(f).

6-16. (canceled)

17. The CPN1 variant of claim 1, wherein the CPN1 variant comprises at least one modification corresponding to a wild type non-human CPN1.

18. The CPN1 variant of claim 1, wherein the CPN1 variant comprises at least one modification corresponding to a wild type human CPN1.

19. The CPN1 variant of claim 18, wherein the CPN1 variant comprises at least one modification corresponding to a wild type CPN1 comprising the amino acid sequence as set forth in SEQ ID NO: 6.

20. The CPN1 variant of claim 1, wherein the CPN1 variant comprises an amino acid sequence having at least 70% sequence identity to SEQ ID NO: 6.

21. The CPN1 variant of claim 1, wherein the CPN1 variant comprises the amino acid sequence as set forth in any one of SEQ ID NOs: 7-8 and 11-12, or an amino acid sequence having at least 70% sequence identity to any one of SEQ ID NOs: 7-8 and 11-12.

22-24. (canceled)

25. The CPN1 variant of claim 1, wherein the CPN1 variant comprises the amino acid sequence of SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 11, or SEQ ID NO: 12, wherein the amino acid sequence further comprises a modification string, and wherein the modification string is selected from the group consisting of the modification strings provided in Table 3A, Table 3B, Table 3C, Table 4A, Table 4B, and Table 4C.

26. (canceled)

27. A fusion construct comprising a carboxypeptidase N catalytic subunit (CPN1) or variant thereof.

28. The fusion construct of claim 27, comprising an amino acid sequence selected from the group consisting of SEQ ID NO: 73-SEQ ID NO: 485.

29. The fusion construct of claim 27, wherein the fusion construct comprises the amino acid sequence of SEQ ID NO: 6, or an amino acid sequence having at least 70% sequence identity to SEQ ID NO: 6.

30. (canceled)

31. The fusion construct of claim 27, wherein the CPN1 variant comprises at least one modification with respect to a wild type CPN1, wherein the CPN1 variant has at least one improved characteristic as compared to the wild type CPN1.

32. The fusion construct of claim 27, wherein the CPN1 variant comprises at least one modification corresponding to a wild type CPN1 comprising the amino acid sequence as set forth in SEQ ID NO: 6.

33. The fusion construct of claim 27, wherein the CPN1 variant comprises the amino acid sequence as set forth in any one of SEQ ID NOs: 7-8 and 11-12, or an amino acid sequence having at least 70% sequence identity to any one of SEQ ID NOs: 7-8 and 11-12.

34-36. (canceled)

37. The fusion construct of claim 27, wherein the CPN1 variant comprises the amino acid sequence of SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 11, or SEQ ID NO: 12, wherein the amino acid sequence further comprises a modification string, and wherein the modification string is selected from the group consisting of the modifications provided in Table 3A, Table 3B, Table 3C, Table 4A, Table 4B, and Table 4C.

38. (canceled)

39. The fusion construct of claim 27, wherein the fusion construct comprises:

- (a) a Glutathione S transferase (GST) amino acid sequence;
- (b) a mammalian maltose binding protein (mMBP) amino acid sequence;
- (c) a small ubiquitin modifying enzyme (SUMO) amino acid sequence;
- (d) a Tobacco Etch Virus protease cleavage site (TEV) amino acid sequence;
- (e) an activation peptide of CBP2 (the N-terminal 96aa of a CBP2 protease) amino acid sequence;
- (f) a Factor Xa protease cleavage site (Xa) amino acid sequence;
- (g) a portion of a regulatory CPN2 subunit amino acid sequence;
- (h) a CD180 amino acid sequence;
- (i) a LR1G1 amino acid sequence;
- (j) at least one non-CPN1 or non-CPN2 domain or component;
- (k) an activation peptide that increases sensitivity of the fusion construct; or
- (l) a half-life extender.

40-63. (canceled)

64. The fusion construct of claim 27, wherein the fusion construct is non-immunogenic, is in a zymogen form, and/or is in an active form.

65-66. (canceled)

67. A method of treating a disease or condition in a subject in need thereof, comprising administering to the subject the CPN1 variant of claim 1.

68-76. (canceled)

77. A nucleic acid encoding the CPN1 variant of claim **1**.

78. A pharmaceutical composition comprising the CPN1 variant of claim **1**, and optionally a pharmaceutically acceptable carrier.

79. A method of treating a disease or condition in a subject in need thereof, comprising administering to the subject the fusion construct of claim **27**.

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